Hydrodynamical analogues of some quantum phenomena

Thought-provoking **wave-particle duality** hydrodynamical experiments

using classical de Broglie-Bohm-like "pilot wave"-coupled **walking droplets,**

"single particles having wave-mediated interaction with their own past"

for double-slit **interference**(?), **tunneling, orbit quantization, Zeeman effect, quantum statics … also Aharonov-Bohm, Casimir effect**

Feynman 1995: "I think I can safely say that nobody understands QM"

So how far can we take intuitions from hydrodynamical analogues?

Which translate to QM, which don't? **What are the proper intuitions?**

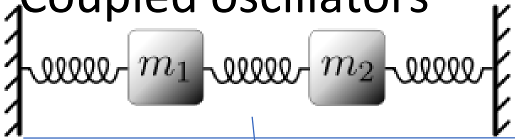
Videos: Veritasium (3.7M views), "Through the wormhole", MIT1, 2, 3, John Bush1, 2, 3

http://dualwalkers.com/ - lots of materials, Yves Couder 2013, 2017 lecture

Jarek Duda

Where is CM-QM boundary? "Classical" vs "Quantum"

Coupled oscillators



$$m\ddot{u}_1 = -Cu_1 + C(u_2 - u_1)$$
$$m\ddot{u}_2 = -Cu_2 + C(u_1 - u_2)$$

"unitary evolution" of normal modes

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \mathrm{Re}\left(c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_1 t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_2 t}\right)$$

Crystal

Phonons – quasiparticles in QFT

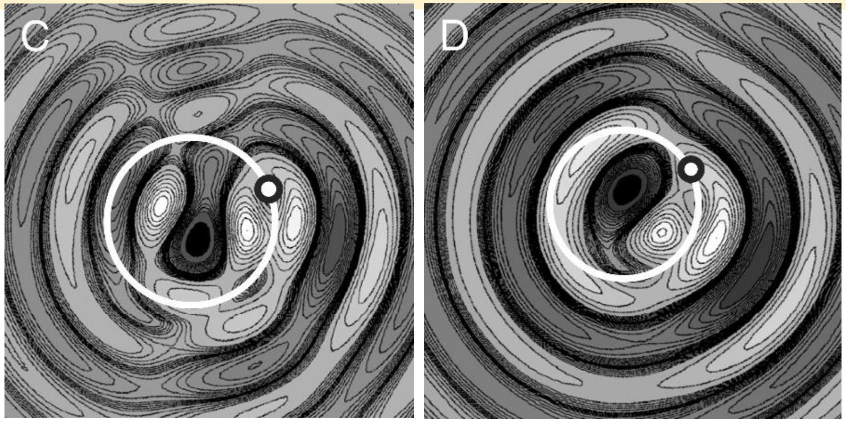
1D lattice of oscillators $\quad m\ddot{u}_x = C(u_{x+1} - 2u_x + u_{x+1})$ $\quad u_x = \mathrm{Re}\left(\sum_k c_k\, e^{i(kx-\omega_k t)}\right)$

Infinitesimal lattice limit → field, like in **hydrodynamics**

e.g. Couder's

Quantized

orbits



Schrödinger equation:

1) coupled **standing wave** $\psi_0 e^{iEt/\hbar}$, resonant with electron's clock,

2) $\rho = |\psi|^2$ **statistics**, e.g. MERW?, arg ψ: expected $\Delta\psi$?

Stable field configurations: **solitons** (topological?)

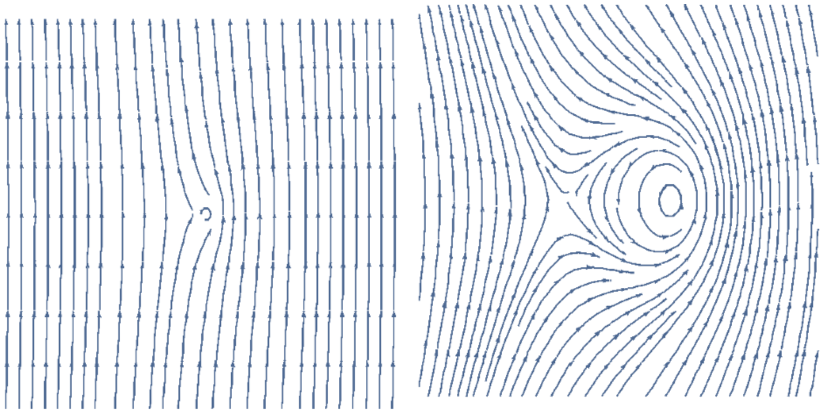
perturbative QFT "algebra for particles"

ensemble of scenarios: Feynman diagrams

Varying

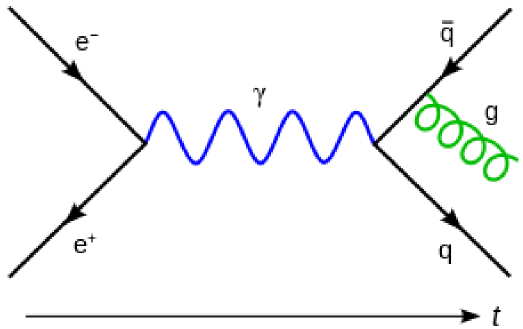
particle

Numbers



$\xrightarrow{\textit{description?}}$

$\xleftarrow[\textit{fields?}]{}$



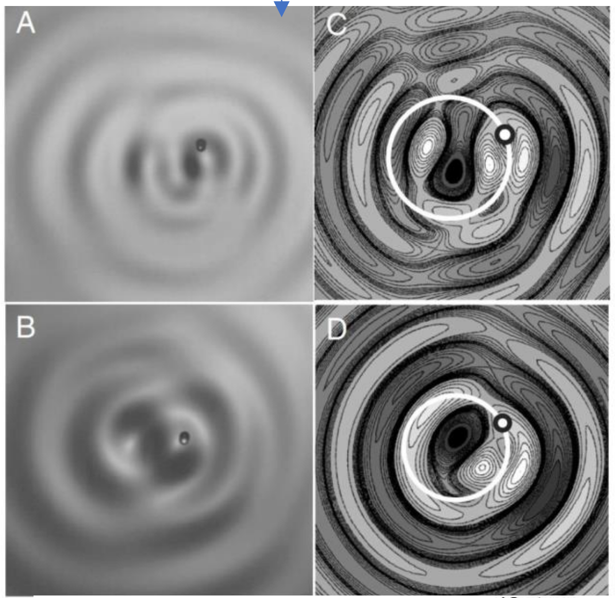
$$\psi = \sqrt{\rho}\exp(i\phi)$$

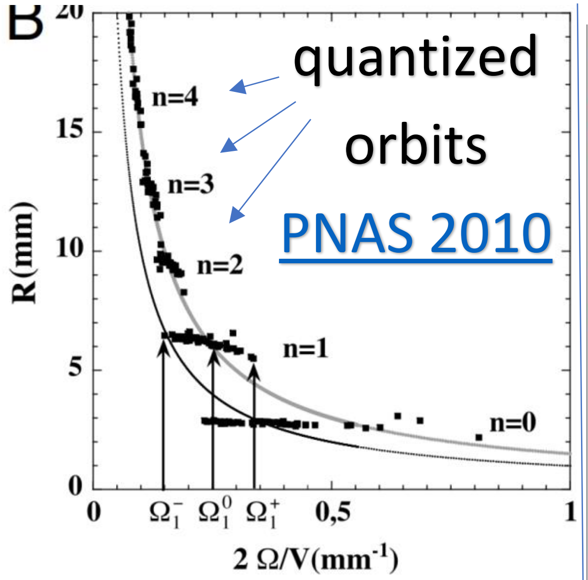
Simultaneous both wave and particle e.g. in [Afshar] or [nature.com/articles/ncomms7407]

What are quantum states? (Meaning: real objects or only a tool?)

Probability distributions represent averaging (e.g. Feynman, mean-field-approximation) and our incomplete knowledge, Schrödinger equation can be derived from Feynman path ensembles - these are tools of **effective models**.

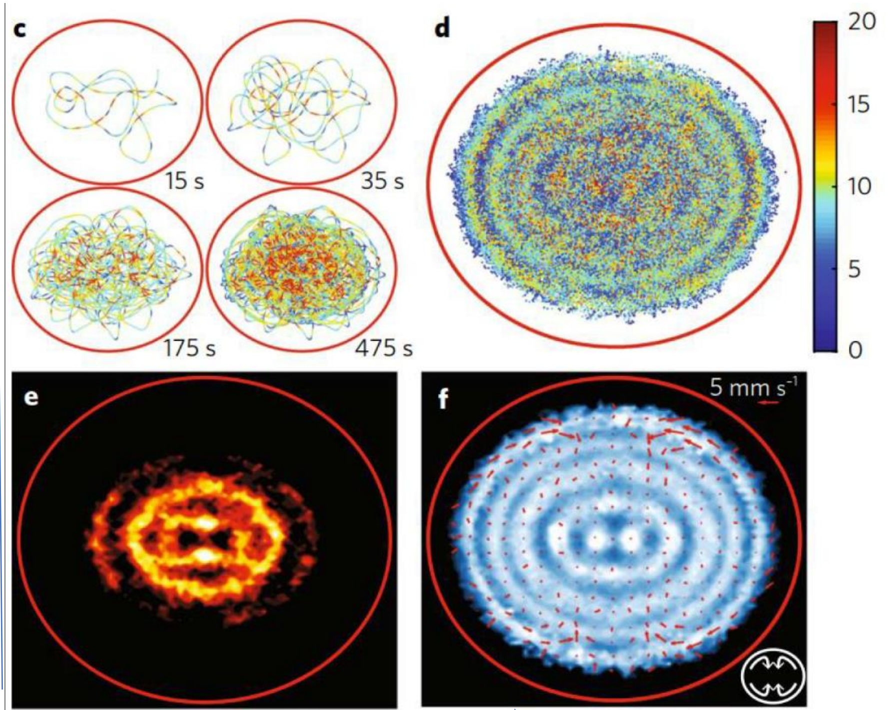
But we have **wave-particle duality**: probability focuses on **corpuscle**, but there is also **coupled "pilot" wave** – which in hydrodynamical analogs e.g. [has to become standing wave for quantization] … **Schrodinger hides both natures.**

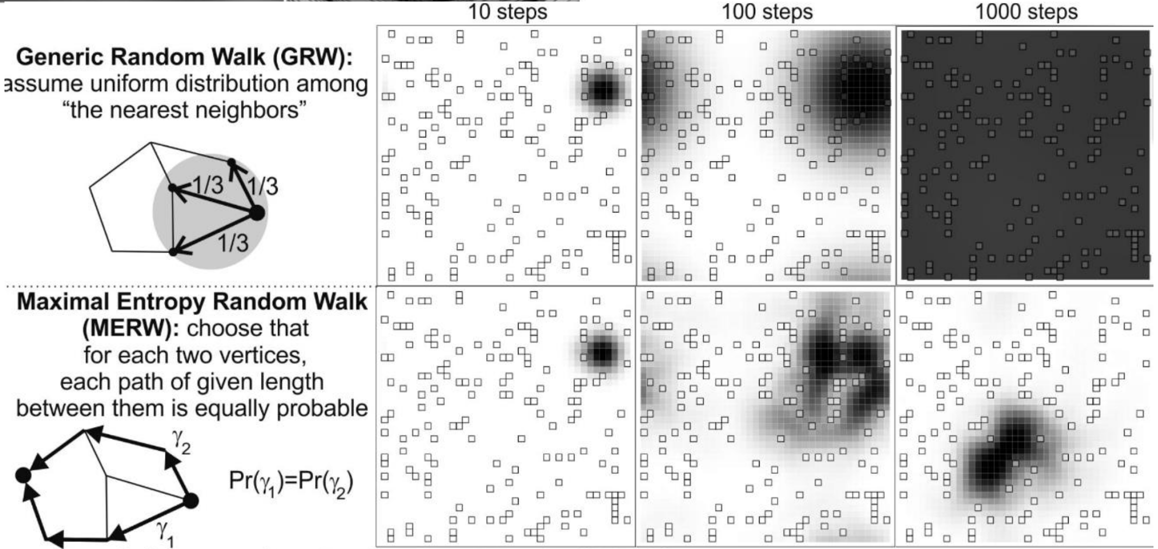




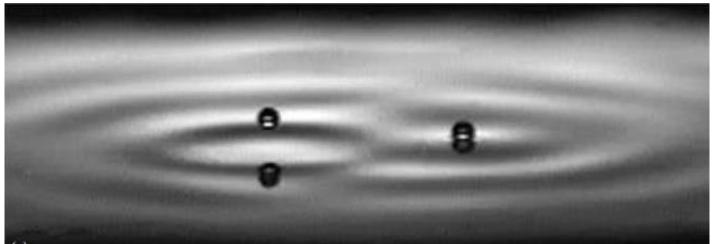
quantized orbits

[PNAS 2010]





[MERW] … droplet trajectories average to QM-like [PRE 2013]



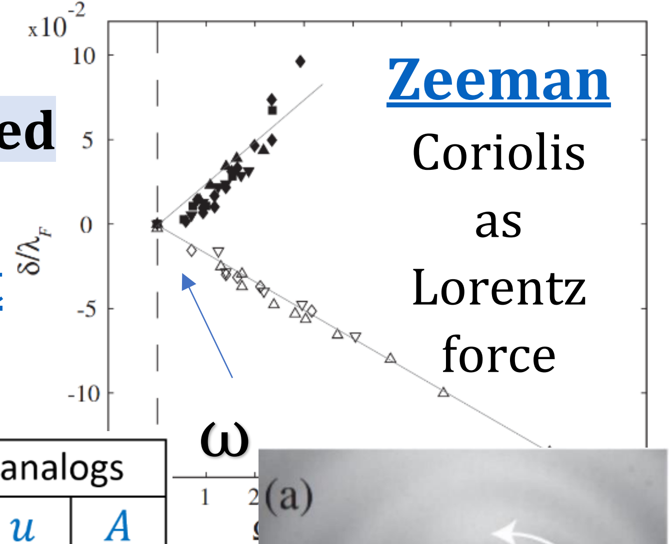
electromagnetism(c) ↔hydrodynamics(c_s) analogy

(superconducting) ... **Barnett effect: B from ω for uncharged**

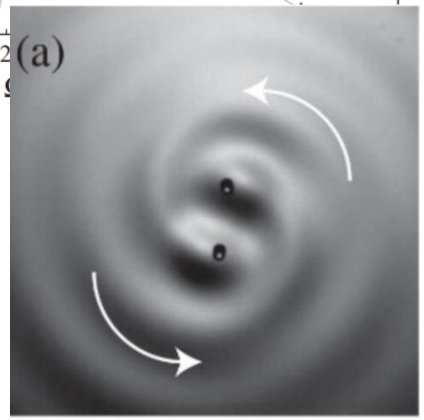
$\boldsymbol{B} = \nabla \times \boldsymbol{A} \leftrightarrow \boldsymbol{\omega} = \nabla \times u$ for $A \leftrightarrow u$ velocity, vorticity ω

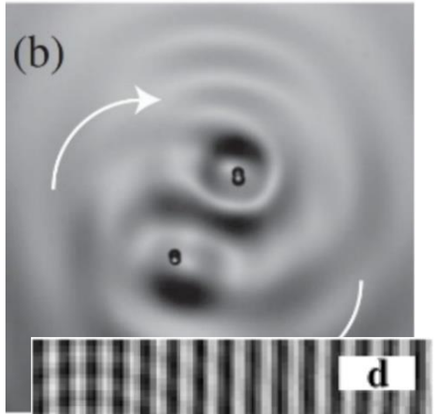
Lorentz force ↔ Coriolis force e.g. for Zeeman effect

current/flow e.g. in superfluid: $\vec{J} \sim \vec{v}_s m = \hbar\vec{\nabla}\varphi - q\vec{A}$



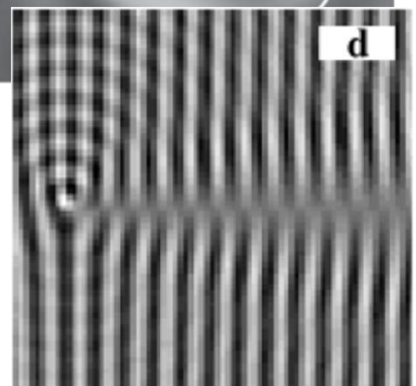
Zeeman
Coriolis
as
Lorentz
force





Hydrodynamics superfluid $\nu = 0$	**Electromagnetism**	analogs	
Navier-Stokes equation $\frac{\partial u}{\partial t} = -l - \nabla\phi_p + \nu\nabla^2 u$, $\phi_p = p/\rho + u^2/2$	$\frac{\partial A}{\partial t} = -E - \nabla\phi$ vector and scalar potential; E - electric field	u, l	A, E
vorticity $\omega = \nabla \times u$ $\nabla \cdot \omega = 0$; Lamb vector $l \equiv \omega \times u$	$B = \nabla \times A$ $\nabla \cdot B = 0$ Coulomb Thomson; magnetic induction	ω	B
vorticity tendency $\frac{\partial \omega}{\partial t} = -\nabla \times l + \nu\nabla^2\omega$	$\frac{\partial B}{\partial t} = -\nabla \times E$ Faraday's law		
Lamb vector tendency and turbulent current (j) $\frac{\partial l}{\partial t} = \nabla \times \eta - j$	$\frac{\partial(\epsilon_0 E)}{\partial t} = c^2\nabla \times H - \mu_0 J$ (zero polarization) Amper's law	η	H
vorticity field strength (η) and magnetization $\eta = \lvert u^2\rvert\omega - M$, $M = u(u\cdot\omega) + \nu\nabla^2 u$	$H = \mu_0^{-1}B - M$ magnetic field strength and magnetization		
turbulent charge density $\nabla \cdot l = u \cdot \nabla \times \omega - \lvert\omega^2\rvert = \rho_n$	$\nabla \cdot (\epsilon_0 E) = \rho_e$ electric charge density link	ρ_n	ρ_e

Theory	Gauge fields	Circulation	Gauge condition	Matter field
Electro-dynamics	φ , $\vec{A}$ four-potential	$\vec{B} = \vec{\nabla} \times \vec{A}$ magnetic f.	$\vec{\nabla} \cdot \vec{A} + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} = 0$	$\vec{E}_e = -\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi$
Hydro-dynamics	$\chi = v^2/2, \vec{v}$ flow velocity	$\vec{\omega} = \vec{\nabla} \times \vec{v}$ vorticity	$\vec{\nabla} \cdot \vec{v} + \frac{1}{c_s^2}\frac{\partial\chi}{\partial t} = 0$	$\vec{E}_h = -\frac{\partial\vec{v}}{\partial t} - \vec{\nabla}\chi$



link

Aharonov-Bohm (Michael Berry) electron~hydro:

Adding particles: localized field configurations – topological solitons (slides)

Liquid crystal **long-range interactions** due to **nontrivial vacuum** like for **Higgs** $V(\vec{n}) = (|\vec{n}|^2 - 1)^2$

$F \sim 1/D$:"**Annihilation** dynamics of topological defects induced by microparticles in nematic liquid crystals" Soft Matter

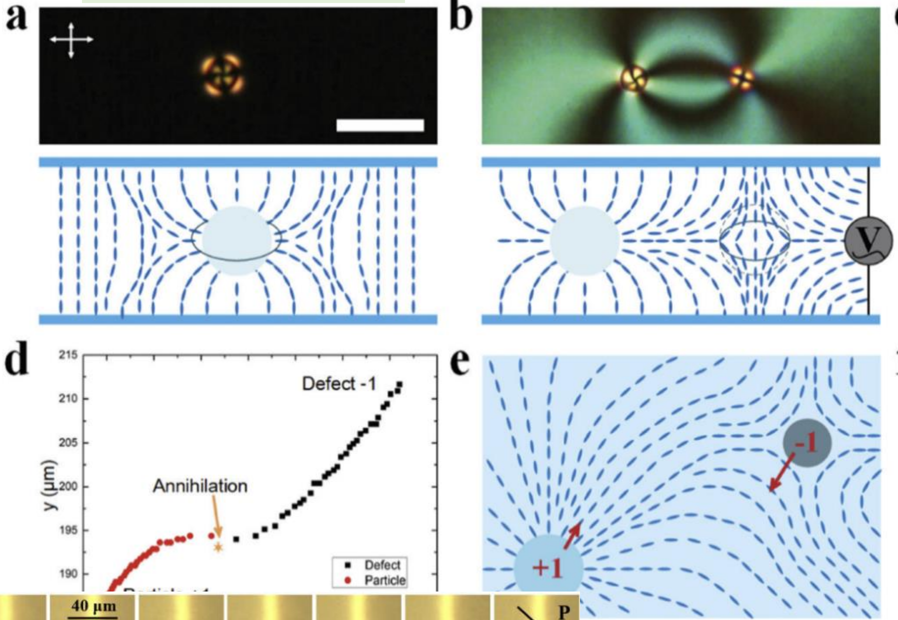
Coulomb: "**Coulomb**-like interaction in nematic emulsions induced by external torques exerted on the colloids" PRE

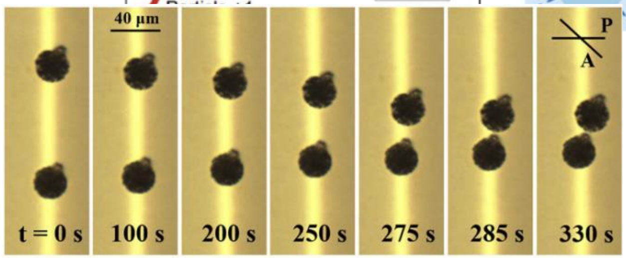
"**Coulomb**-like elastic interaction induced by symmetry breaking in nematic liquid crystal colloids" Scientific Reports

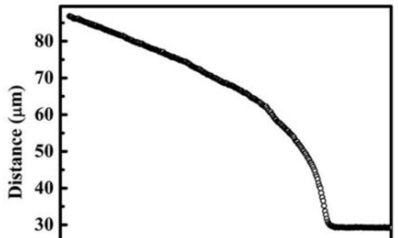
dipole-dipole: "Novel Colloidal Interactions in Anisotropic Fluids" Science **repulsion** Nature

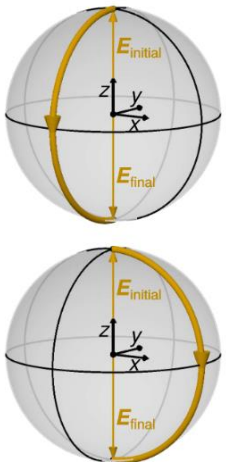
quadrupole-quadrupole: "**Long-range forces** and aggregation of colloid particles in a nematic liquid crystal" PRE

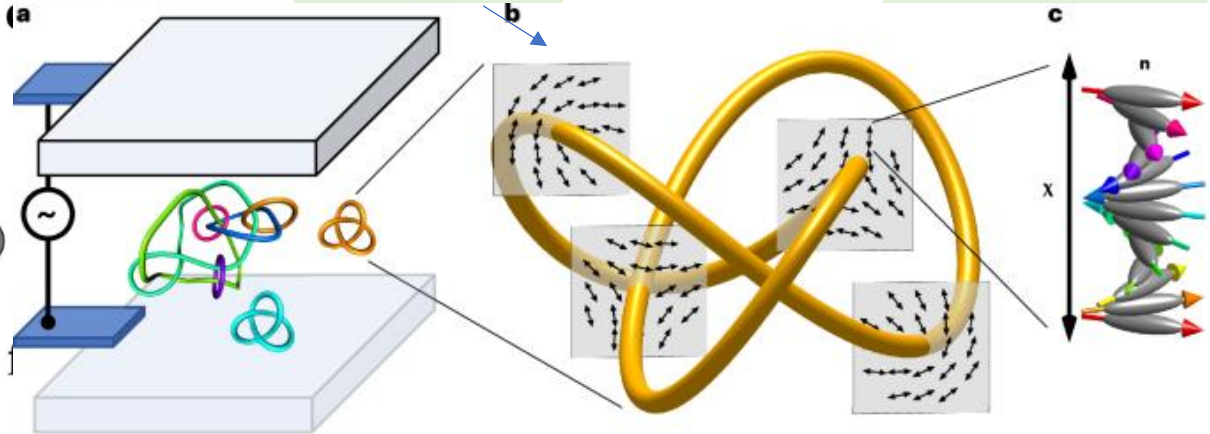
"**Fusion and fission** of particle-like chiral nematic vortex knots", baryon number (Nature Physics) **EM-hydro analogy**

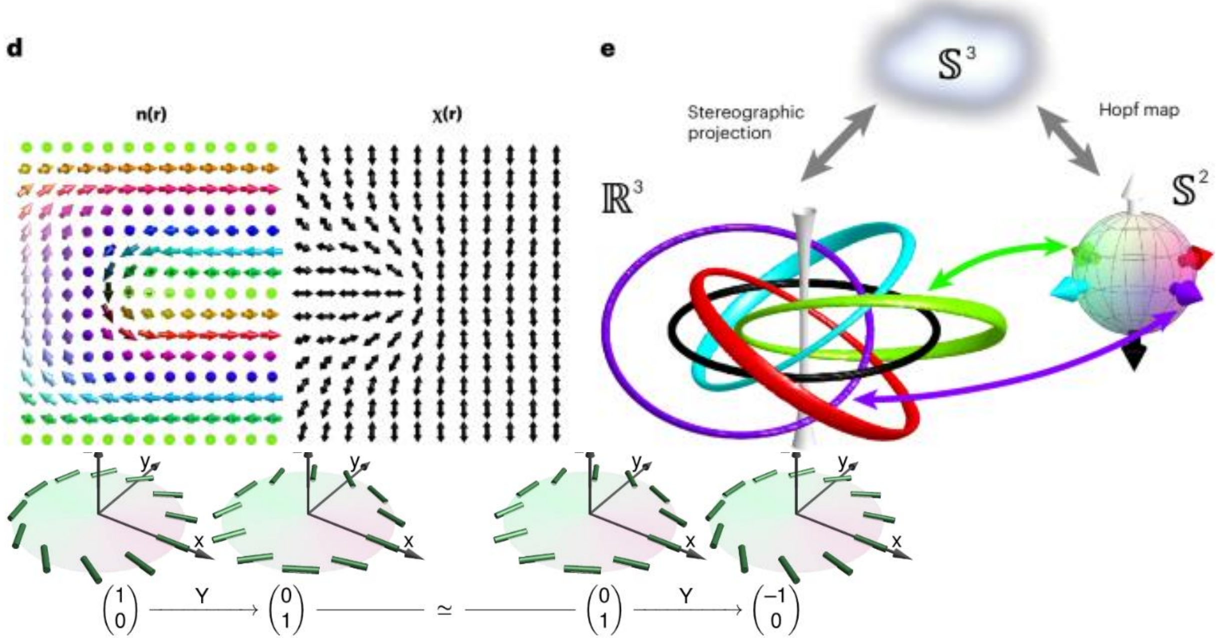






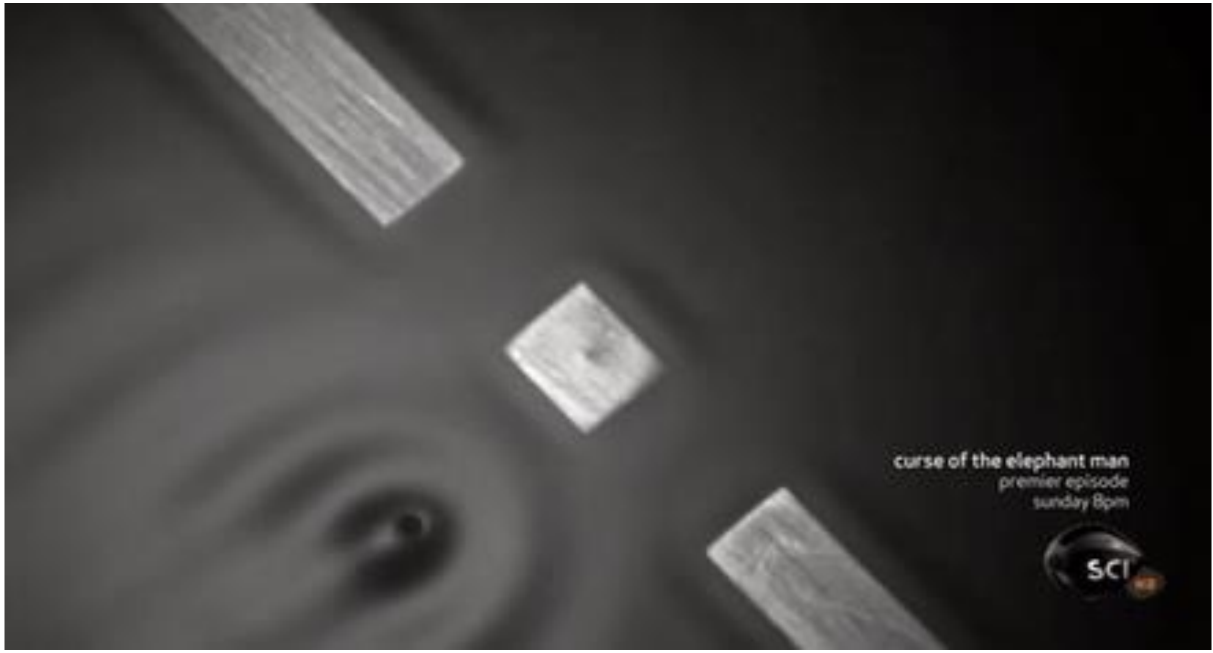






de Broglie-Bohm interpretation

(http://en.wikipedia.org/wiki/**Pilot_wave**)



Madelung substitution:

$$\psi(\boldsymbol{x},t)=\sqrt{\rho(\boldsymbol{x},t)}\cdot\exp\left(i\,\frac{S(\boldsymbol{x},t)}{\hbar}\right)$$

into Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}\psi=\left(-\frac{\hbar^2}{2m}\nabla^2+V\right)\psi$$

getting continuity equation:

$\frac{\partial\rho}{\partial t}+\nabla\cdot(\rho\boldsymbol{v})=0$ for particle density, of velocity: $\boldsymbol{v}=\frac{\nabla S}{m}=\frac{\hbar}{m}\,\mathrm{Im}\left(\frac{\nabla\psi}{\psi}\right)$

and Hamilton-Jacobi equation for phase/wave $\left(H+\frac{\partial S}{\partial t}=0\quad,\quad H=\frac{mv^2}{2}+V\right)$

$$-\frac{\partial S}{\partial t}=-\frac{(\nabla S)^2}{2m}+V+Q$$

where

$$Q=-\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}$$

QM = classical mechanics $(\hbar=0)$ + interaction with "**pilot wave**" (Q)

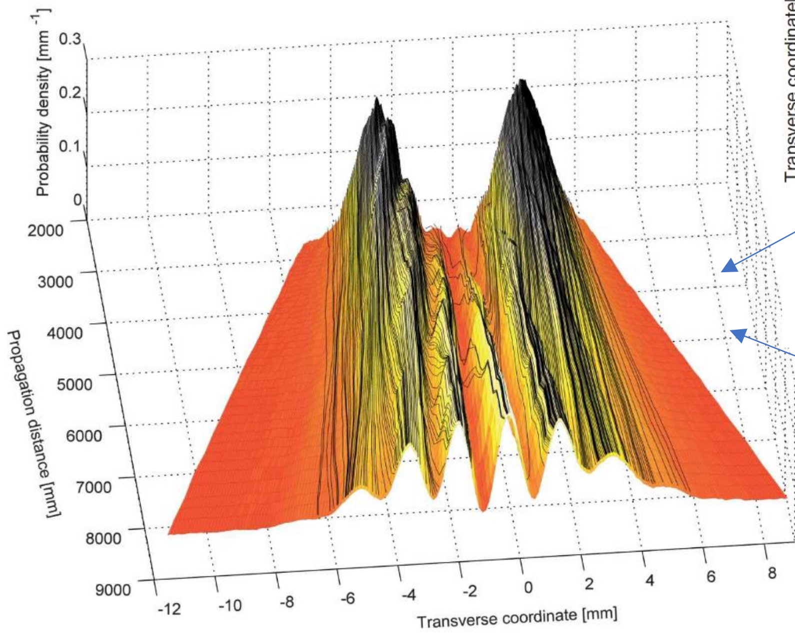
WKB semiclassical approximation – expand around small $\hbar$

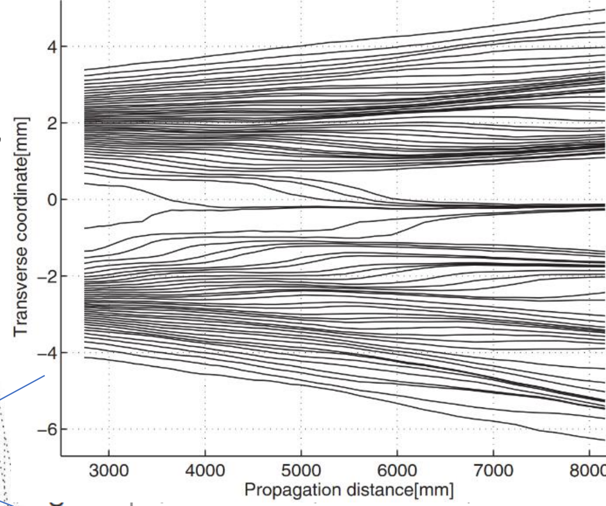
Observing the Average Trajectories of Single Photons in a Two-Slit Interferometer,

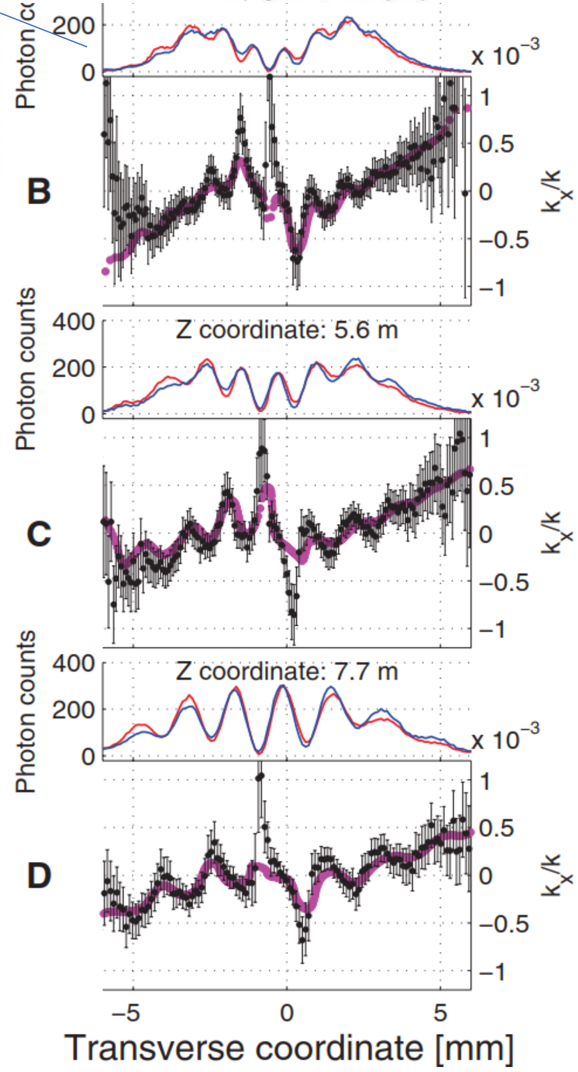
Science 2011 (500+ citations)

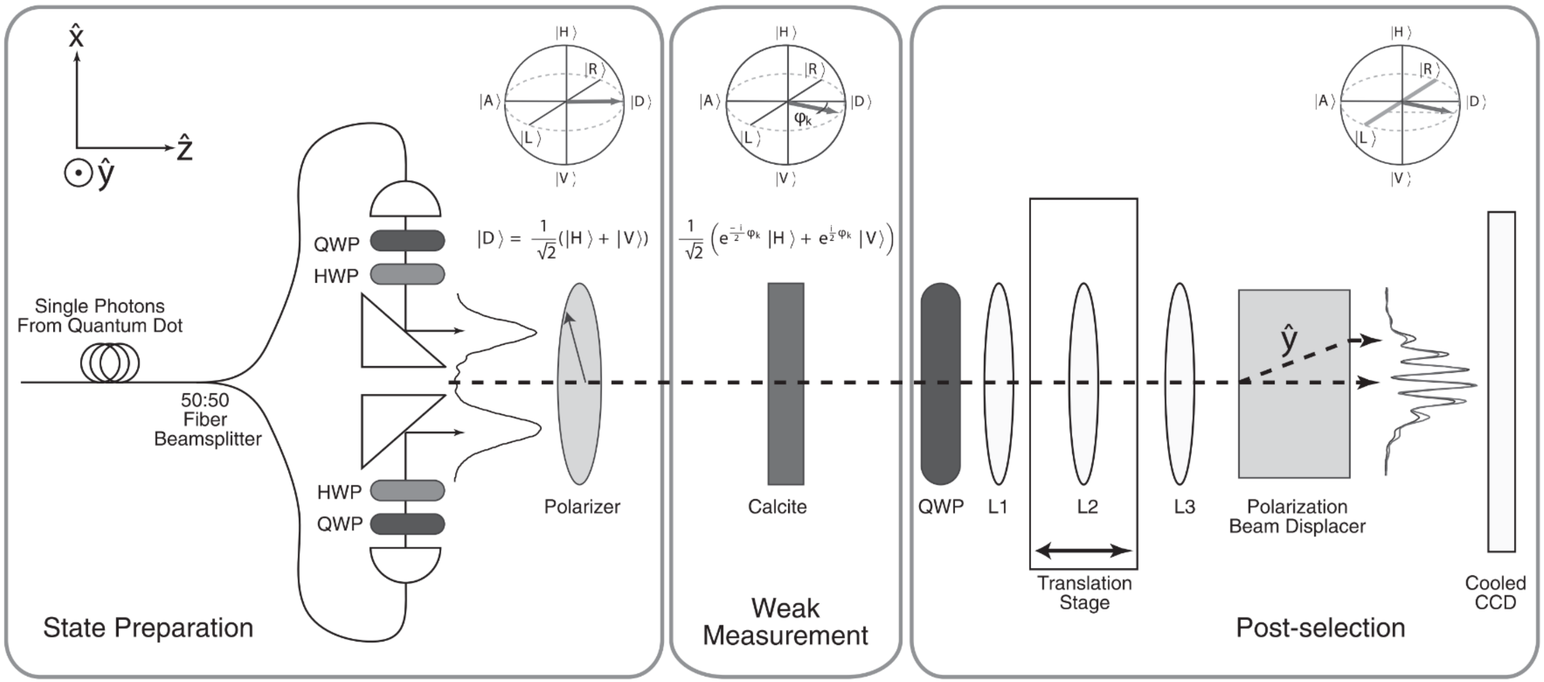
"Single-particle trajectories measured in this fashion **reproduce those predicted by the Bohm-de Broglie interpretation of quantum mechanics**, although the reconstruction is in no way dependent on a choice of interpretation."

$$\frac{k_x}{|\mathbf{k}|} = \frac{1}{\zeta}\left[\sin^{-1}\left(\frac{I_R - I_L}{I_R + I_L}\right)\right]$$









What propels these coupled "pilot" waves?

Observation of **electron's clock** ($\approx 10^{21} Hz$) – resonating with crystal lattice

De Broglie: $\boldsymbol{E} = mc^2 = hf$ Schrödinger: $\psi = \psi_0 e^{i\boldsymbol{E}t/\hbar} = \psi_0 e^{2\pi i f t}$

Compton wavelength $\lambda_C = \frac{c}{f_0} \approx \frac{r_{Bohr}}{2\pi \cdot 137} \approx 2\mathrm{pm}$ **light distance in "one tick"**

Time dilation: $f = \frac{f_0}{\gamma}$ for $\gamma = 1/\sqrt{1 - v^2/c^2}$

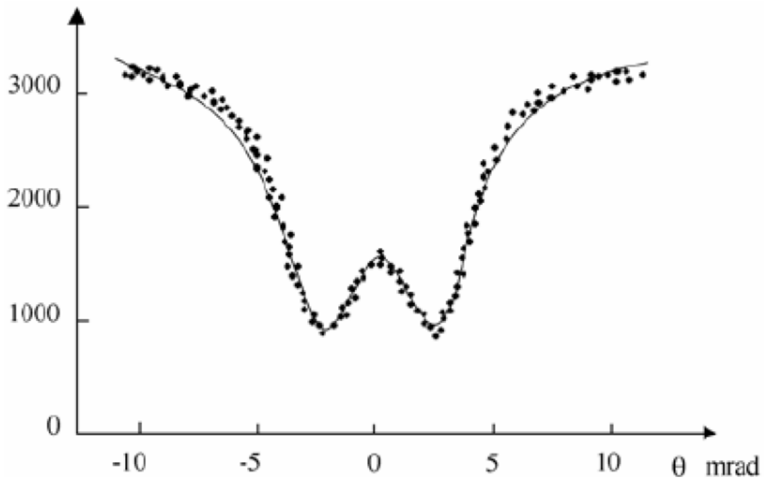
$$L = \frac{v}{f} = \frac{hv\gamma}{m_0 c^2} = \frac{hp}{(m_0 c)^2}$$

<011> axis of silicon crystal has $L = 3.84$ Å

$$cp = m_0 \gamma v = \frac{L (m_0 c)^2}{hc} \approx \frac{3.84 \text{Å} (0.511 \text{MeV})^2}{0.0124 \text{ MeV Å}} \approx 80.87 \text{ MeV}$$

Along <110> axis of silicon crystal atomic spacing correspond to 'internal clock' period for $E \approx 81 MeV$ electrons – observed resonance:

P. Catillon, N. Cue, M.J. Gaillard, R. Genre, M. Gouanère, R.G. Kirsch, J.-C. Poizat, J. Remillieux, L. Roussel, M. Spighel, *A Search for the de Broglie Particle Internal Clock by Means of Electron Channeling,* Found Phys (2008) 38: 659–664



A maritime analogy of the Casimir effect, AJP 1996

A water wave analog of the Casimir effect, AJP 2009

video of experiment

Restricted waves between plates – lower energy - attraction

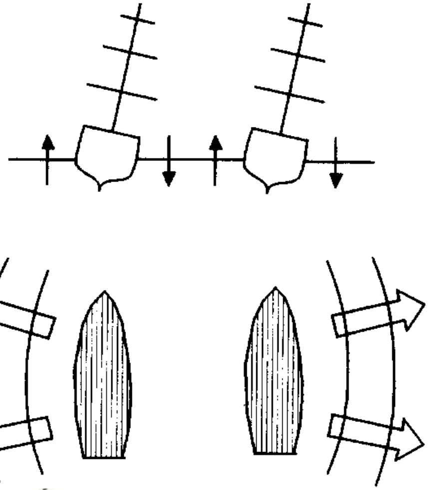
However, in water we need energy source,

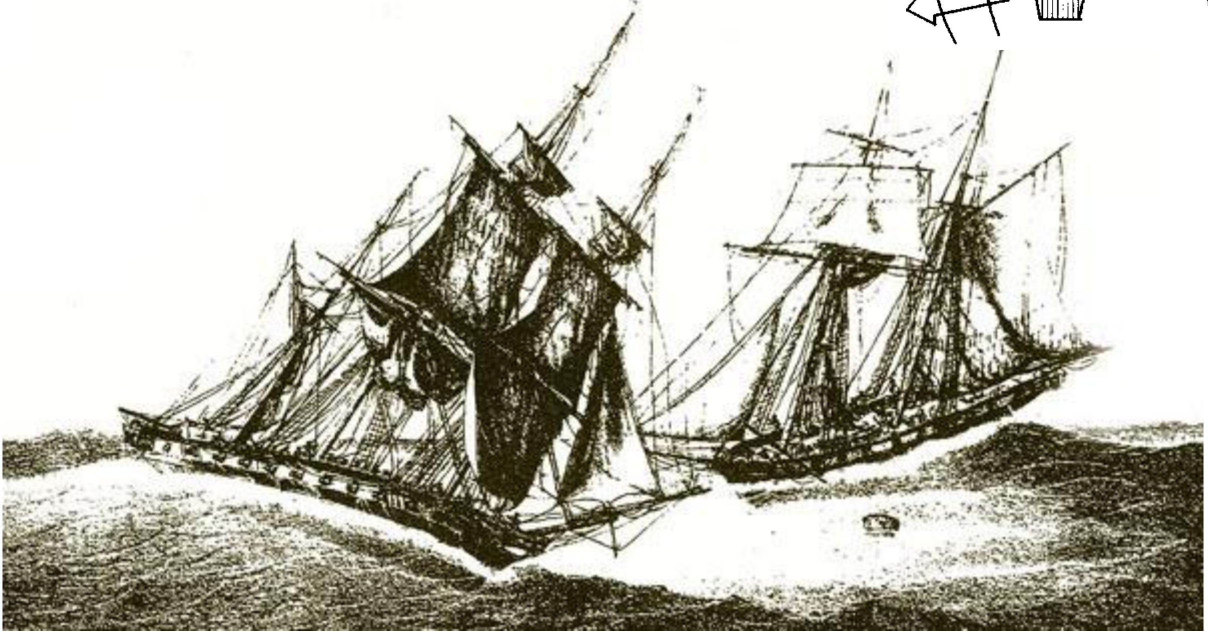
in QED “vacuum energy” (clock?),

regularization needed

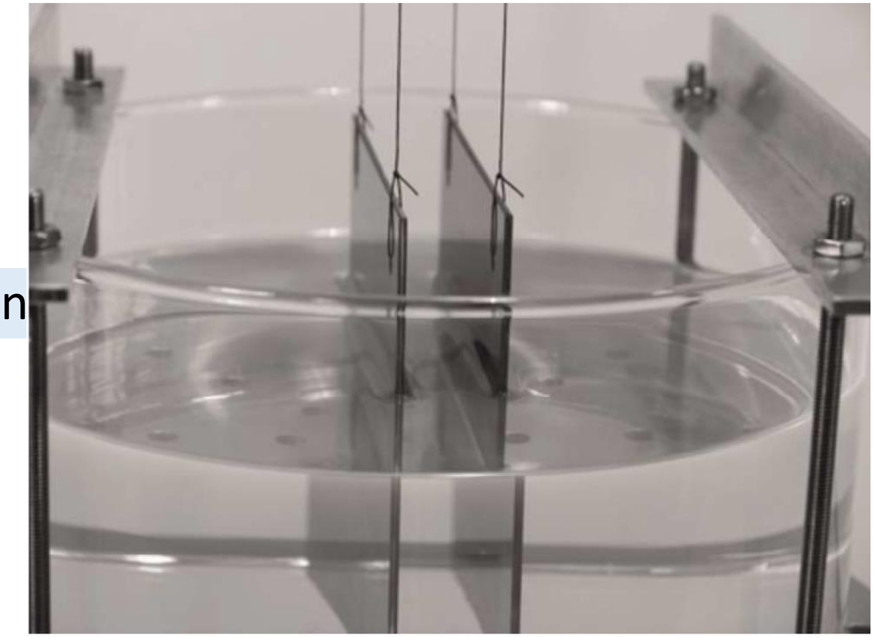
e.g. $\sum_{n=1}^{\infty} n = -\frac{1}{12}$

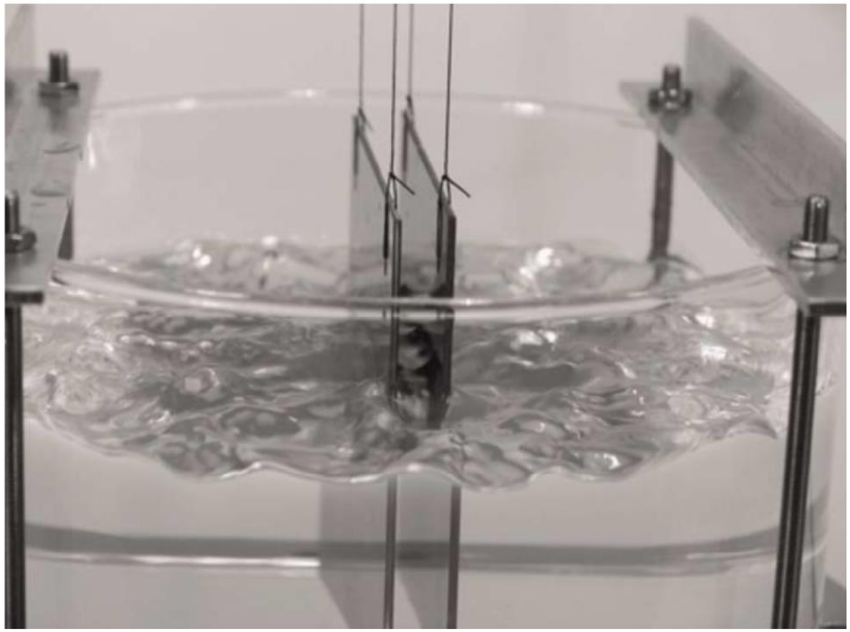
(a)





(b)

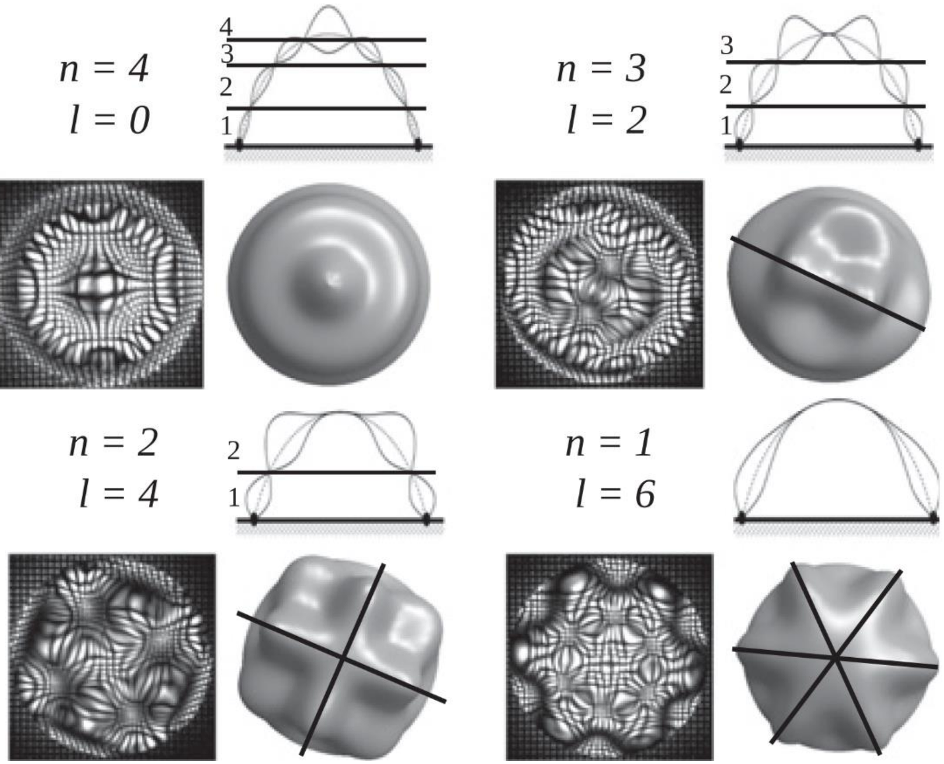


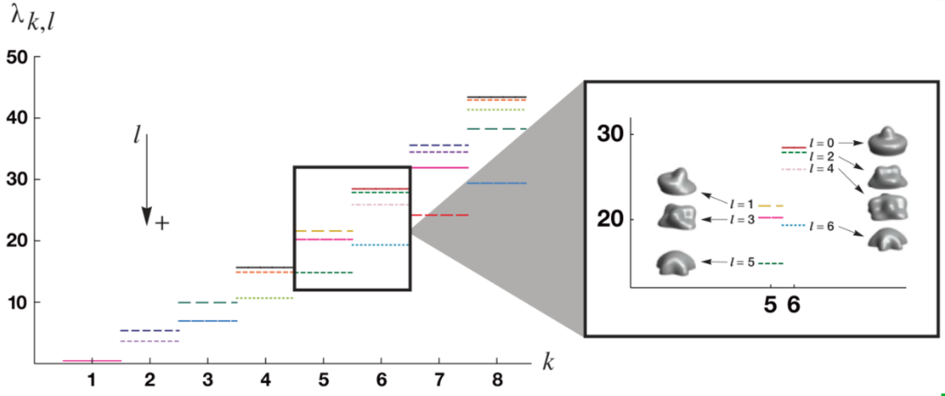


[Droplet motion fill a periodic table](#)

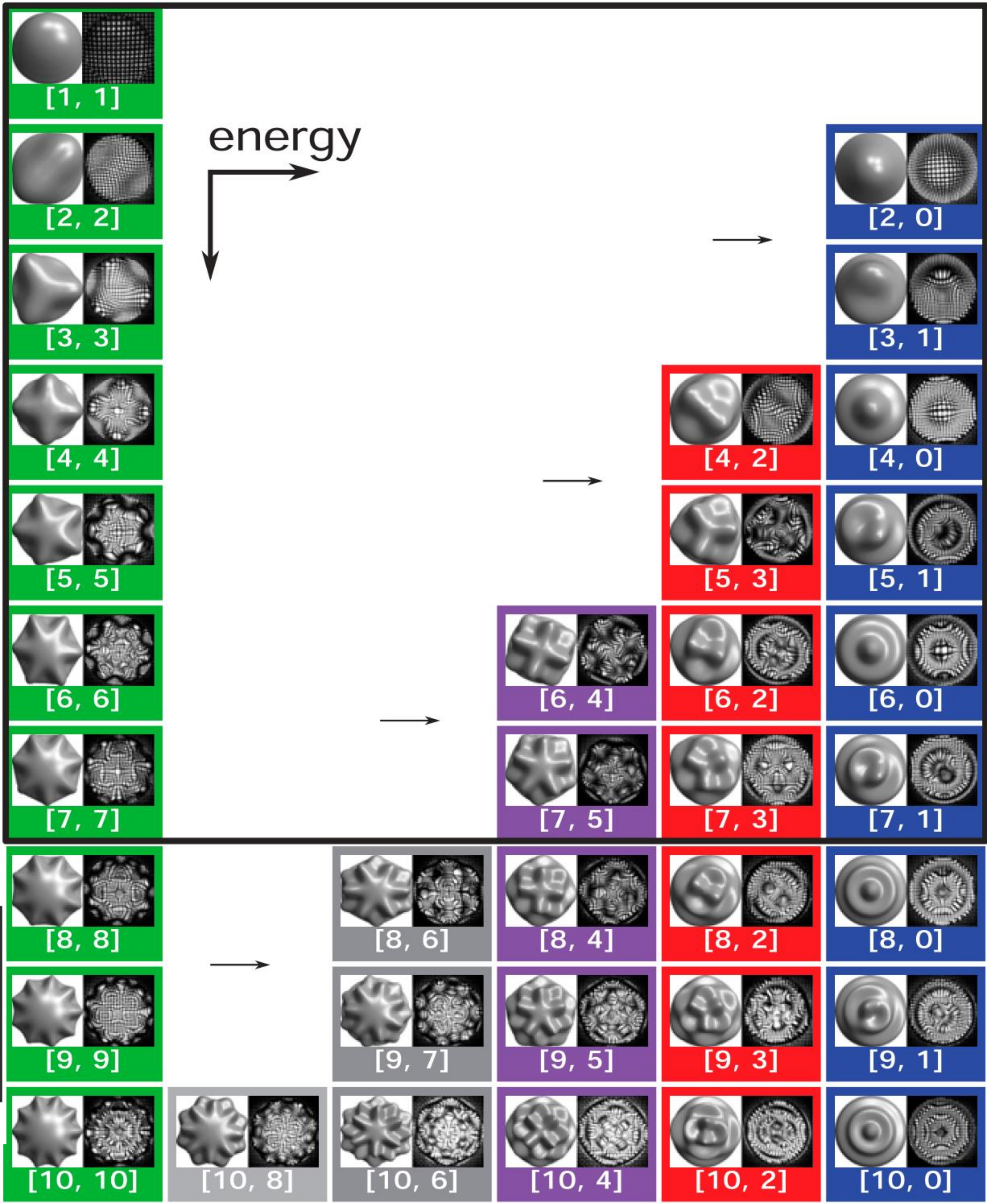
[Steen](#), Chang, Bostwick, **PNAS** 2019

double quantization of vibrations





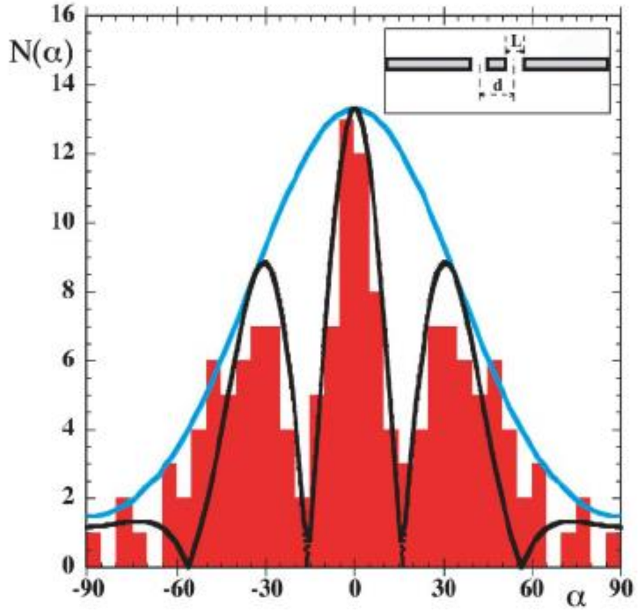
$k = l + 2\,(n - 1)$



Wave-particle duality realized with walking droplets

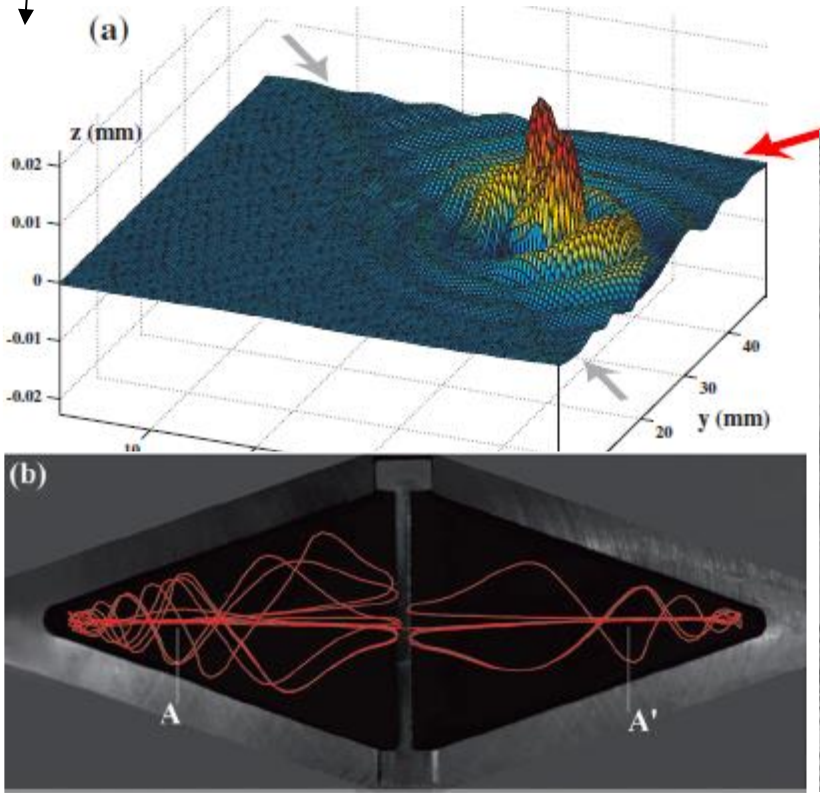
Bouncing droplet on vertically vibrated bath is **coupled to the surface waves it generates**. Becomes a “walker” moving at **constant velocity**.

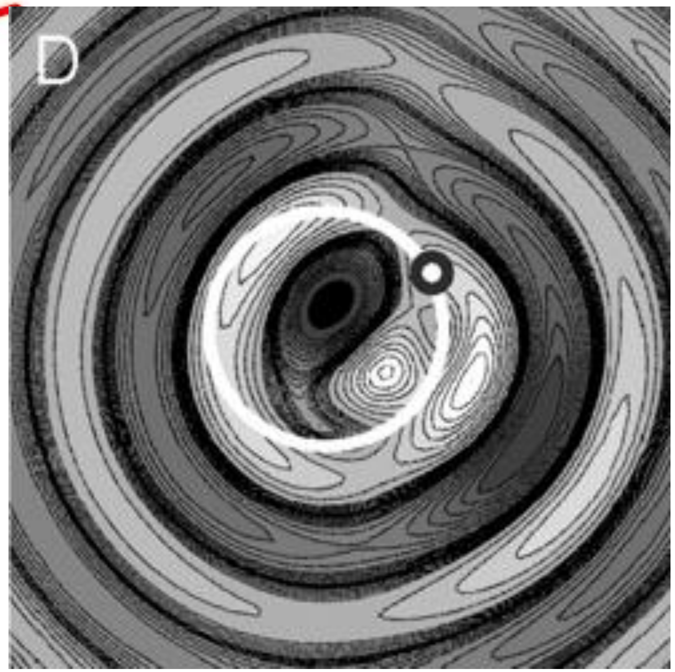
Y. Couder and E. Fort, **Single-Particle Diffraction and Interference at a Macroscopic Scale,** Phys. Rev. Lett. 97 (2006)

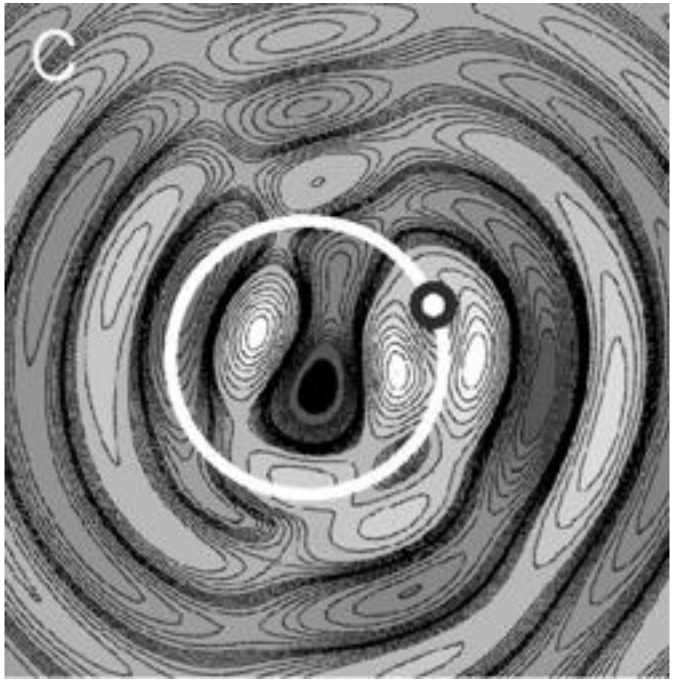


A. Eddi, E. Fort, F. Moisy, and Y. Couder, **Unpredictable Tunneling of a Classical Wave-Particle Association**, Phys. Rev. Lett. 102 (2009)

E. Fort, A. Eddi, A. Boudaoud, J. Moukhtar, and Y. Couder, **Path-memory induced quantization of classical orbits,** PNAS vol. 107 (2010)







Silicon oil + droplets

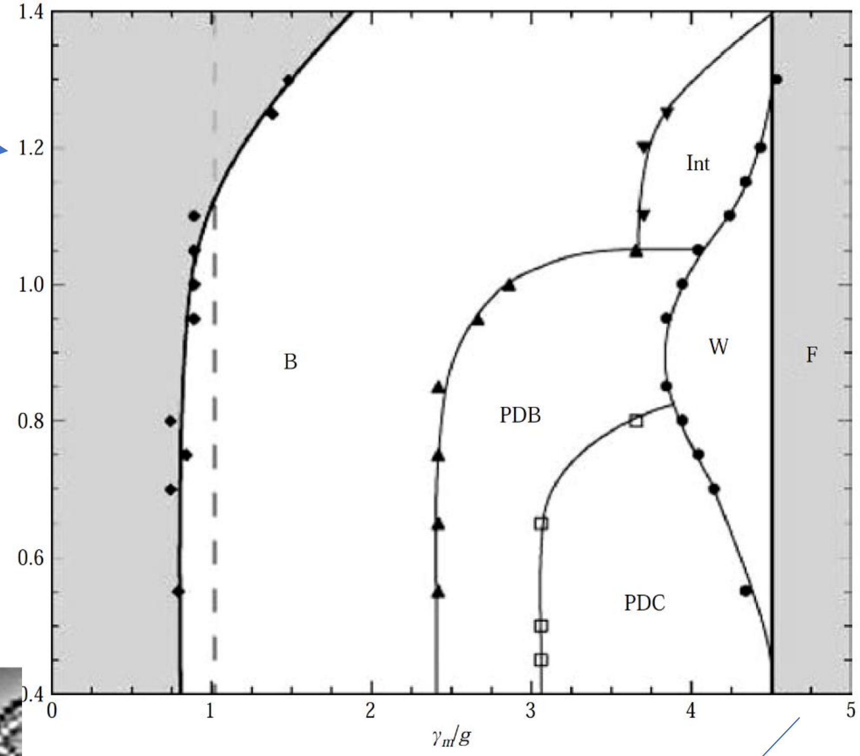
Diameter $\approx 0.8mm$

$$f \approx 80Hz$$

$$\ddot{z}(t) = \gamma \cos(2\pi f t)$$

$$z(t) = -A\cos(2\pi f t)$$

$$A = \frac{g}{4\pi^2 f^2}\frac{\gamma}{g} \approx 0.15mm$$



$\lambda_F \approx \frac{200\ mm/s}{40Hz} \approx 5mm$ **Faraday wavelength**

B: simple bouncing

PDB: period-doubling

PDC: PD chaos

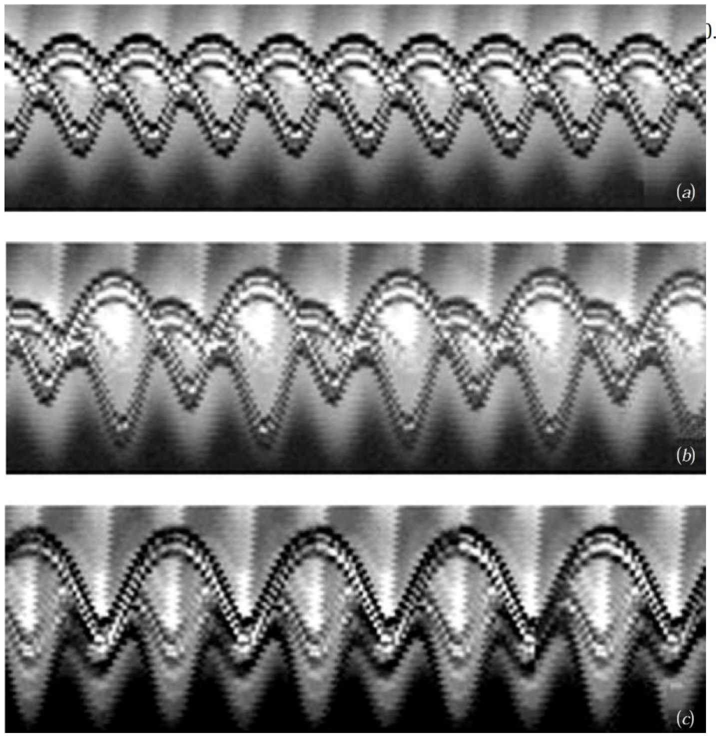
Int: intermittent

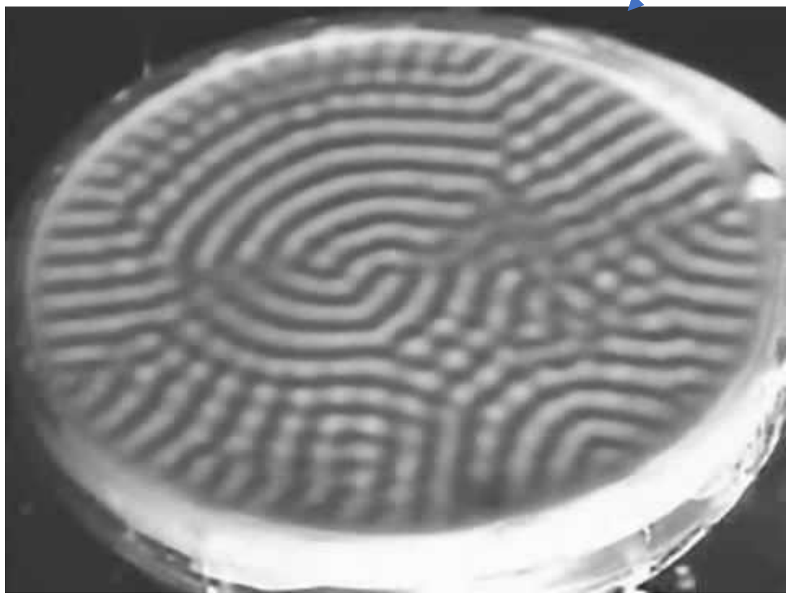
W: walking droplets

"memory" $|\gamma - \gamma_F|^{-1}$

F: Faraday instability

[Source: JFM 2006]

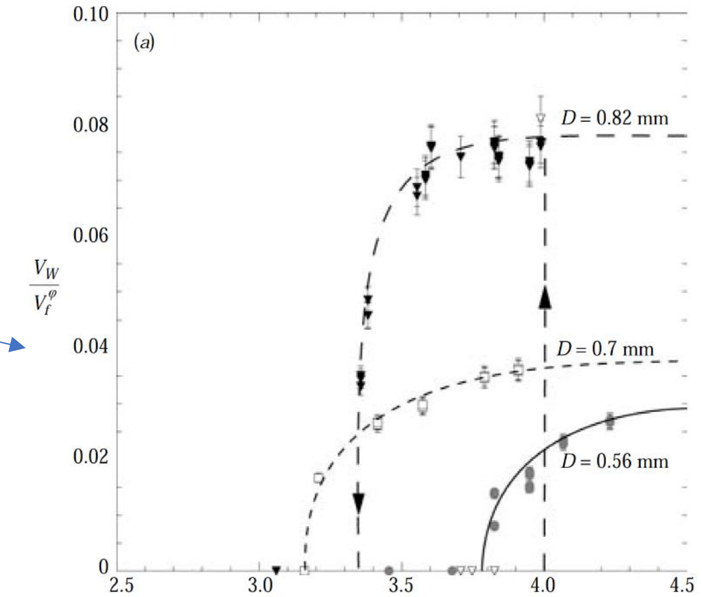


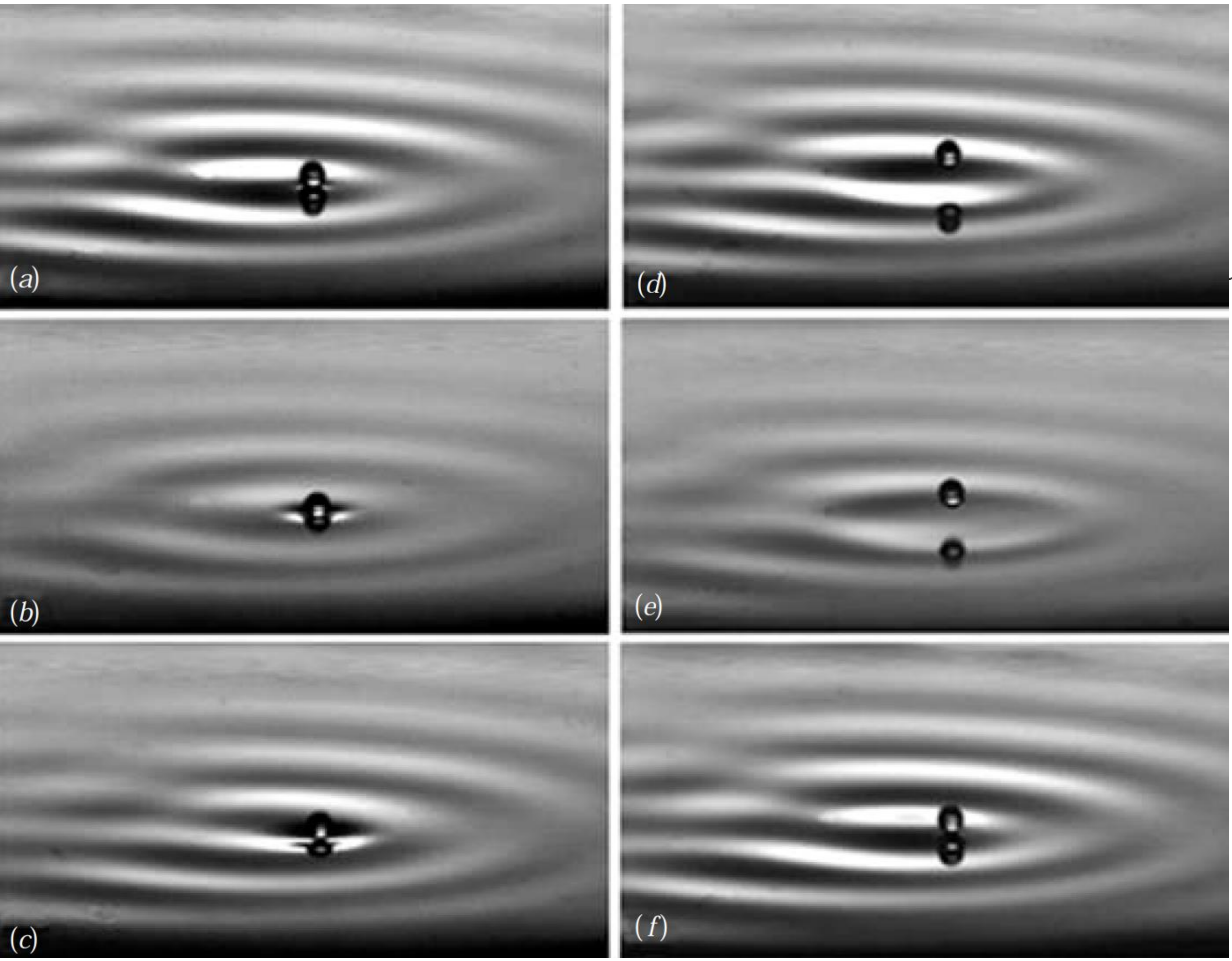


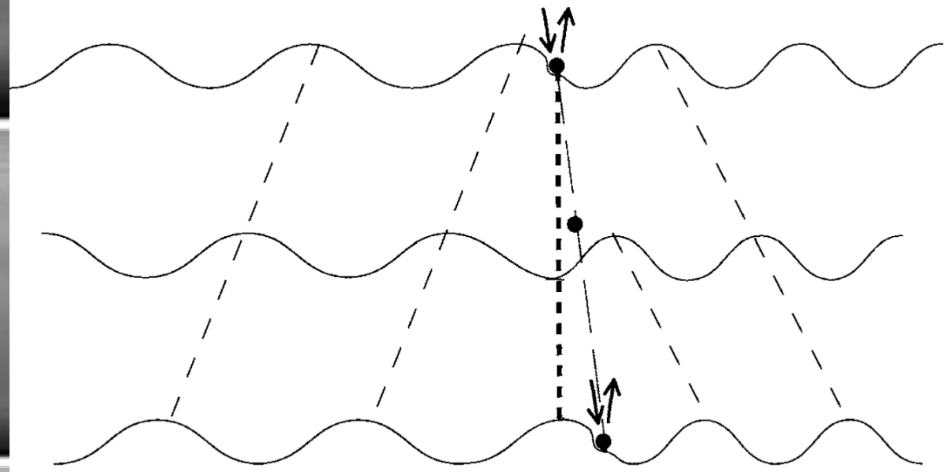
Particle–wave association on a fluid interface, JFM 2006

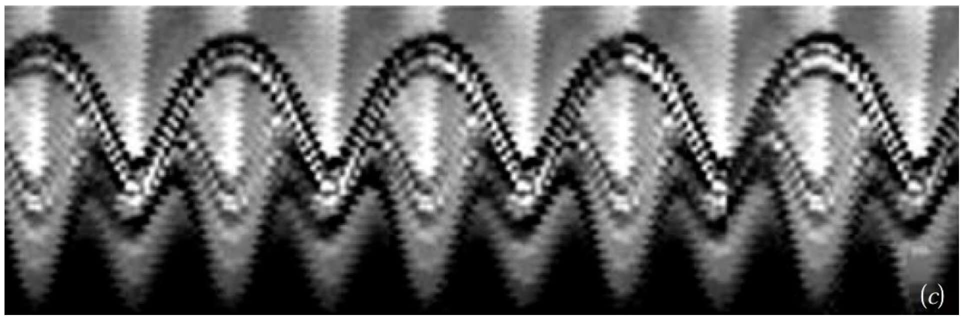
Near Faraday instability "memory" → ∞ : $\tau \propto |\gamma - \gamma_F|^{-1}$

Walkers: $f_F = f/2 = 40Hz, \quad V_F = 189\,\frac{mm}{s}$ forced by droplet, fixed walker's velocity (γ/g) $\;0 < V_W < 20\ \frac{mm}{s}$

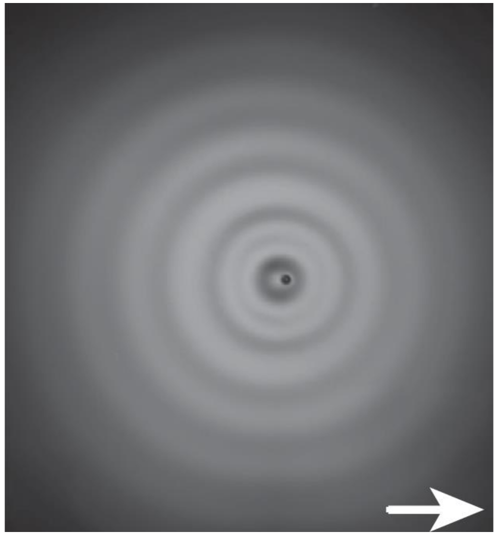


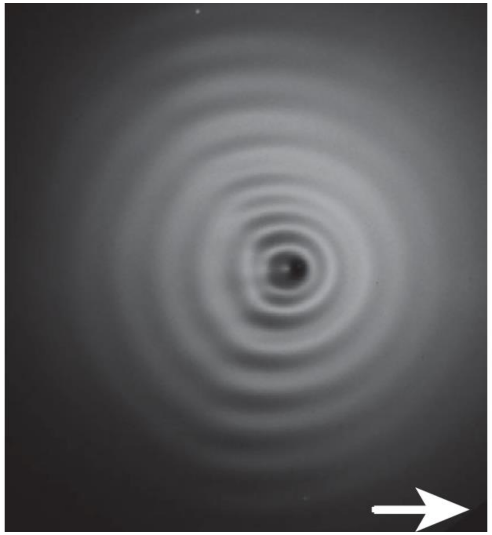


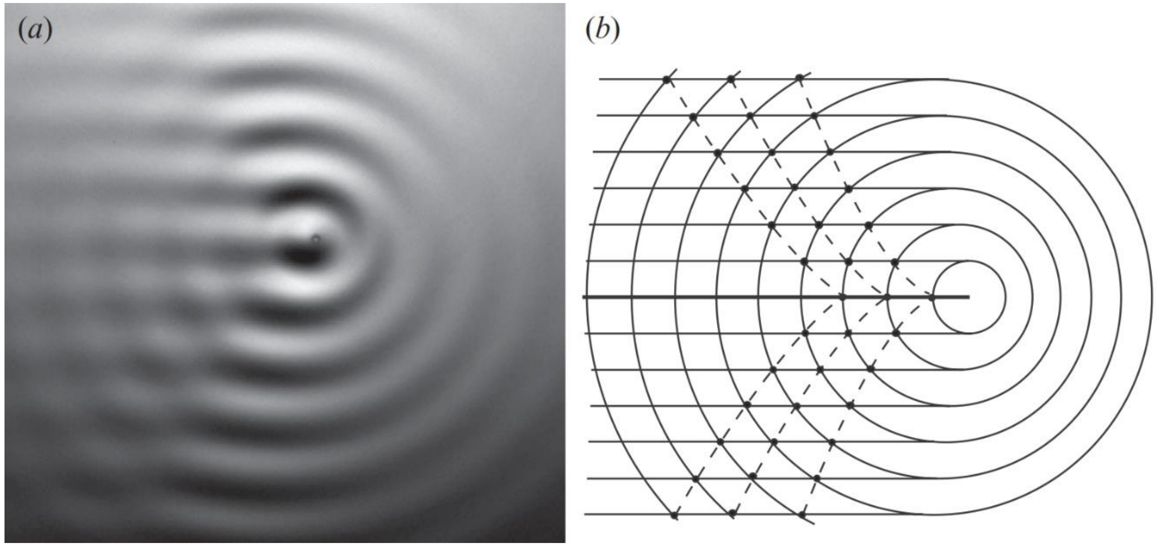


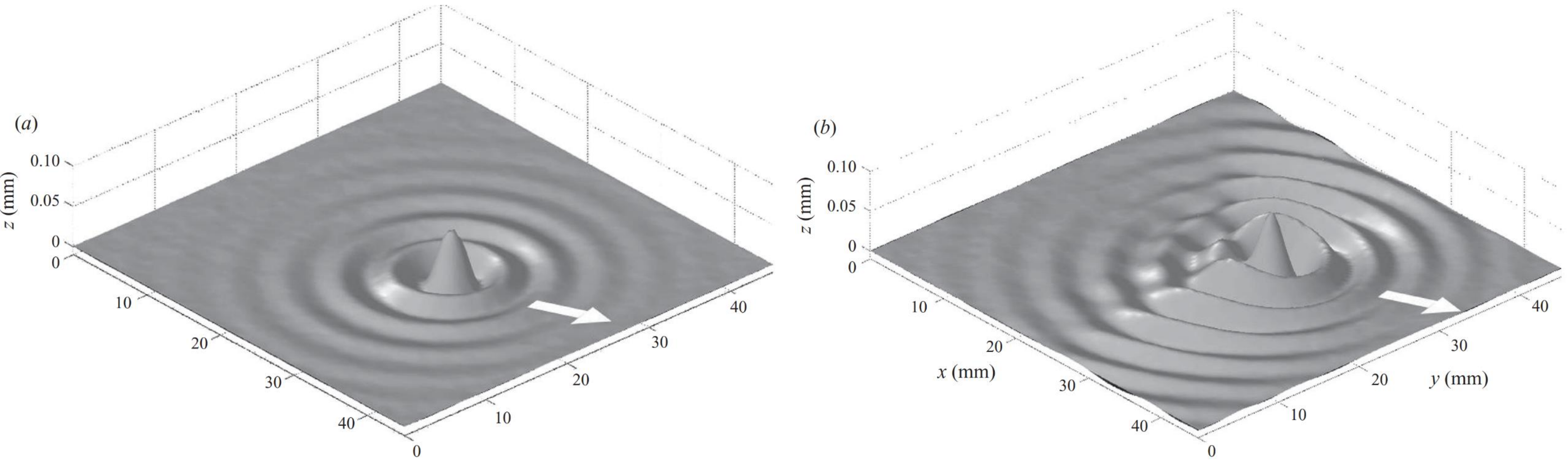


Information stored in Faraday waves: the origin of a path memory, JFM 2011 **Constructive/destructive interference:**

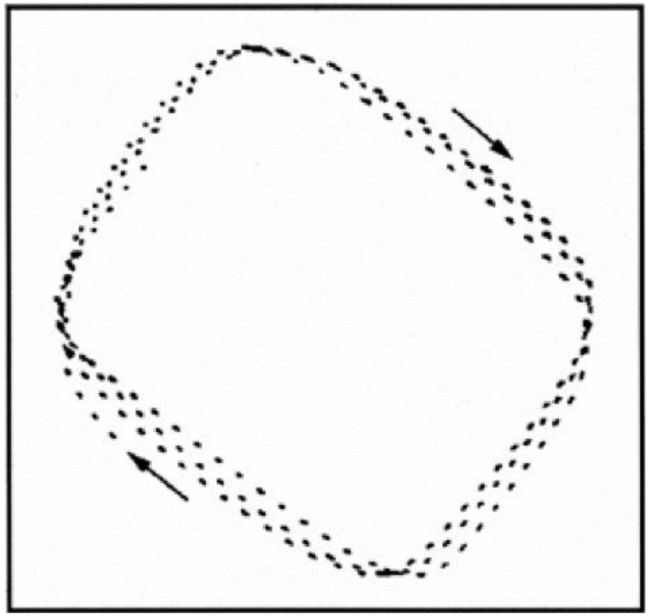


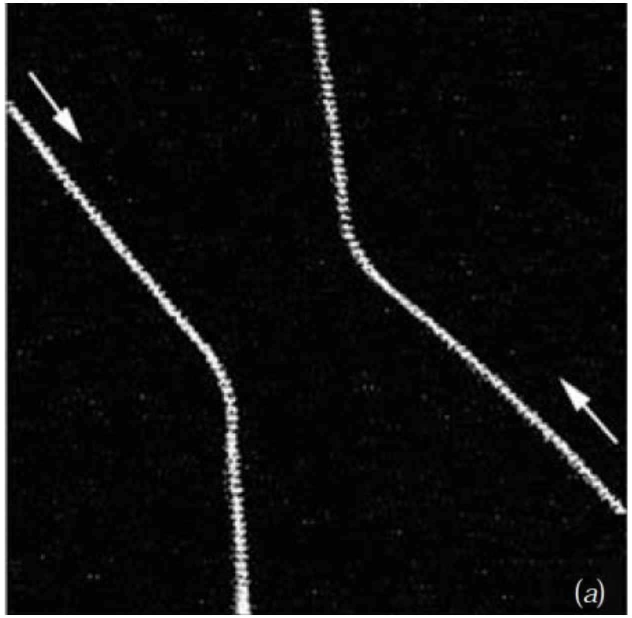


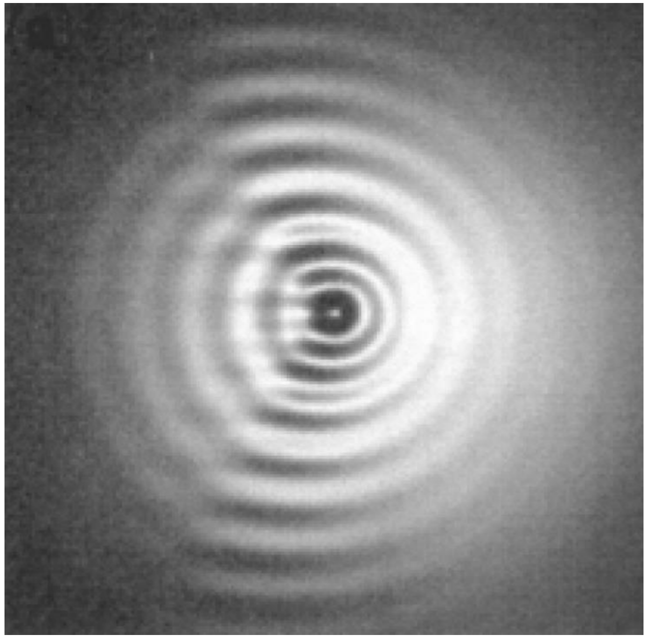


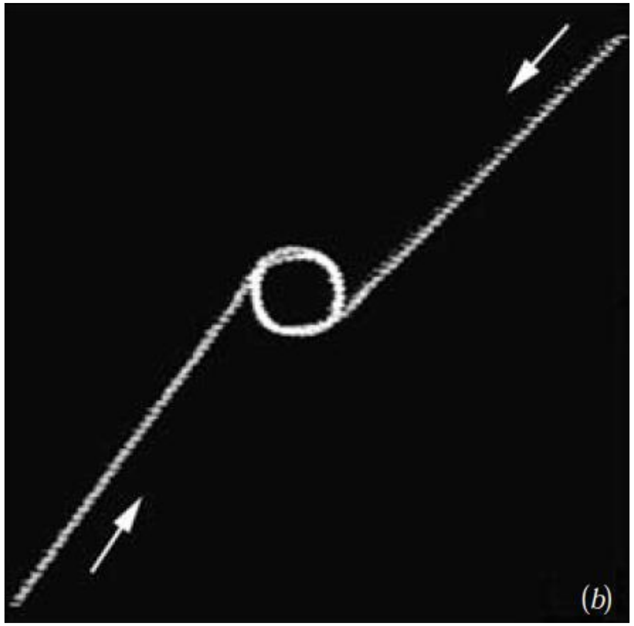


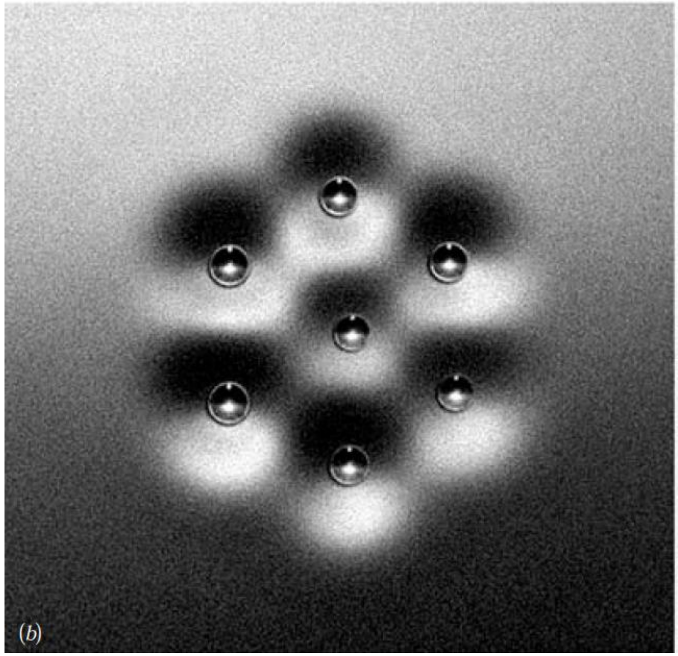
Wave coupled with droplet allows it to **interact** with boundary, other droplets, e.g. to synchronously **self-organize** into **"crystals"**



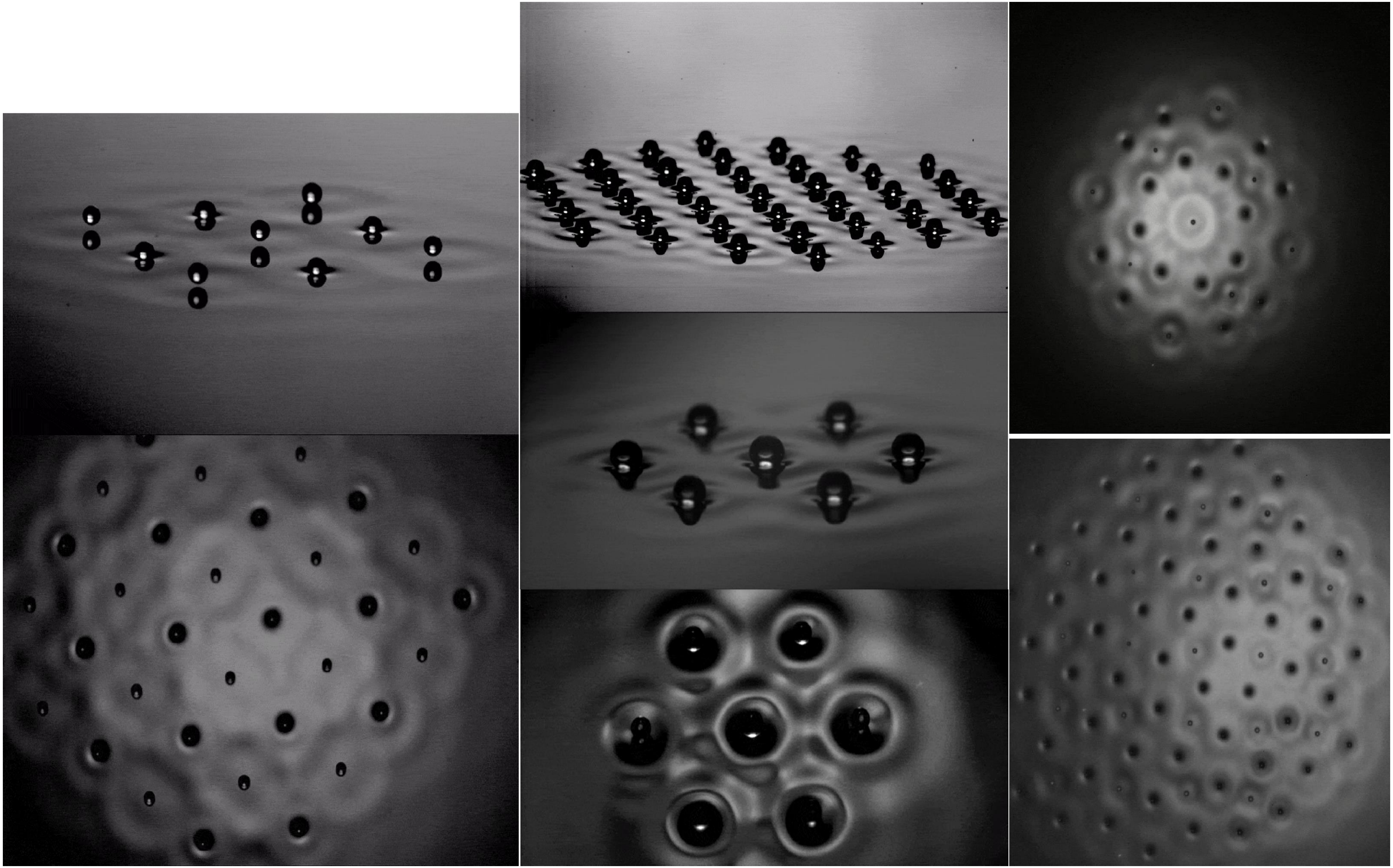








Movies: http://dualwalkers.com/crystals.html



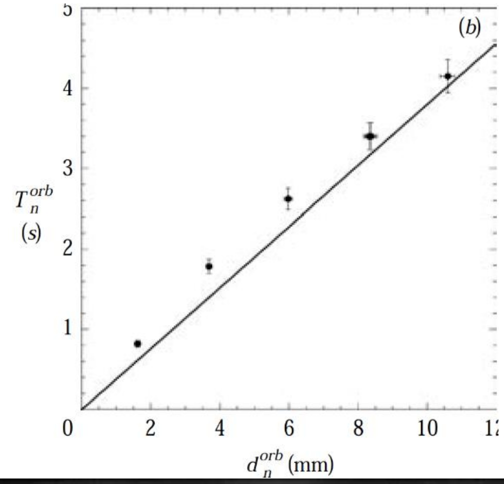
Two walkers have quantized orbits:

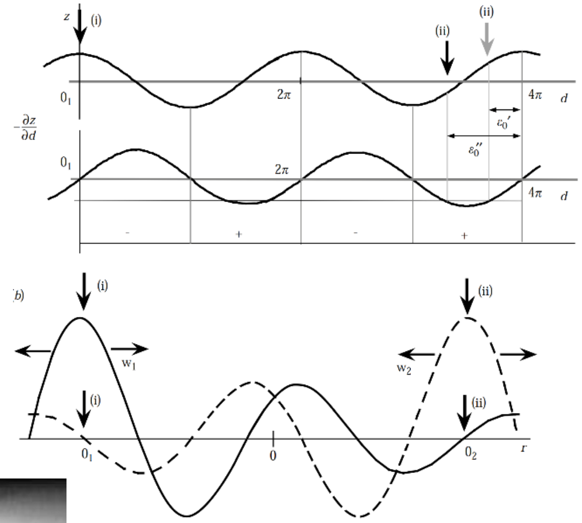
(Nature 2005) $d_n^{orb} = (n - \epsilon_0)\lambda_F$

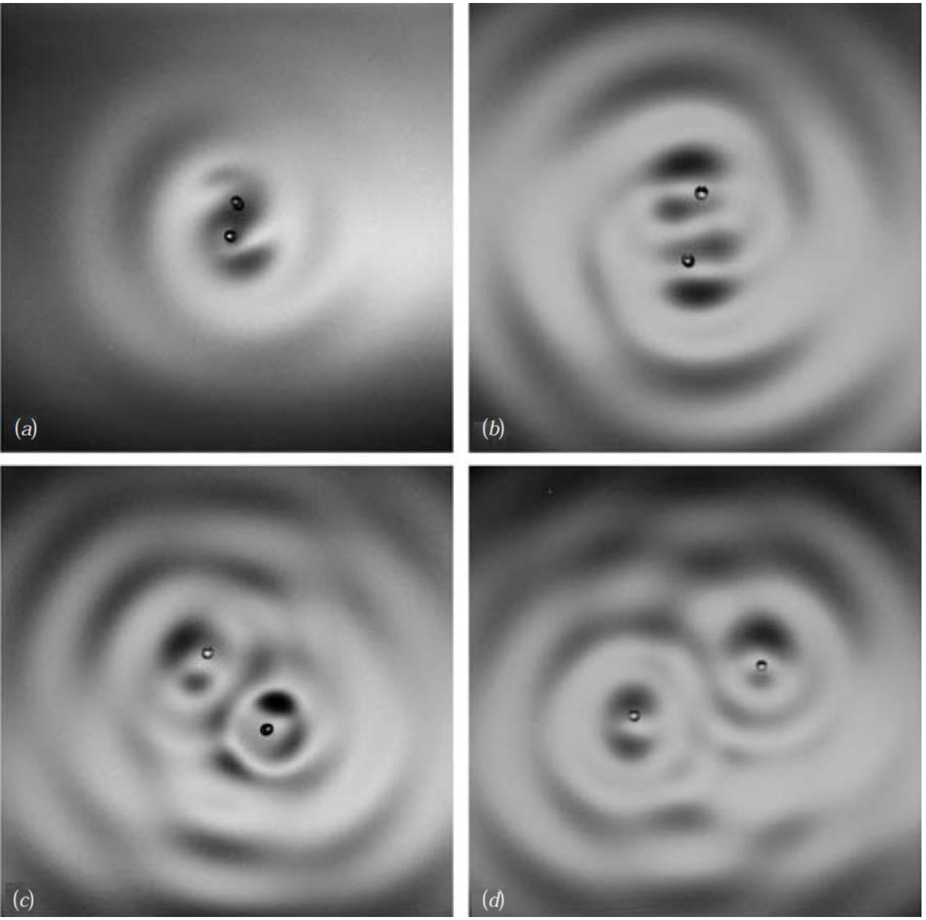
for $n = 1,2,3,\ldots$ in phase,

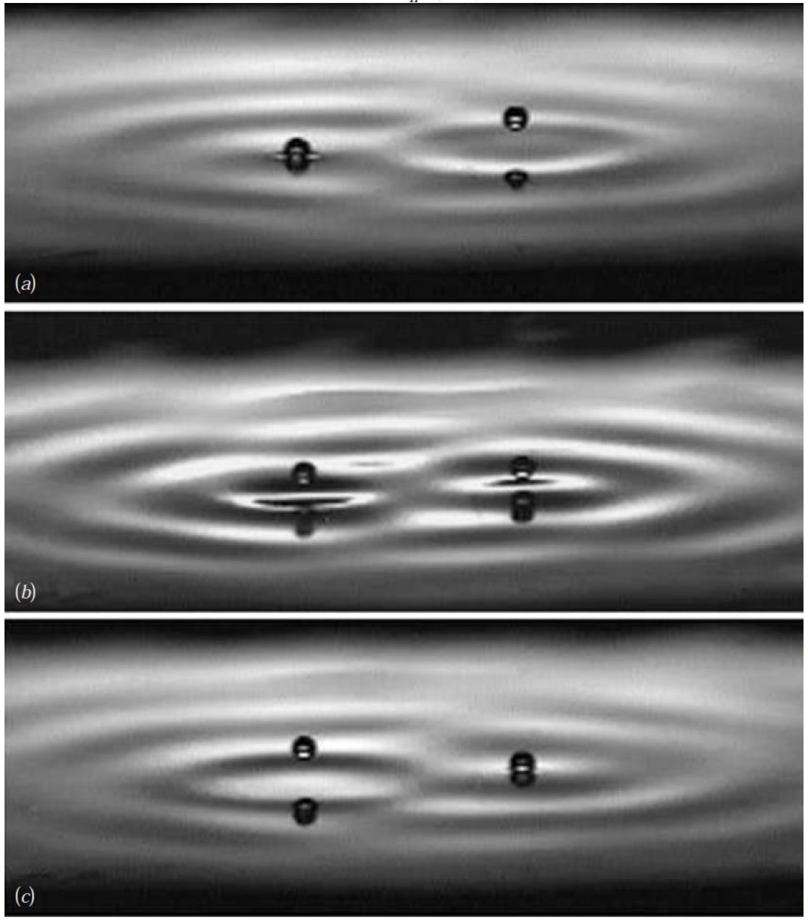
$n = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$ opposite phase

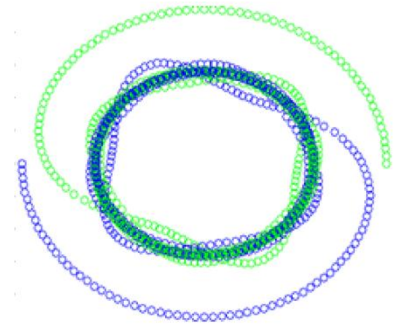
$\epsilon_0 \approx 0.2 \qquad \lambda_F \approx 4.5mm$

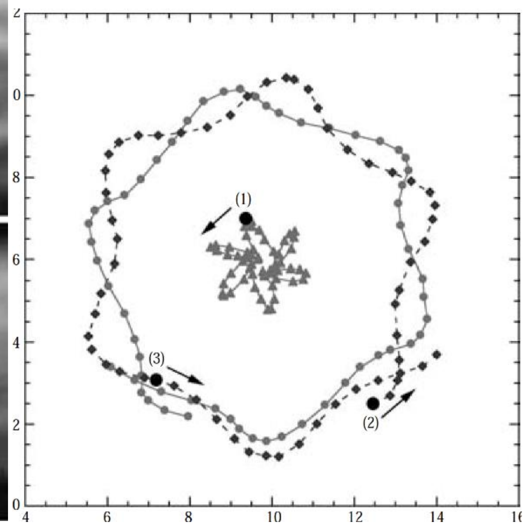




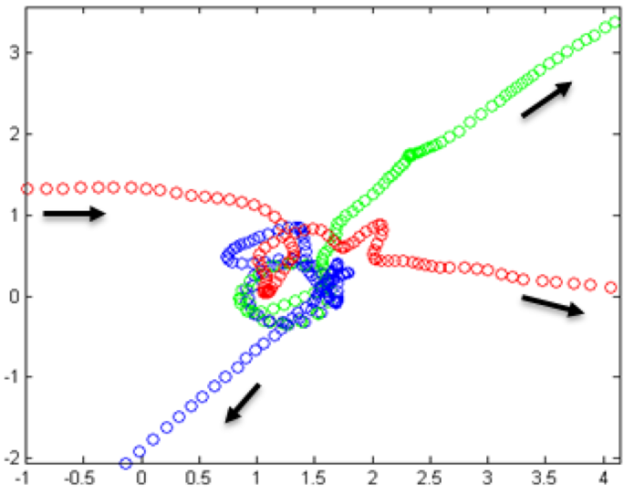




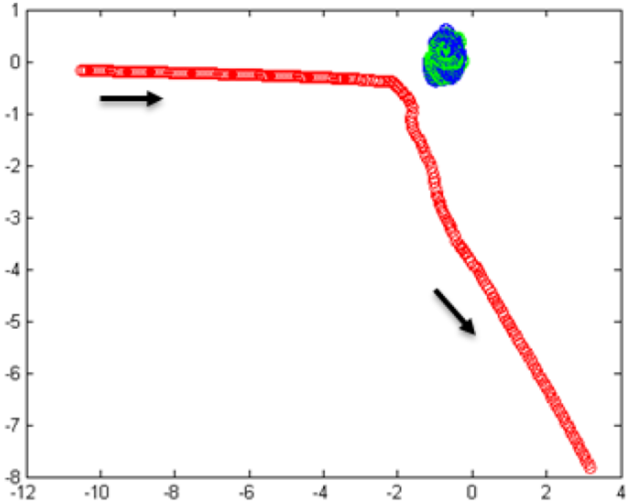




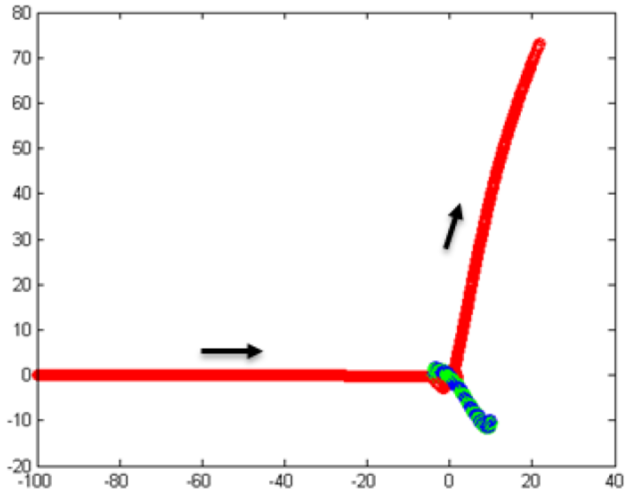
Wave particle duality in multiple bouncing fluid droplets, “Feynman diagrams” to calculate scattering statistics?



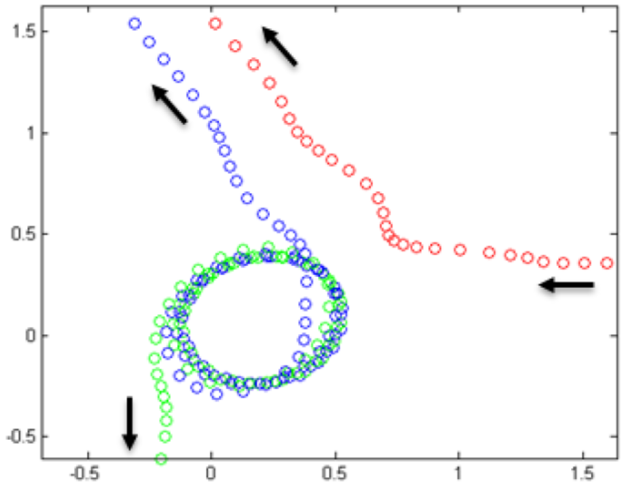
(a) Droplets in different directions



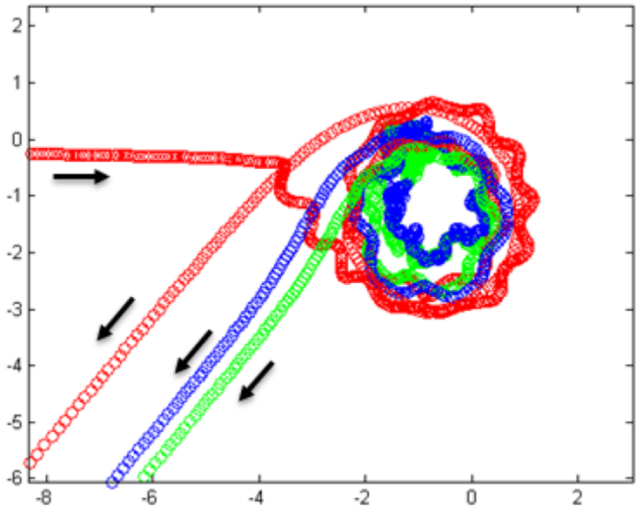
(b) Scattering



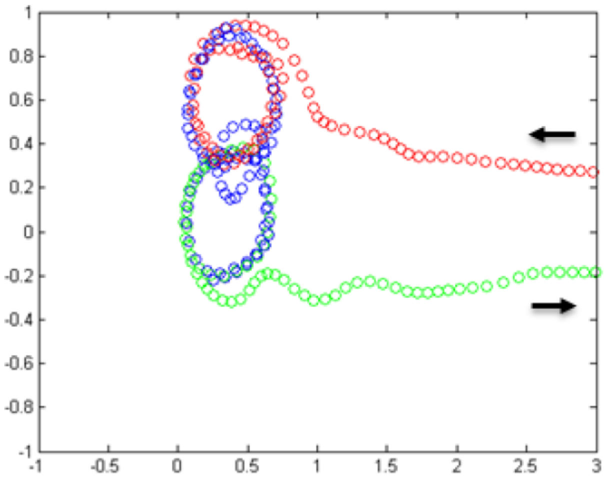
(c) Slingshot



(d) Two parallel walkers



(e) Three parallel walkers



(f) Switching orbits

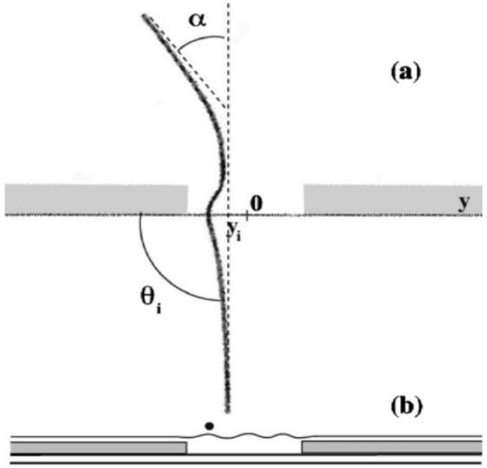
Feynman about double-slit: "contains all of the mystery of QM"

Single-Particle Diffraction and Interference at a Macroscopic Scale, Couder, Fort, **PRL** 2006

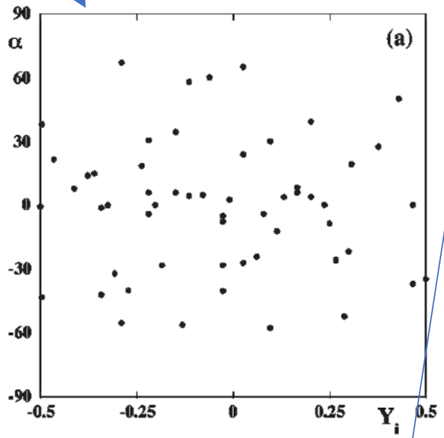
"(…)a given single walker seems randomly scattered. However, diffraction or **interference patterns** are recovered in the **histogram** of the deviations of many successive walkers."

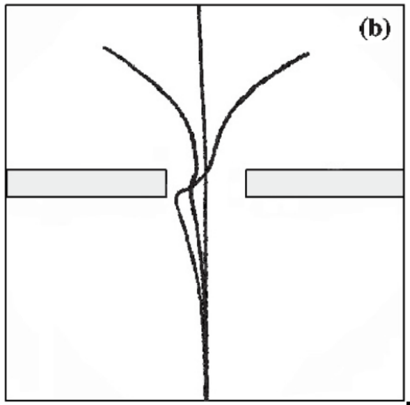
Using "strips glued to the bottom of the cell" – **reducing depth** from $h_0 = 4mm$ to $h_1 \approx 1$

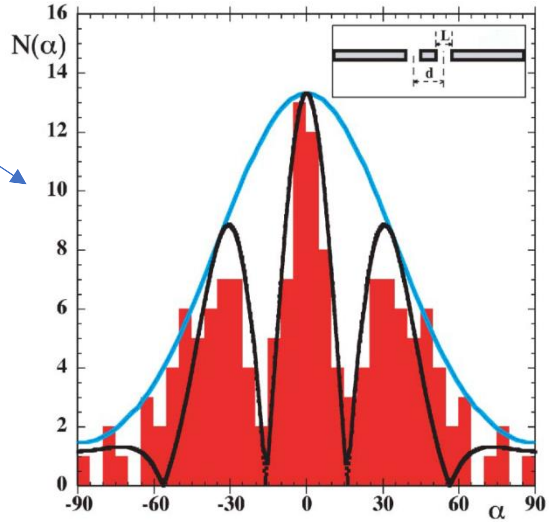
Single slit of length L (video) **double slit**

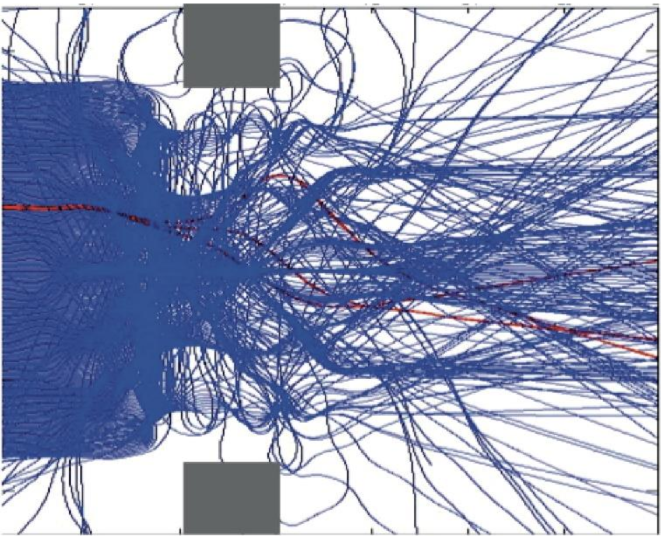


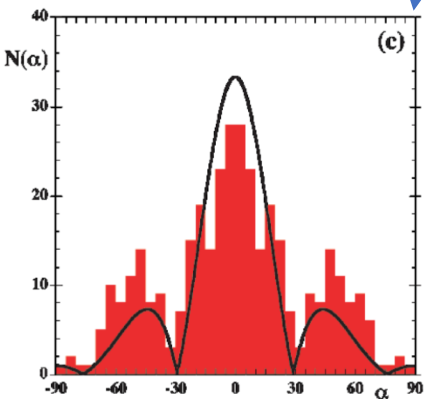
$Y_i = y_i/L$, different $\frac{L}{\lambda_f} = 2.3$ or 3.1 or 1.1

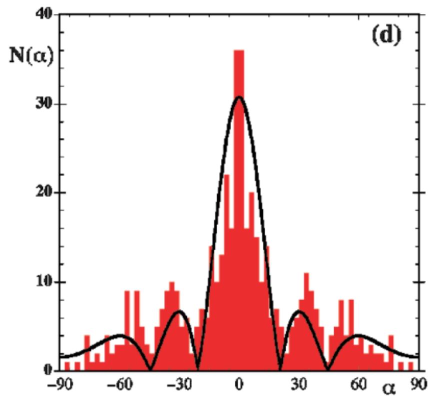


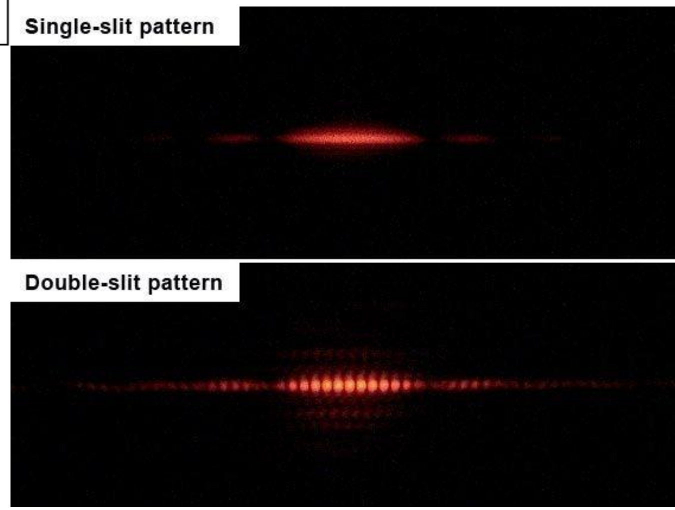






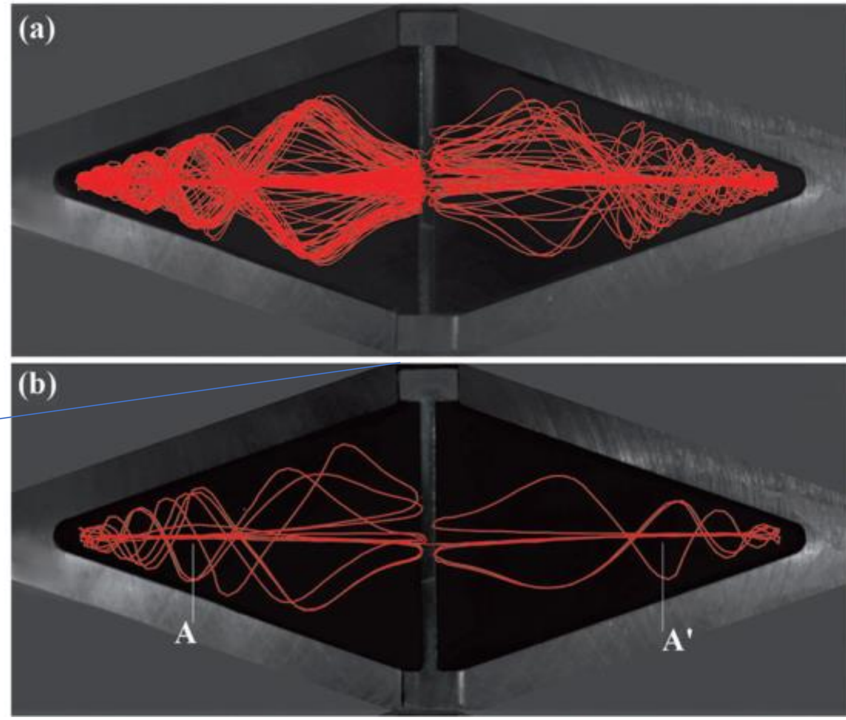


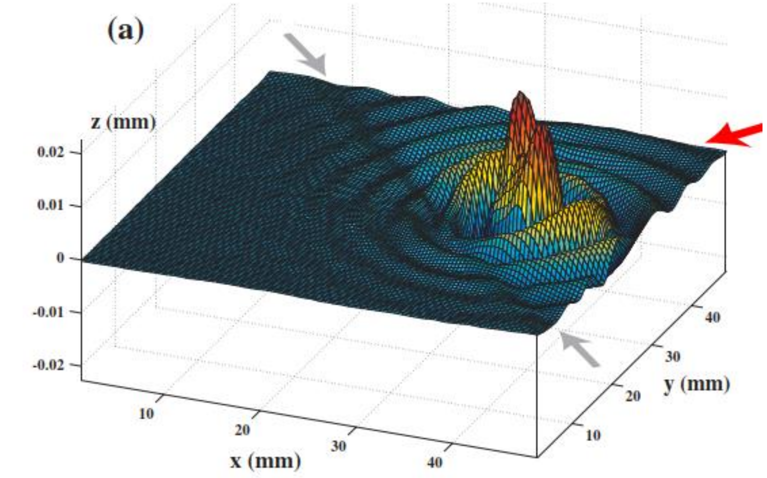


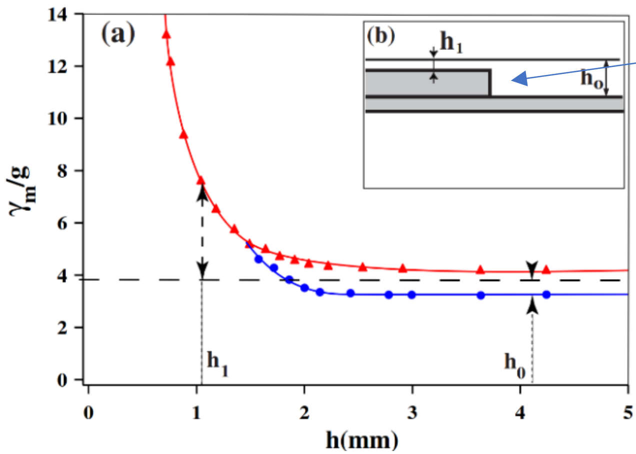


[Unpredictable **Tunneling** of a Classical Wave-Particle Association](), A. Eddi, E. Fort, F. Moisy, and Y. Couder, **PRL** 2009

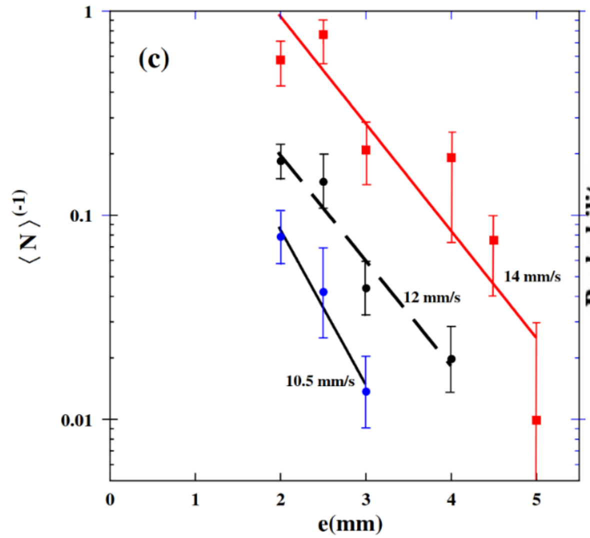
Due to complex history dependence from coupled waves

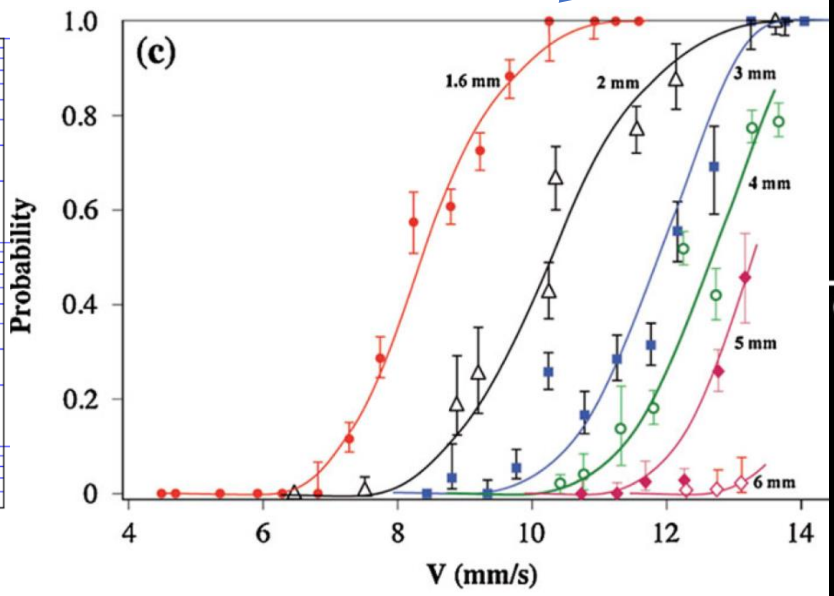


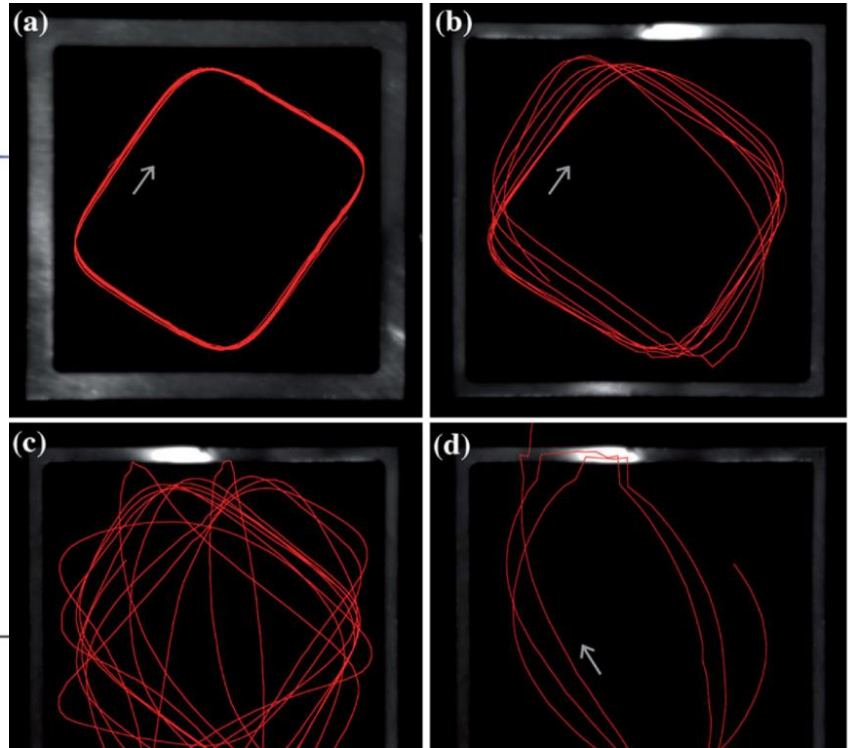




Tunneling probability (width, velocity)

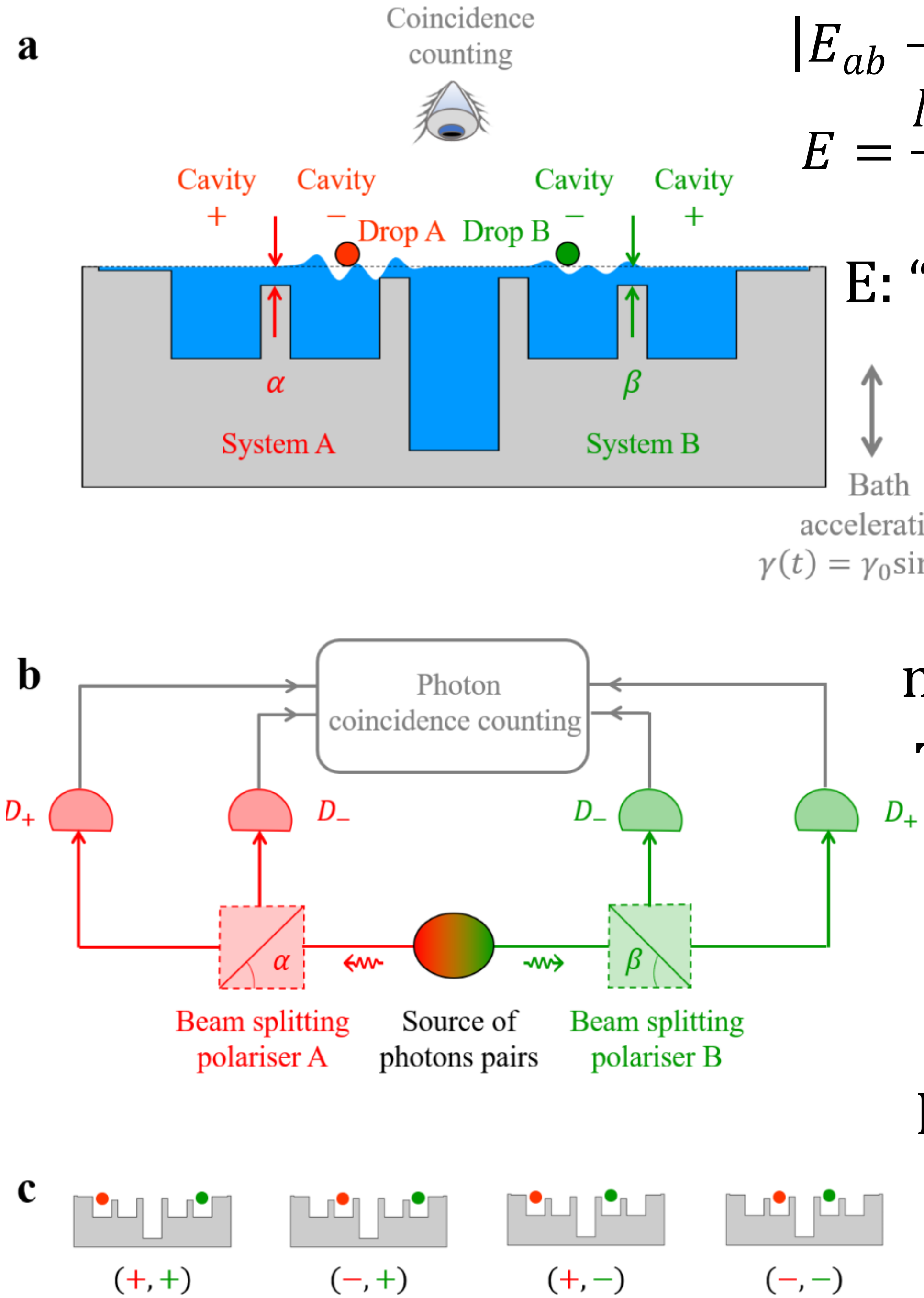






CHSH violation

Static Bell test in pilot-wave hydrodynamics (PRF):



$$|E_{ab} - E_{ab} + E_{ab} + E_{ab}| \leq \mathbf{2}$$

$$E = \frac{N_{++} - N_{+-} - N_{-+} + N_{--}}{N_{++} + N_{+-} + N_{-+} + N_{-}}$$

E: “quantum correlation”

$$\alpha = a, a' \qquad \beta = b, b'$$

barrier depth

$$a = b, a' = b'$$

maximal violation~**2.49**

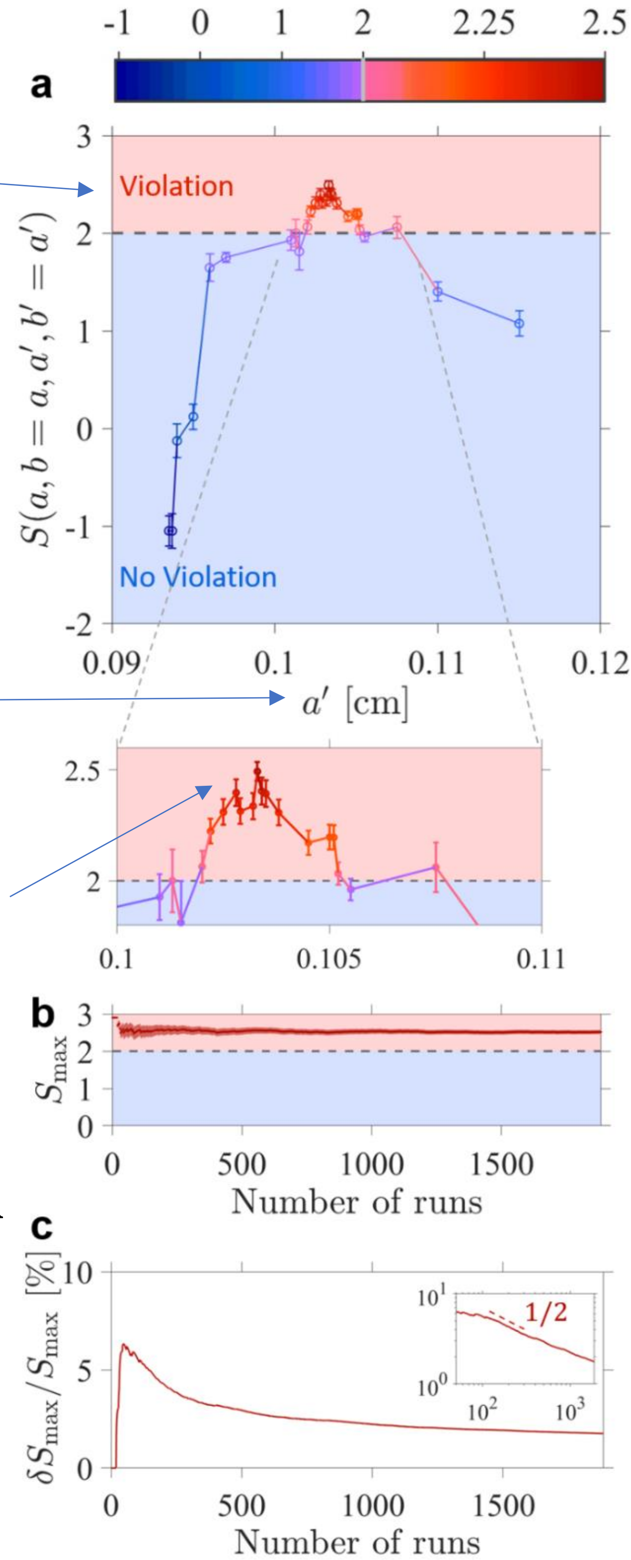
Tsirelson: $2\sqrt{2} \approx \mathbf{2.83}$

$$a = b = 0.099 \text{ cm}$$

$$a' = b' = 0.1033 \text{ cm}$$

Loopholes?

E.g. communication ...

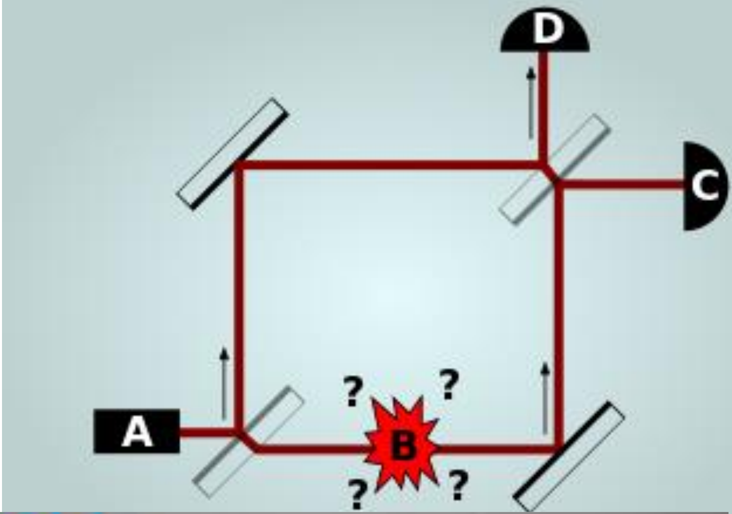


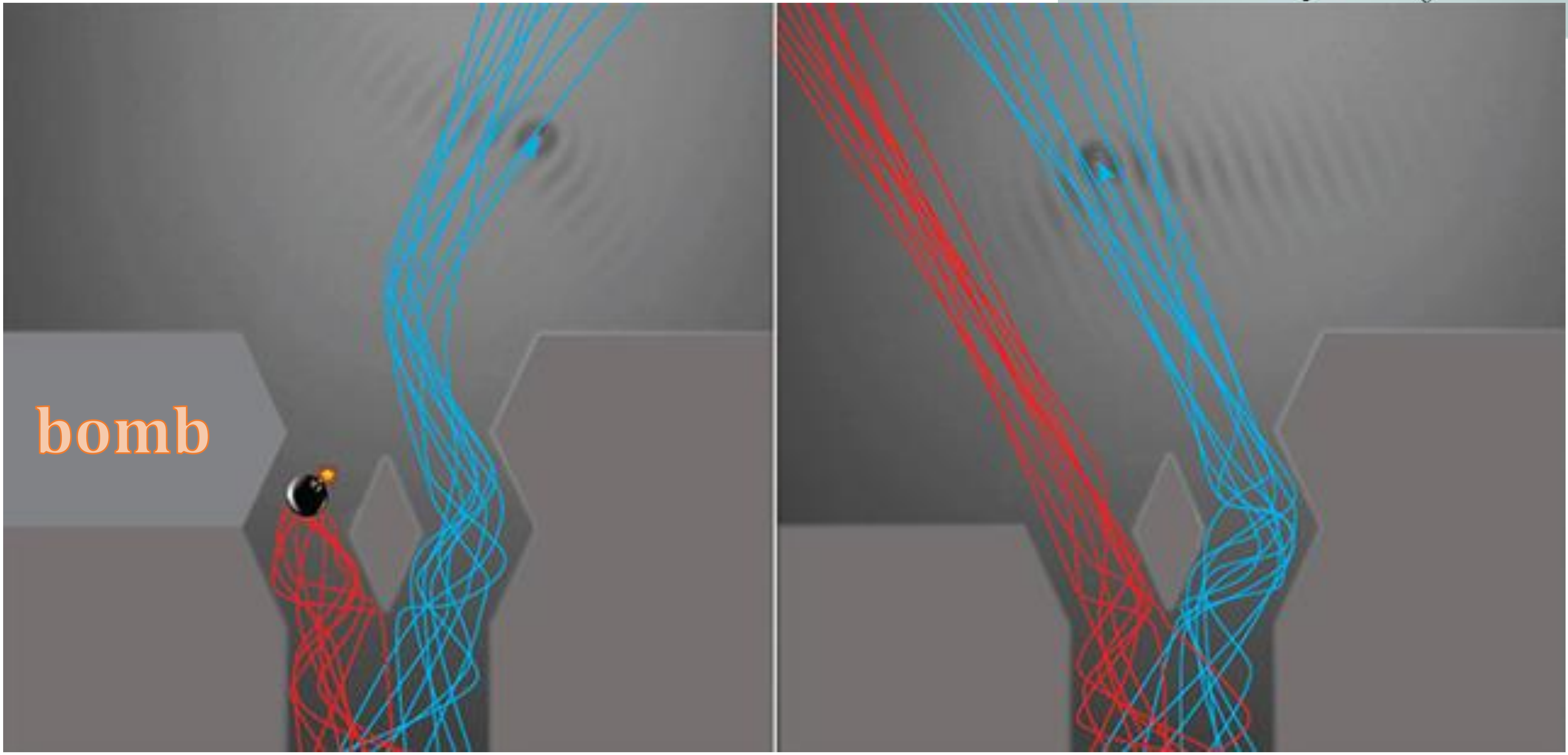
Elitzur-Vaidman bomb tester (PRA2023)

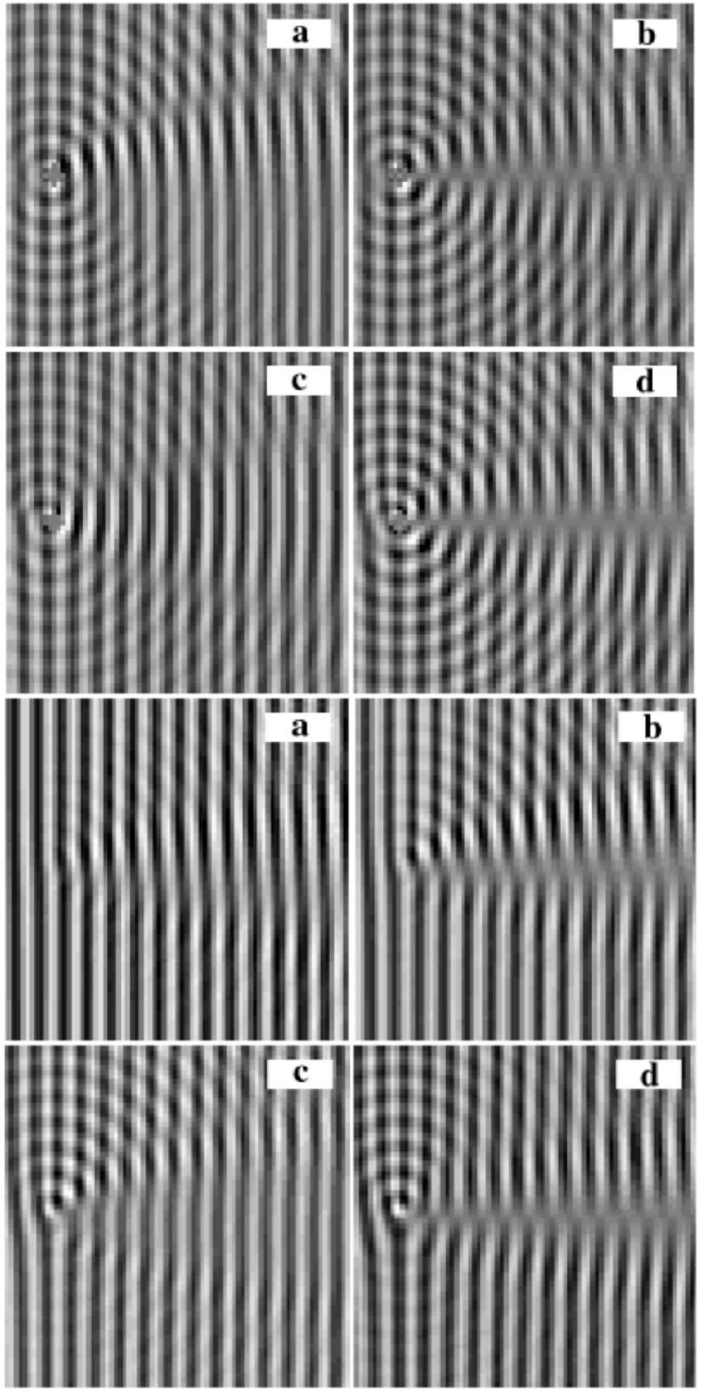
Going left detonates the bomb

Going right: bomb if later turns right (left: no bomb)

Only pilot wave visits bomb (not particle)







vorticity $2\vec{\Omega} = \vec{\nabla} \times \vec{U}$ equivalent to $\vec{B} = \vec{\nabla} \times \vec{A}$

$$\Phi = \oint A(r) \cdot dr = \int\int B(r) \cdot dS$$

Wavefront dislocations in the **Aharonov-Bohm** effect and its water wave analogue, Berry et al., EJP, 1980

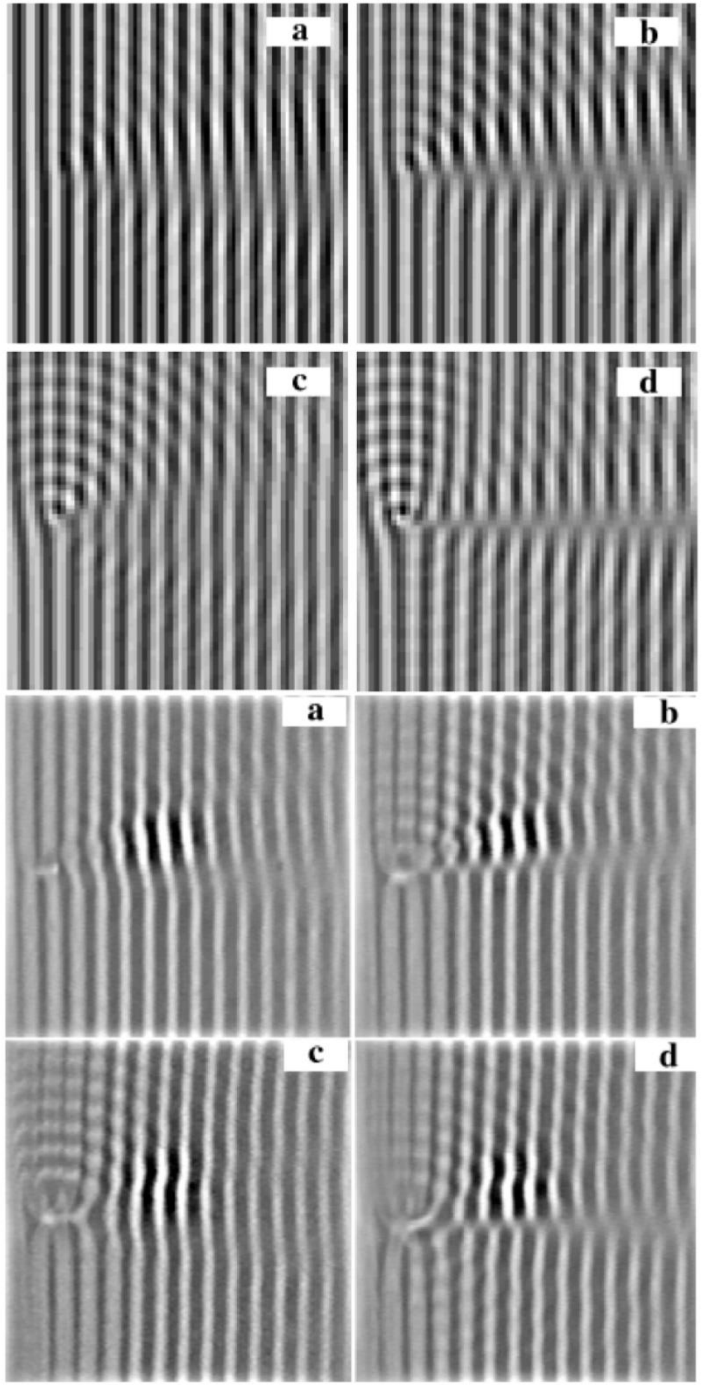
Surface Wave Scattering by a **Vertical Vortex** and the Symmetry of the Aharonov-Bohm Wave Function, **PRL**, 1999

Electron and **solenoid**, various r

Simulation, top – impenetrable

Surface elevation and **vortex**

Top – theory, bottom – experiment



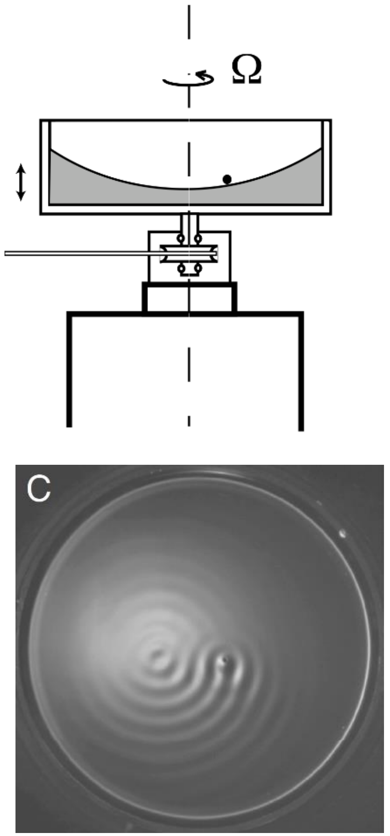
[Path-**memory** induced **quantization** of classical orbits](), Fort, Eddi, Boudaoud, Moukhtar, Couder, PNAS 2010

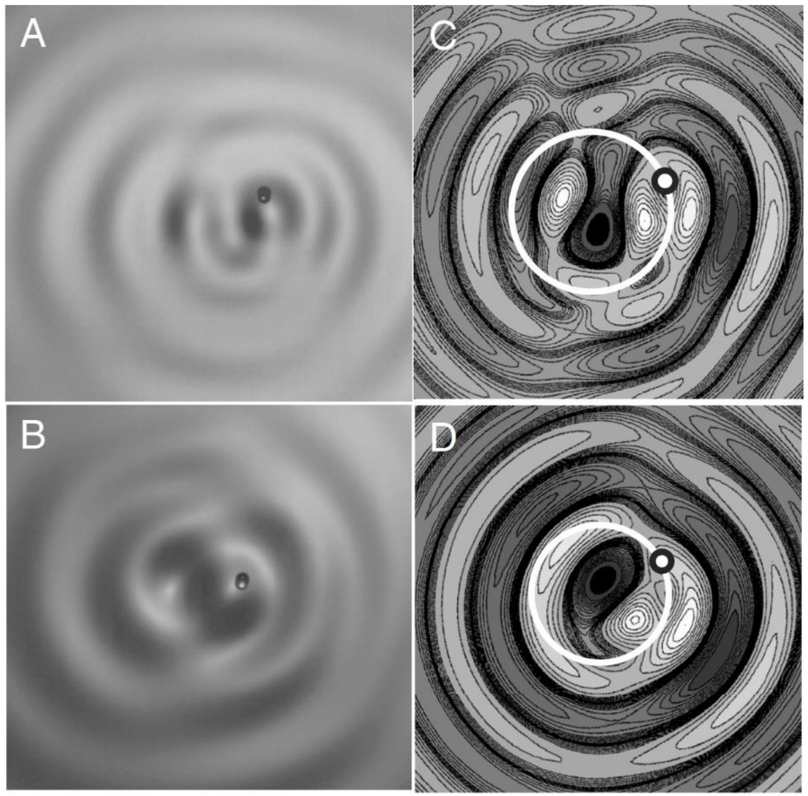
[Landau quantization](): of cyclotron orbits: charged particle in plane perpendicular to fixed magnetic field

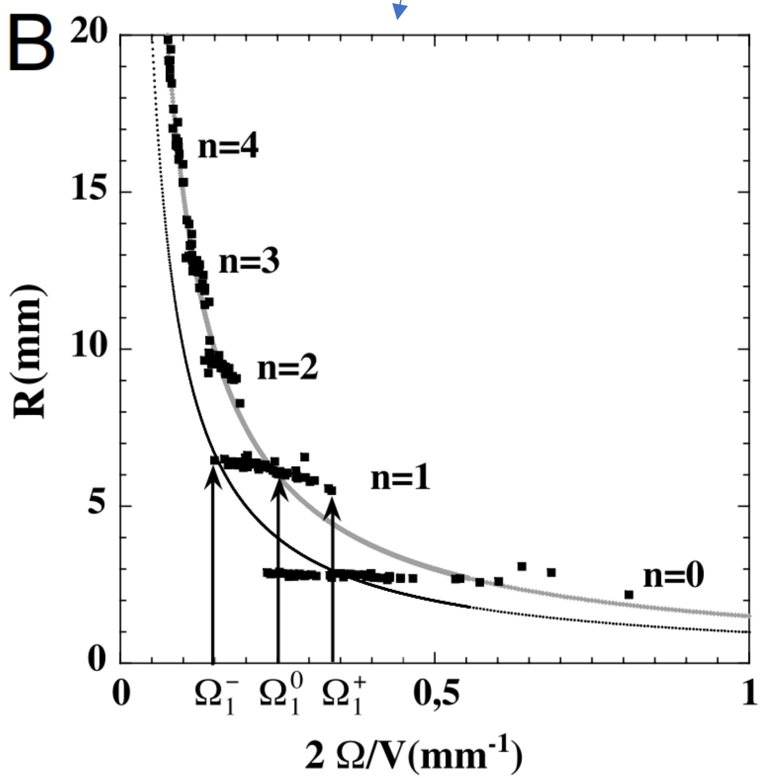
[[Berry 1980]()] **vorticity** $\mathbf{2\vec{\Omega} = \vec{\nabla} \times \vec{U}}$ equivalent to $\boldsymbol{\vec{B} = \vec{\nabla} \times \vec{A}}$ $\qquad R = \frac{mV}{qB} = \sqrt{\left(n+\frac{1}{2}\right)\frac{h}{qB\pi}}$

Coriolis force: $\boldsymbol{\vec{F}_\Omega = -m(\vec{V} \times 2\vec{\Omega})}$ equivalent to Lorentz: $\vec{F}_B = q(\vec{v} \times \vec{B})$ $\qquad R = V/2\Omega$

[Landau](): $\pi R_n = \left(n+\frac{1}{2}\right)\lambda_{dB} = \frac{1}{mv}\oint \vec{p}\cdot d\vec{r}$ $\qquad \lambda_{dB} = \frac{h}{mV}$ $\qquad$ **droplets**: $2R_n \approx \left(n+\frac{1}{2}\right)\lambda_F$



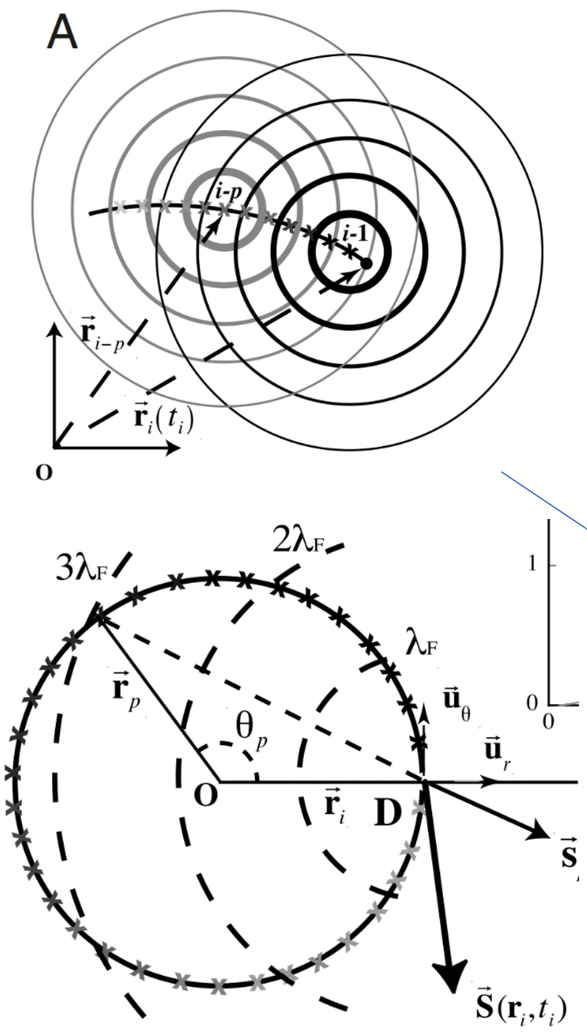


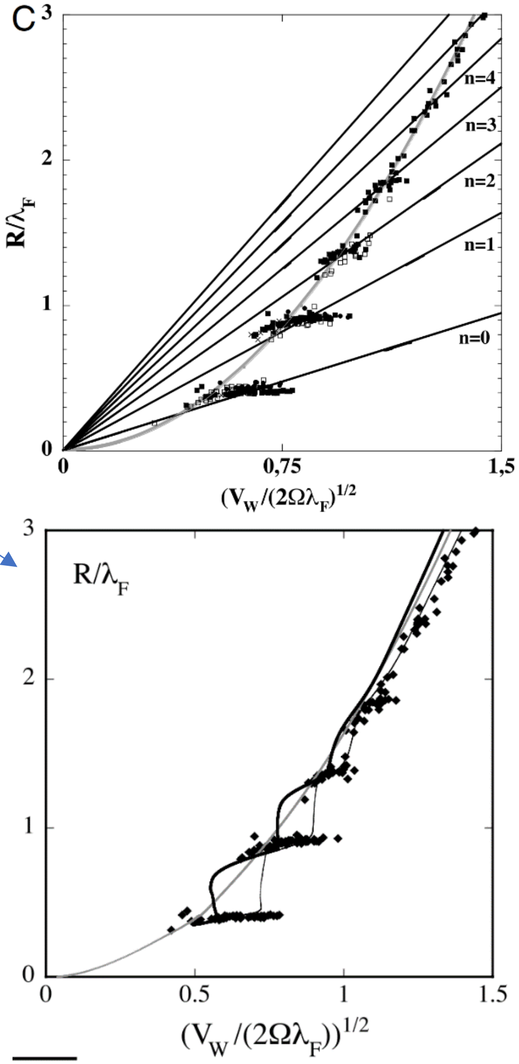


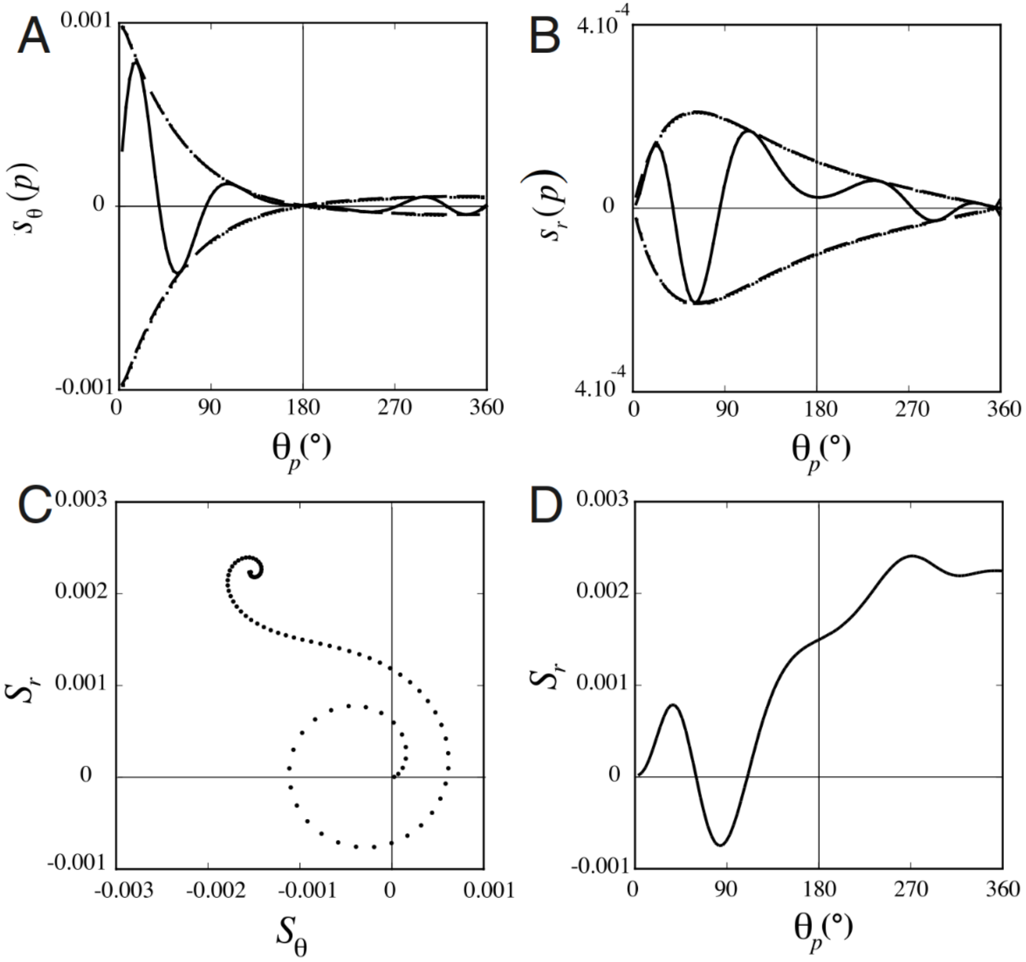
Quantization of angular momentum, clock finds resonance with the field – [standing wave]()

Simulations: free-flight with Coriolis + bouncing from $\vec{S} = \nabla h = S_r \vec{u}_r + S_\theta \vec{u}$ slope in $t_i = i\,T_F$

$$h(\vec{r}, t_i) = \sum_{p=i-1}^{-\infty} \frac{A}{\left|\vec{r} - \vec{r}_p\right|^{1/2}} \exp\left(-\frac{|\vec{r} - \vec{r}_p|}{\delta} - \frac{t_i - t_p}{\tau}\right) \cos\left(\frac{2\pi|\vec{r} - \vec{r}_p|}{\lambda_F} + \phi\right)$$







Level Splitting at Macroscopic Scale, **PRL** 2012 – **Zeeman**

Two droplets: $\boldsymbol{d_n = (n - \epsilon_0)\lambda_F}$ quantization

$\boldsymbol{n = 1, 2, 3}$ … in phase, $\boldsymbol{1/2, 3/2, 5/2,}$ … opposite phase

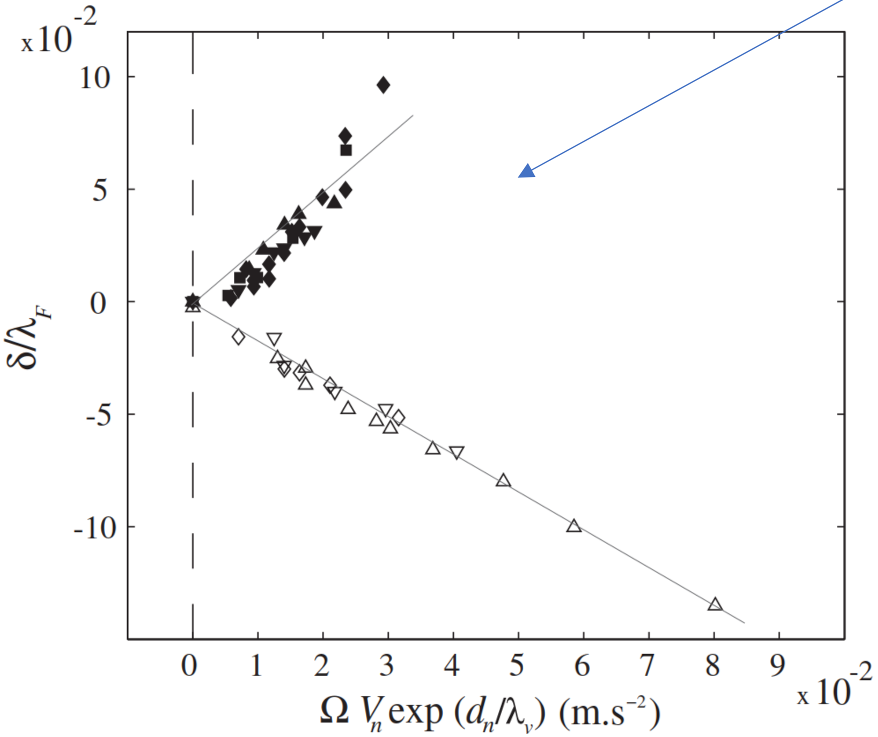
In counterclockwise rotating cell $(\boldsymbol{\Omega})$,

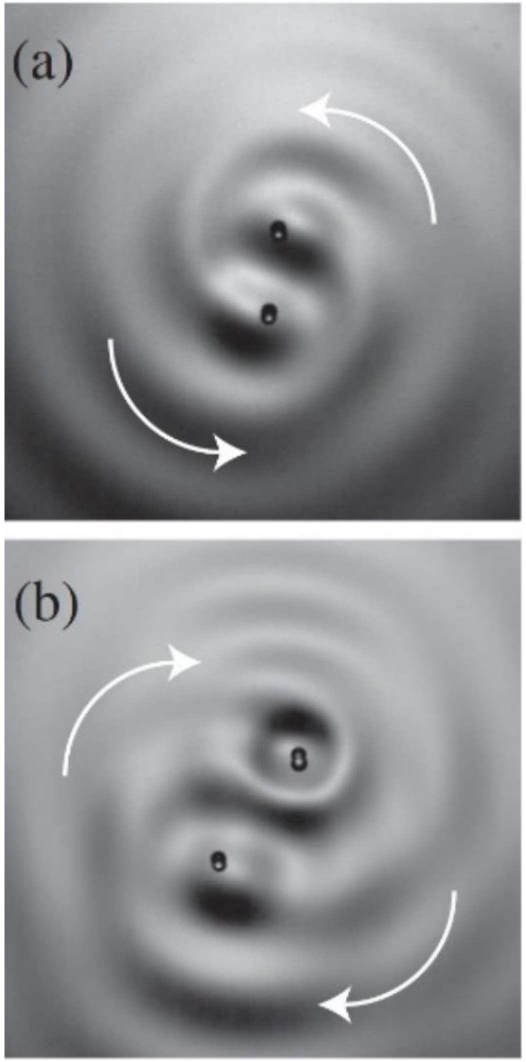
$\boldsymbol{n^+}$ cc, $\boldsymbol{n^-}$ clockwise

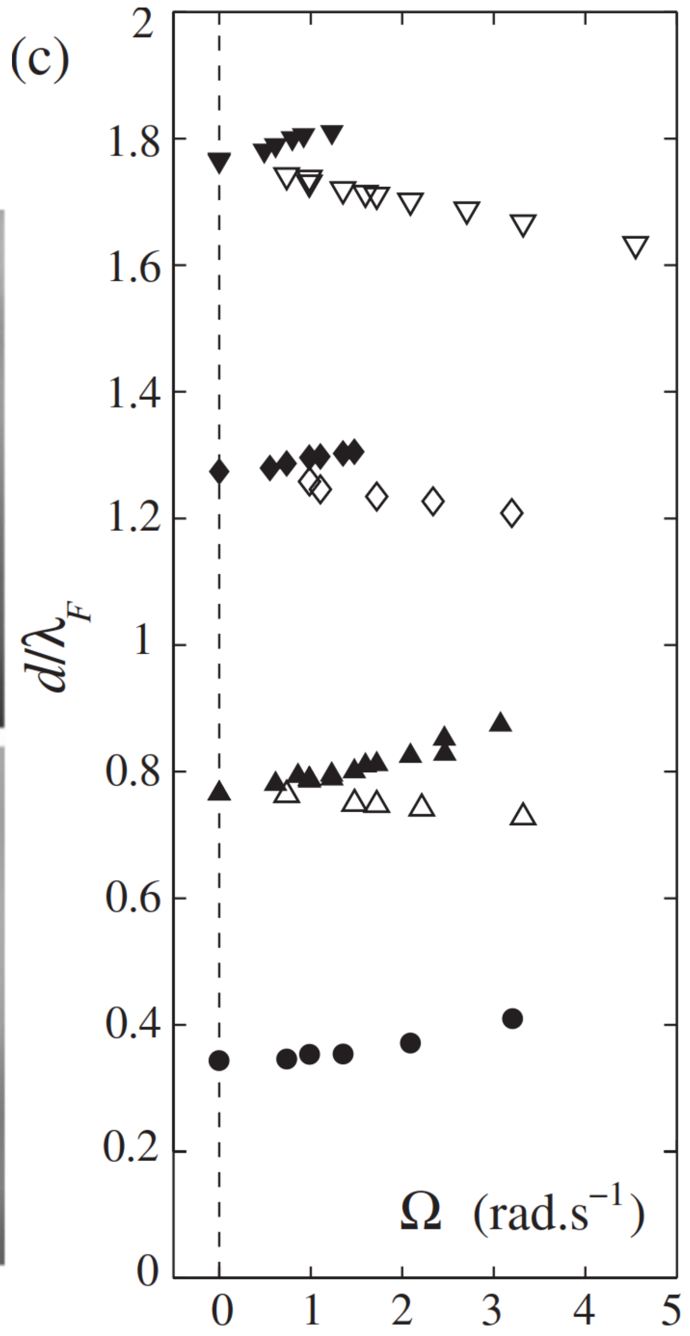
Lorentz force $\rightarrow$ **Coriolis force**

$$\vec{\boldsymbol{F}}_{\boldsymbol{B}} = \boldsymbol{q}(\vec{\boldsymbol{v}} \times \vec{\boldsymbol{B}}) \rightarrow \vec{\boldsymbol{F}}_{\boldsymbol{\Omega}} = -\boldsymbol{m}(\vec{\boldsymbol{V}} \times 2\vec{\boldsymbol{\Omega}})$$

"Zeeman": $\boldsymbol{\delta_E \propto B} \rightarrow \boldsymbol{\delta_d = C\,\Omega}$

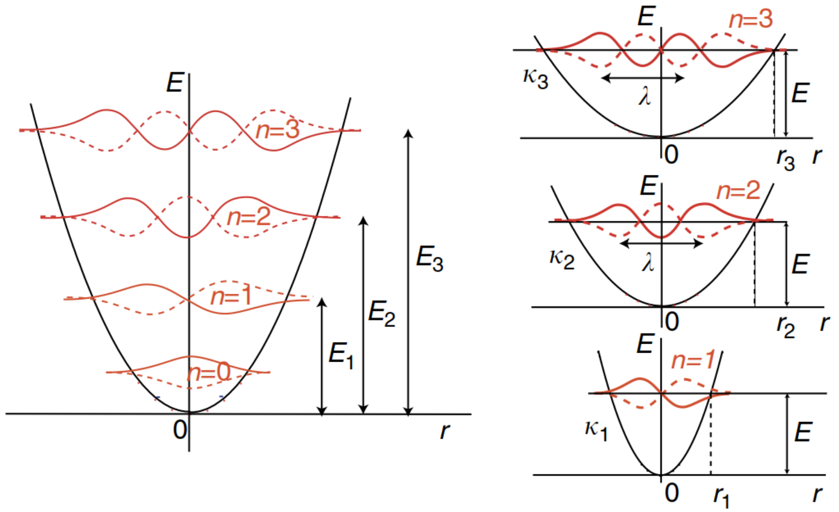




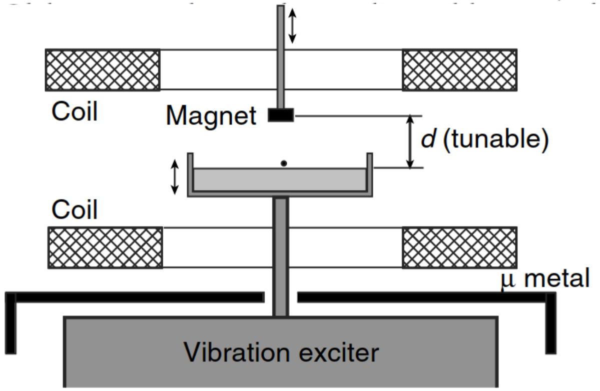


[Self-organization into quantized eigenstates of a classical wave-driven particle](), Perrard, Labousse, Miskin, Fort, Couder, **Nature** C. 2014

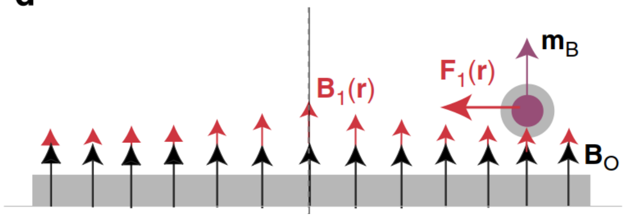
No rotation, but using **ferrofluid**: droplet is tiny magnet, getting **harmonic potential** “spring”: $F_m(d) \approx -\kappa(d)\vec{r}$

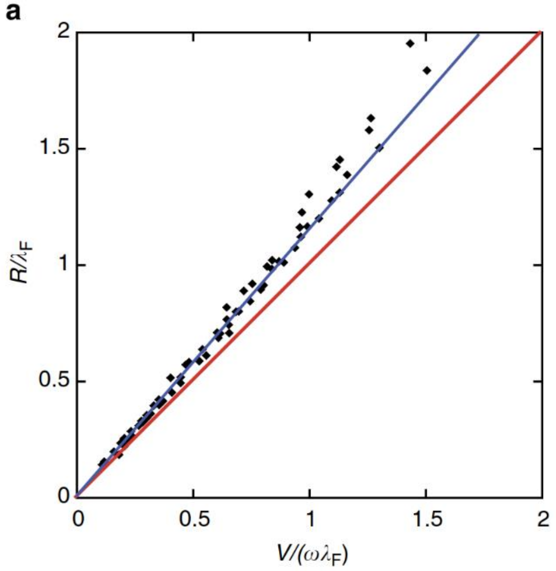


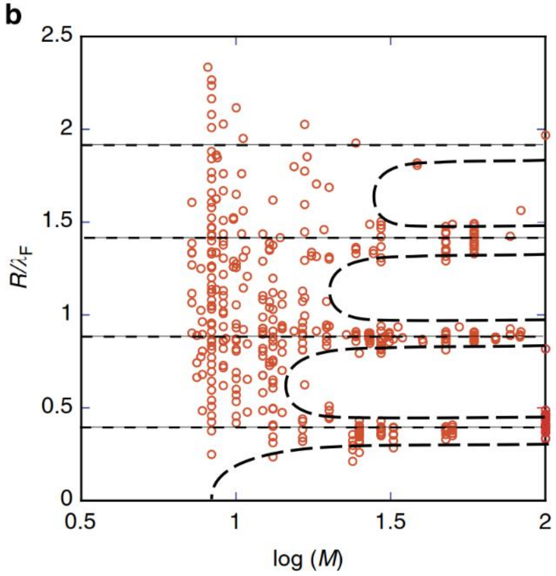
dimensionless width of the potential well $\Lambda = (V\sqrt{m_W/\kappa})/\lambda_F$



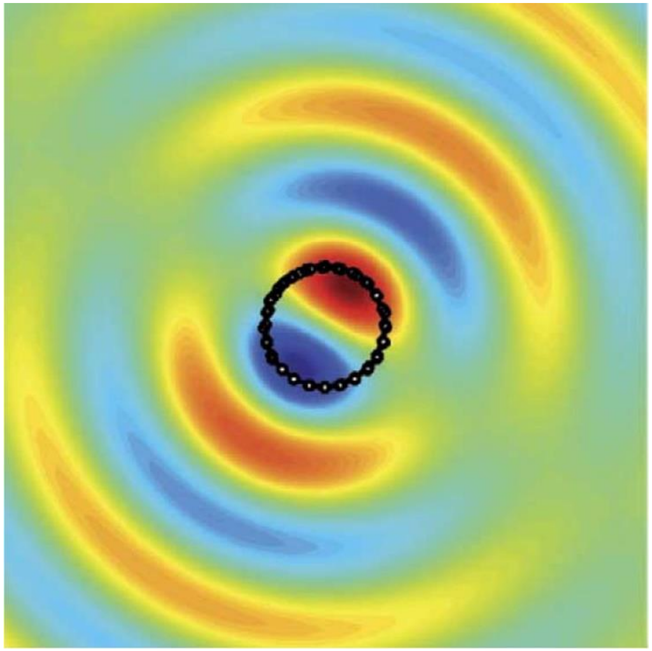
d

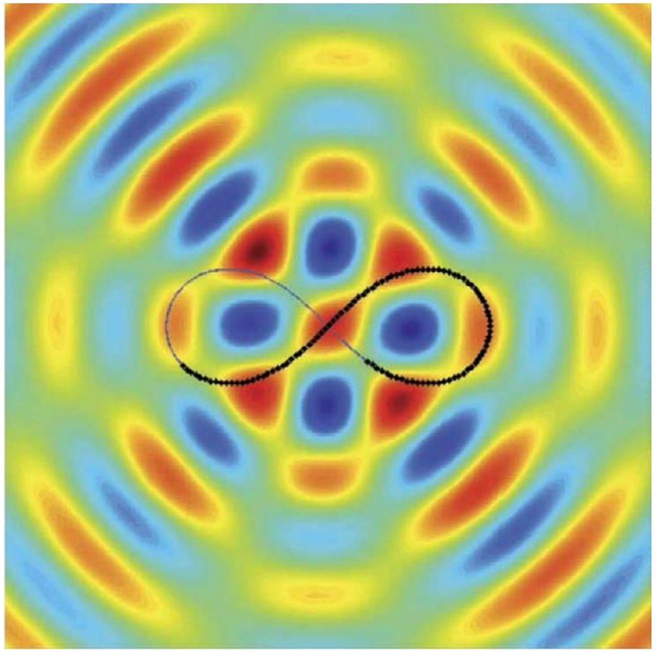






$M = \tau/T_F$

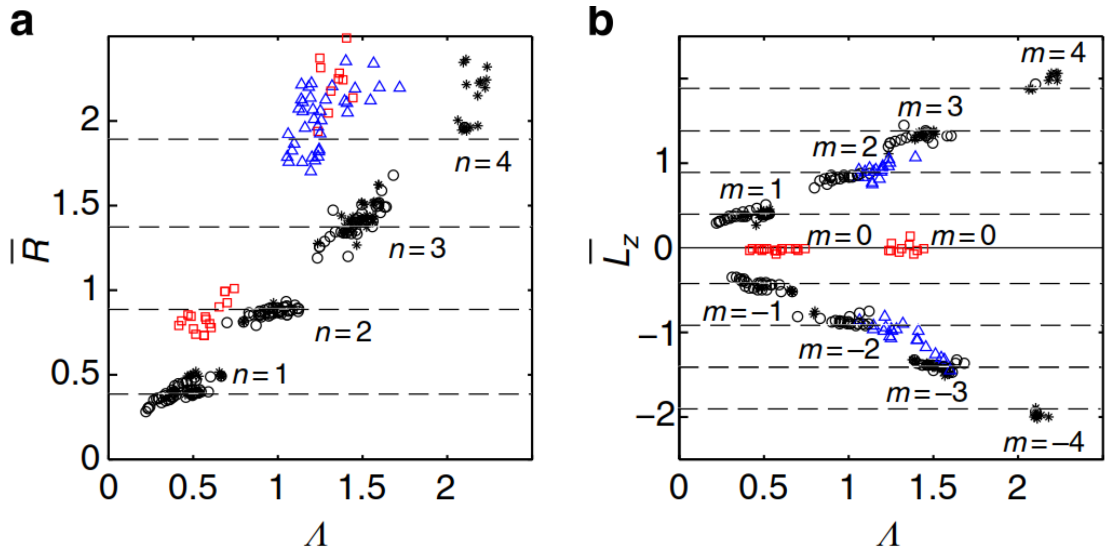




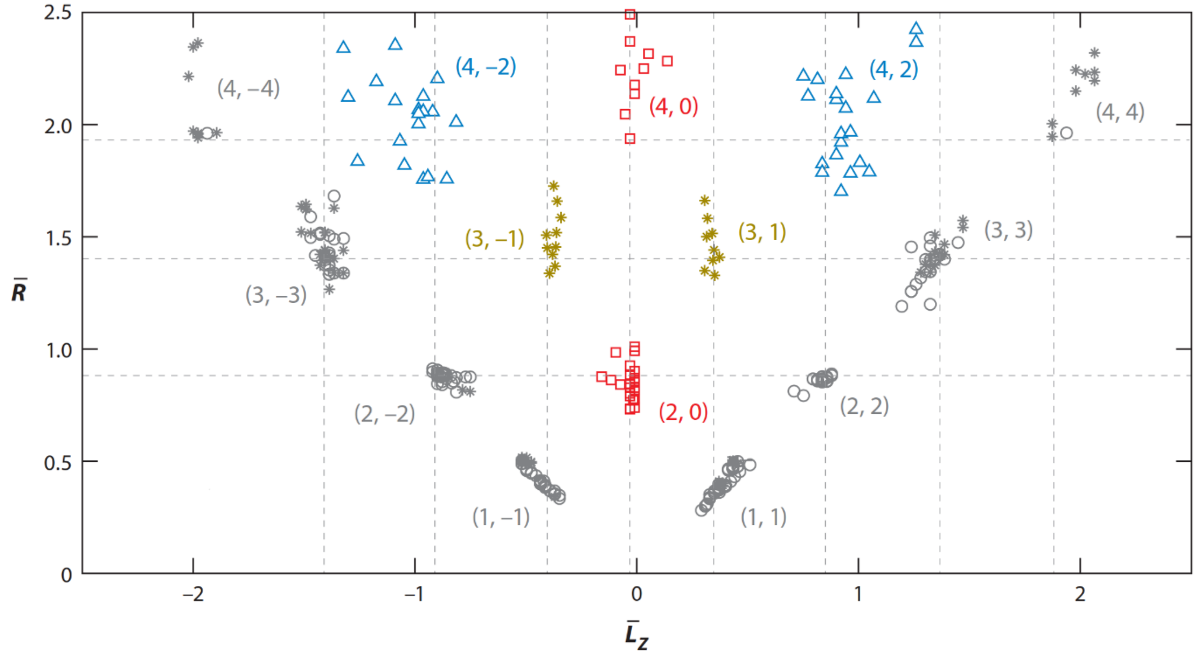
Double quantization – of **radius** (n) and **angular momentum** (m)

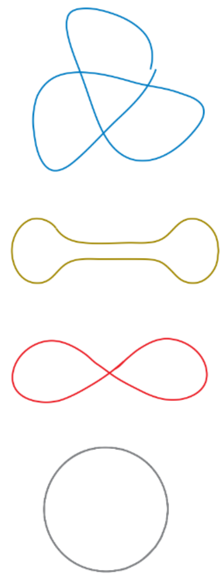
a,b: $n = 1, m = \pm 1$ c,d: $\boldsymbol{n = 2, m = \pm 2}$ e,f: $\boldsymbol{n = 2, m = 0}$

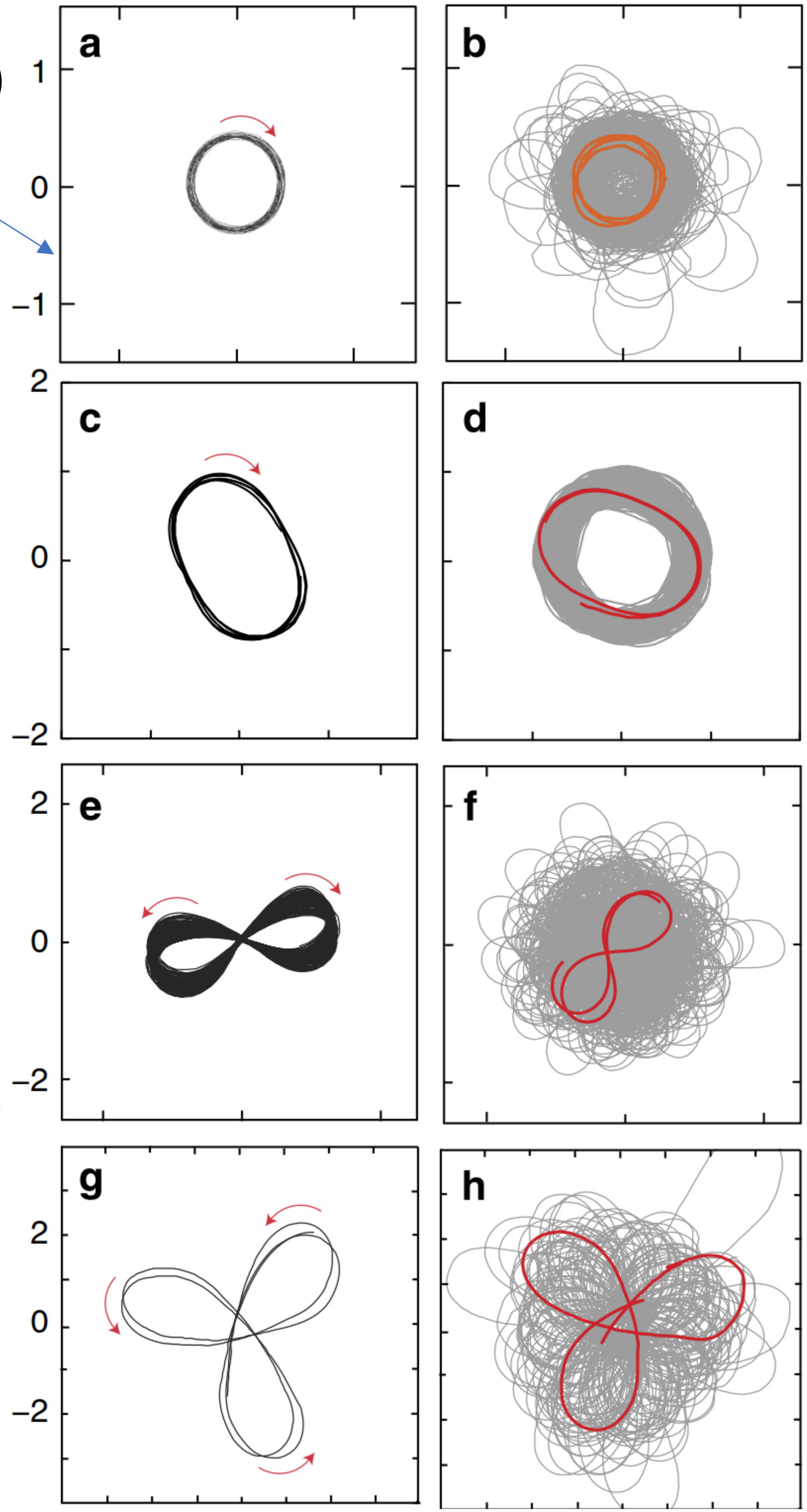
Cassini oval (lemniscate, trefoil, …): $|PP_1| \cdot |PP_2| \cdot \ldots = const$



dimensionless width of the potential well $\Lambda = (V\sqrt{m_W/\kappa})/\lambda_F$





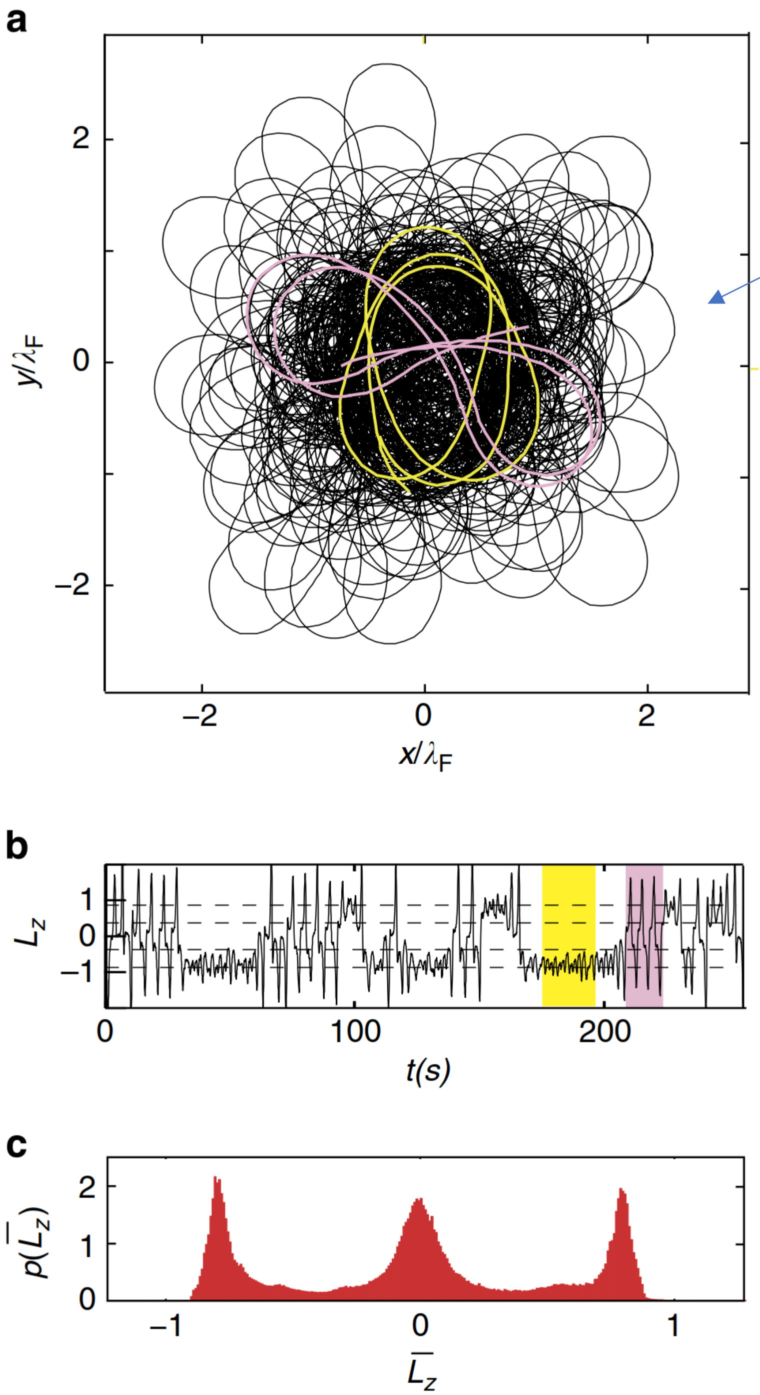


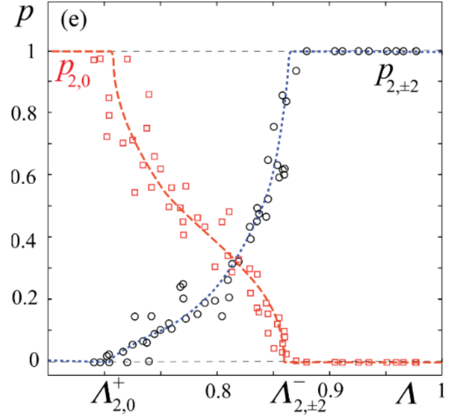
[Chaos Driven by Interfering Memory](#), **PRL** 2014

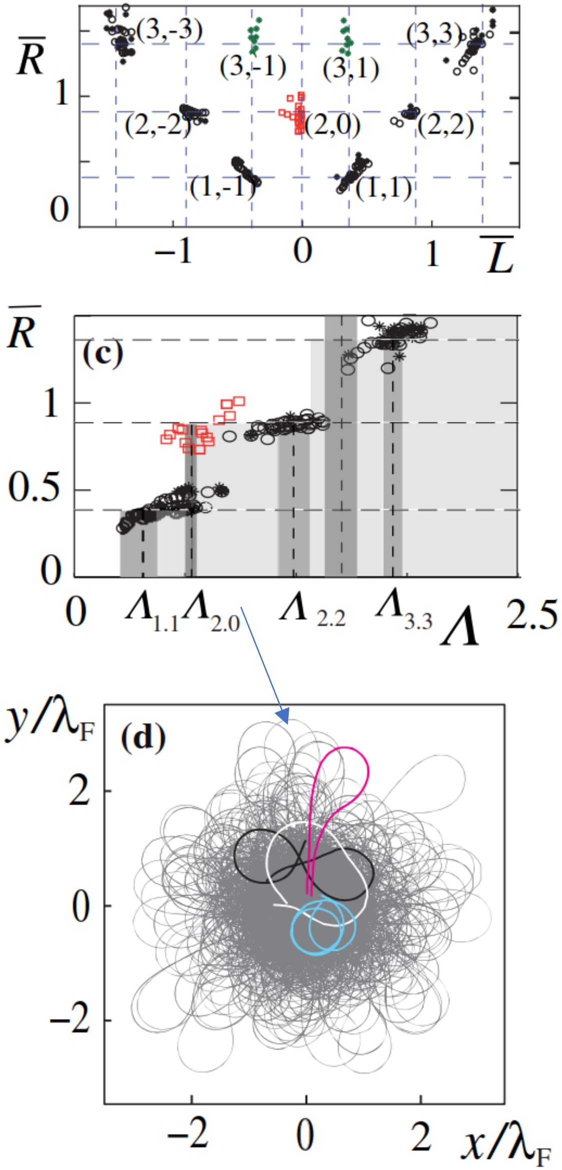
http://dualwalkers.com/eigenstates.html

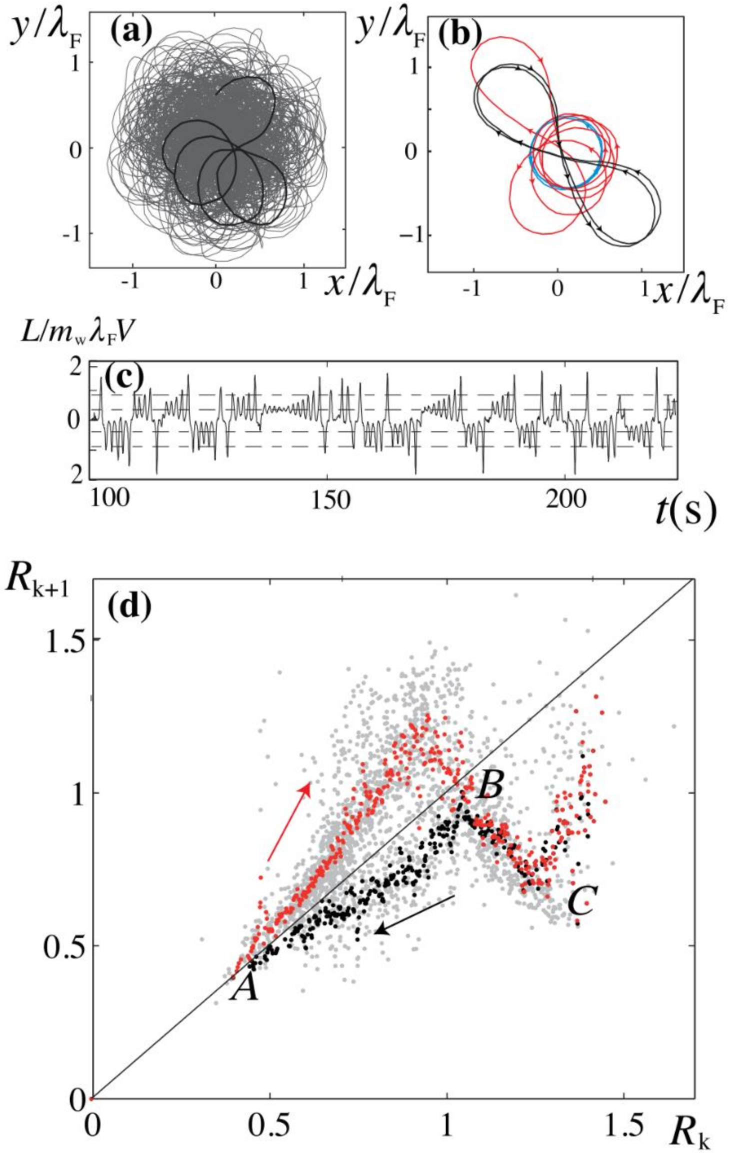
Intermittency between $n = 2, m = \pm 2, 0$

Collapse/**measurement of m?**







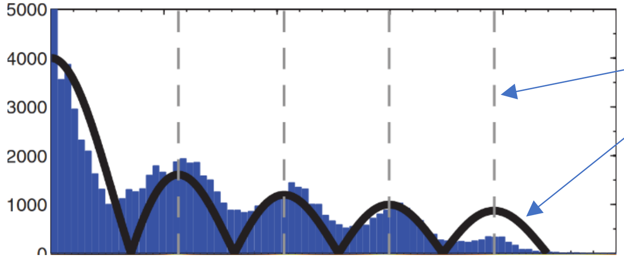


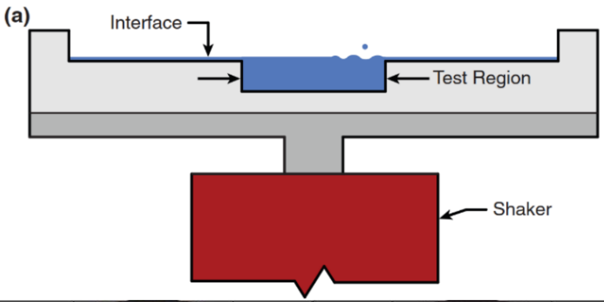
Wavelike statistics from pilot-wave dynamics in a **circular corral**,

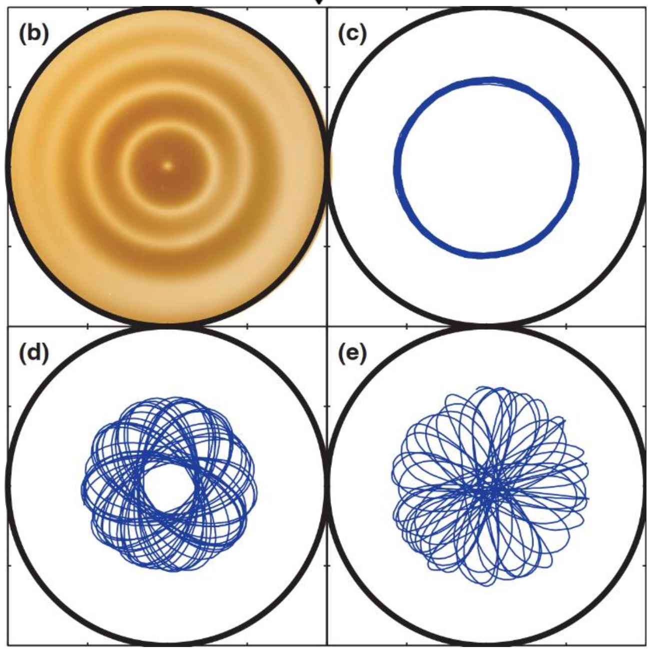
Harris, Moukhtar, Fort, Couder, Bush (MIT), PRE, 2013, video

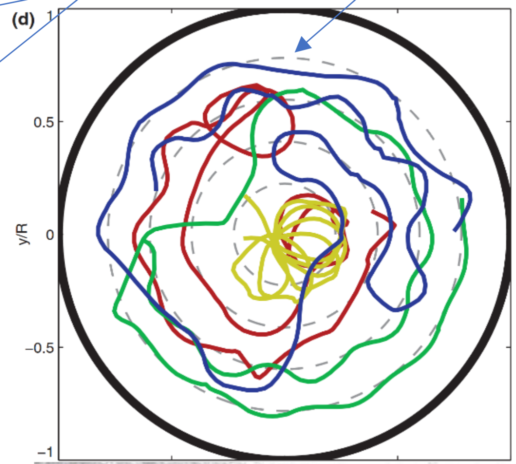
Statistics of walker's trajectory recreates Faraday mode

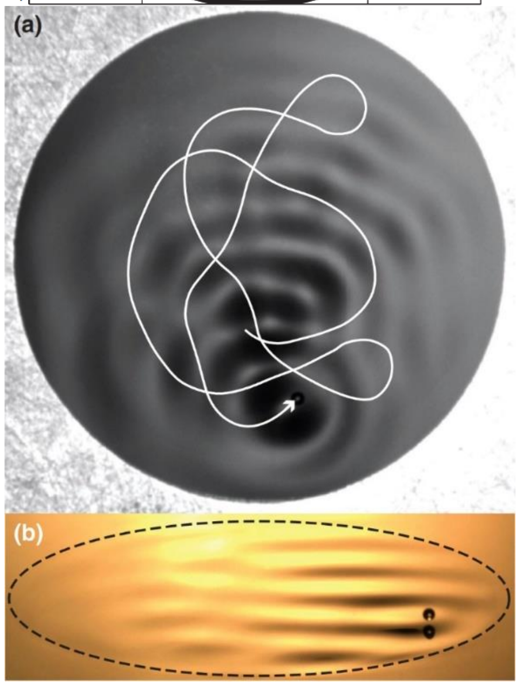
its maxima

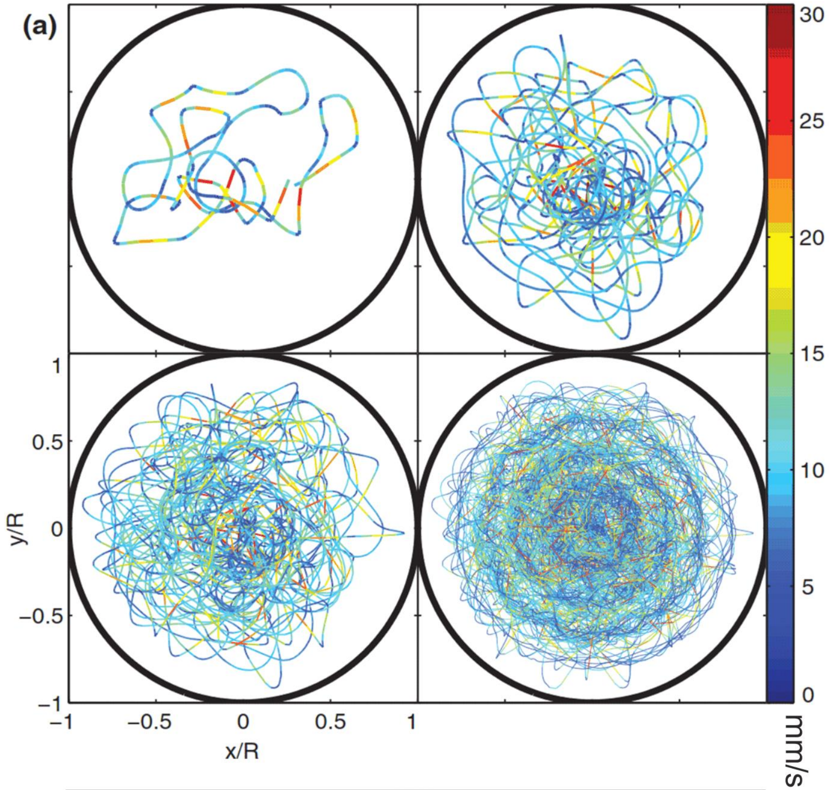


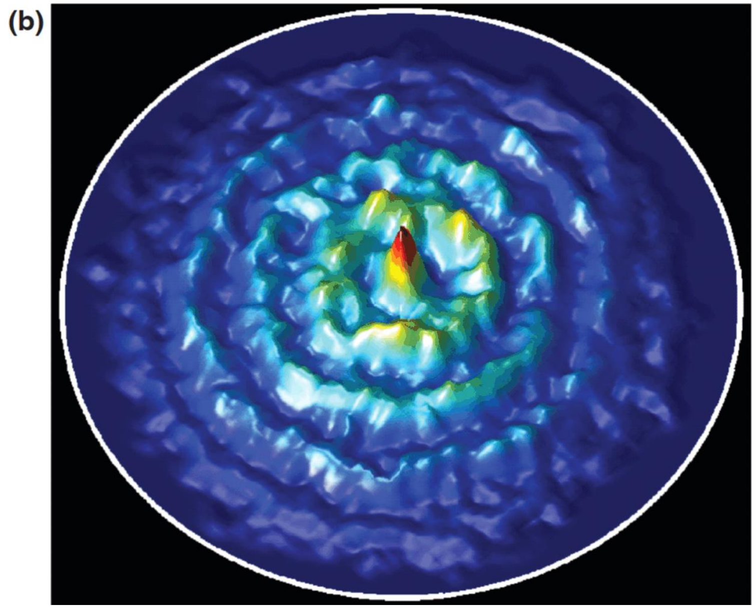






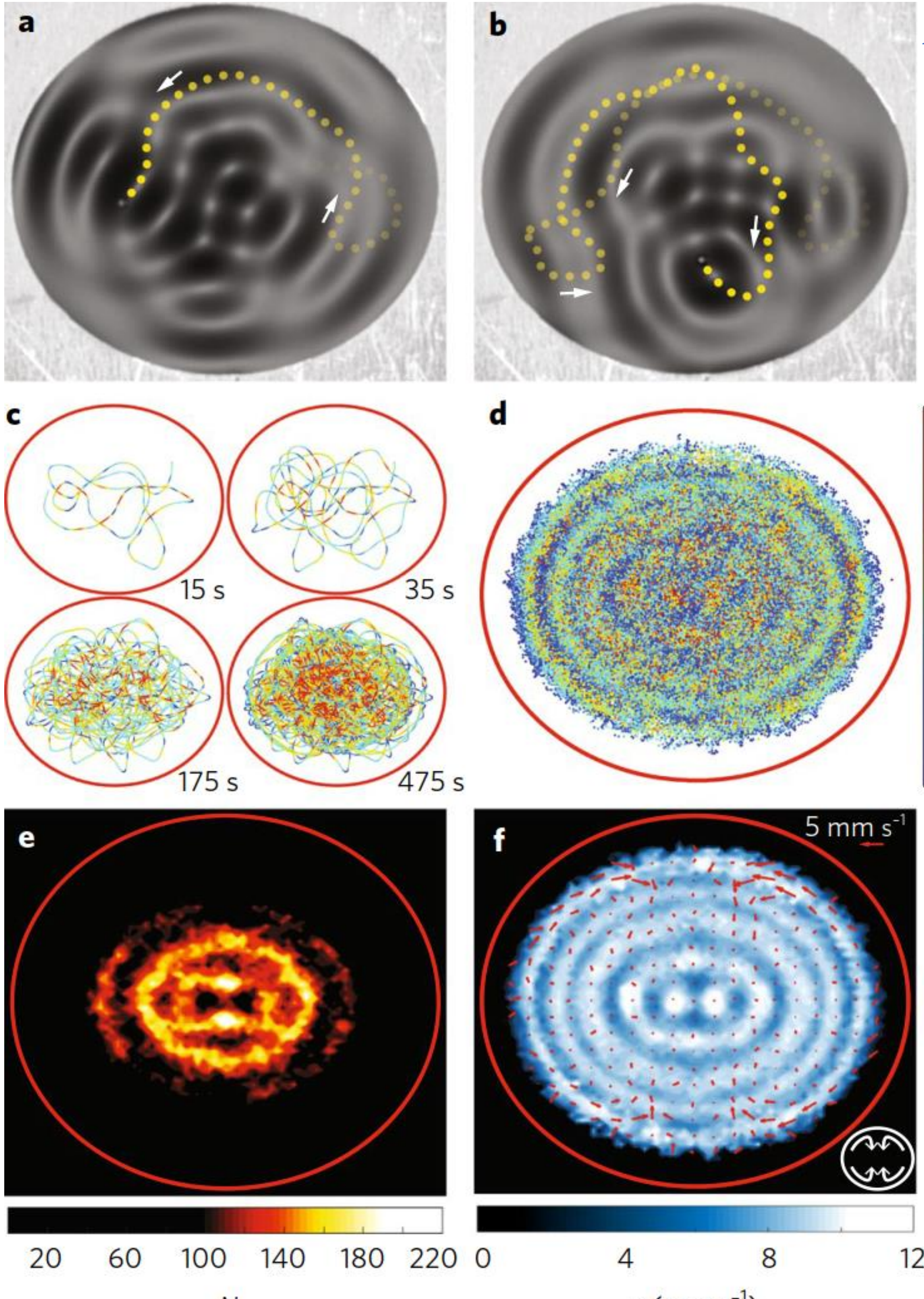


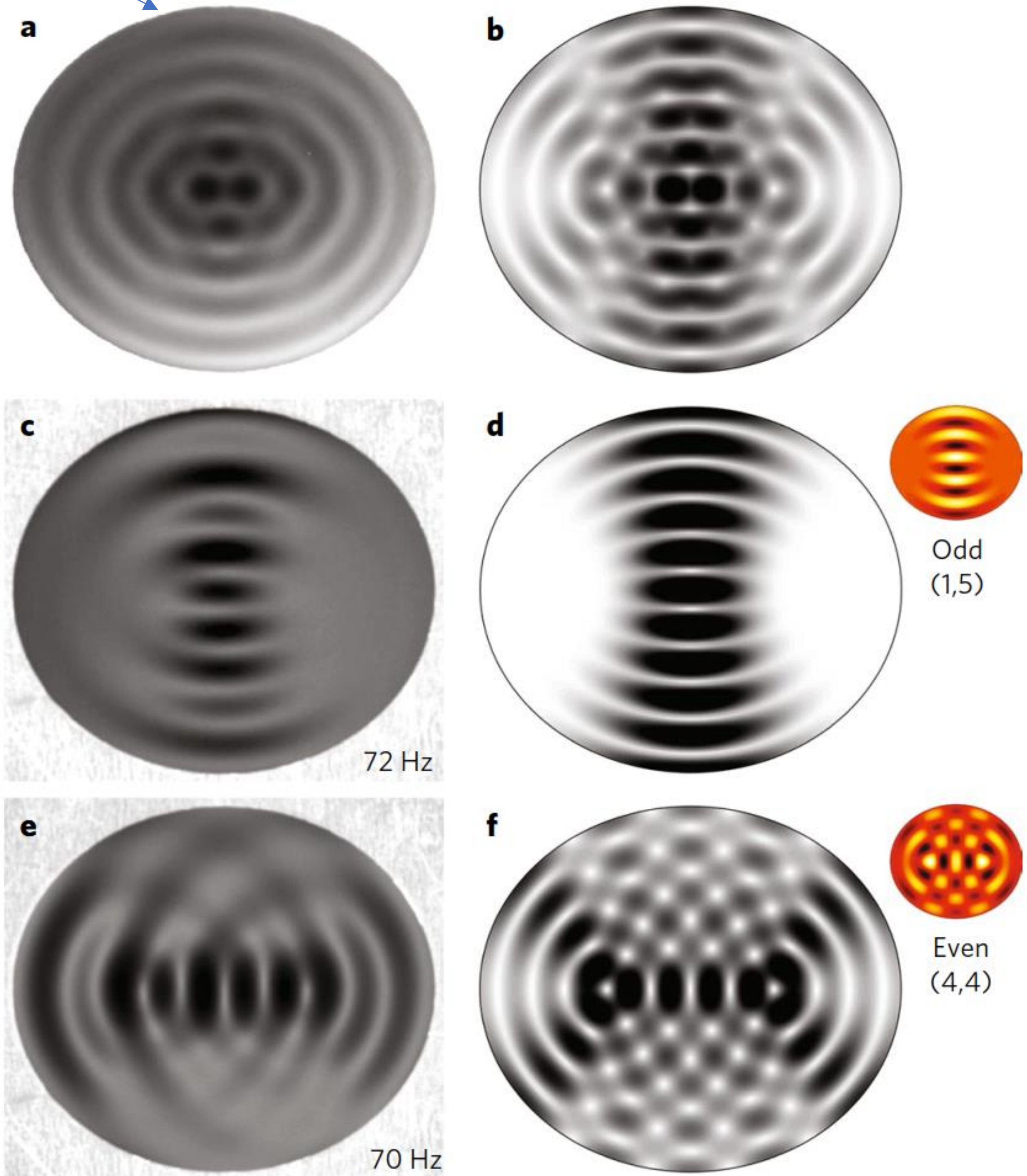


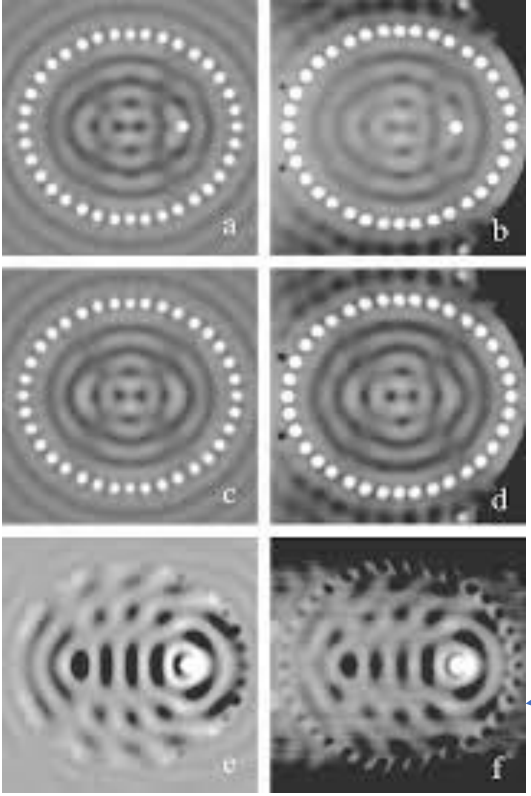


Statistical projection effects in a hydrodynamic pilot-wave system, MIT, **Nature** Physics, 2017

averaged (a), Faraday (c,e), analytic (d+f → b)







With circular defect:

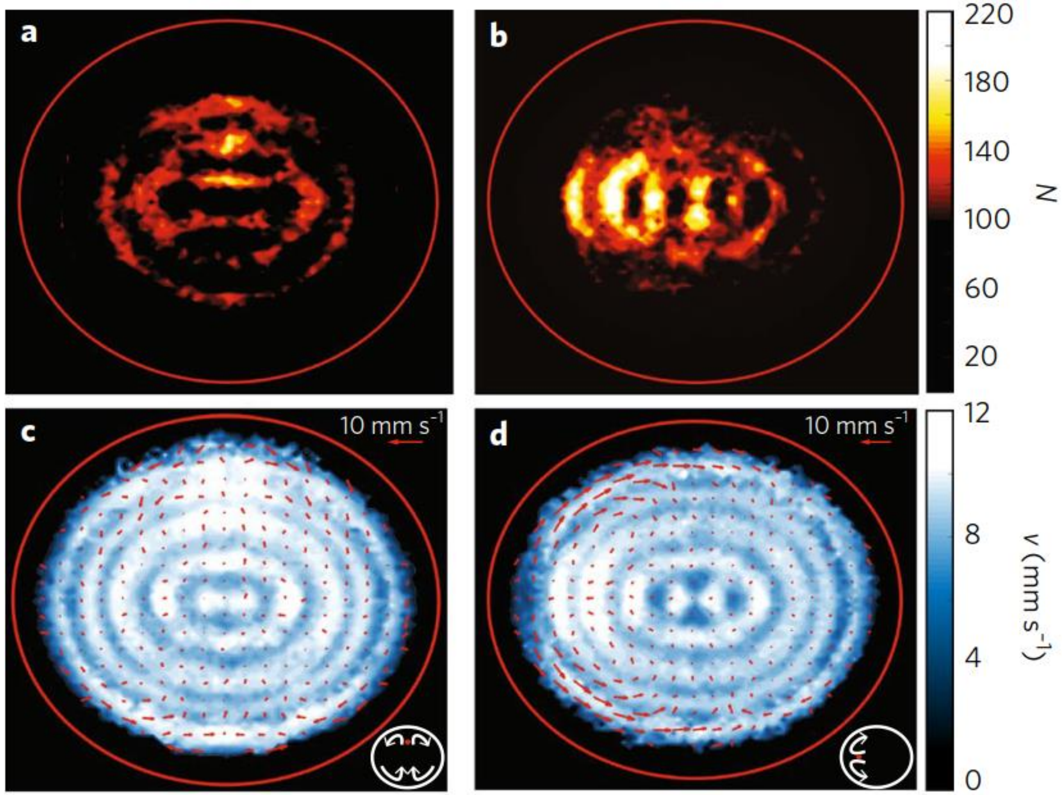
histogram (a,b)

velocities (c,d)

averaged -

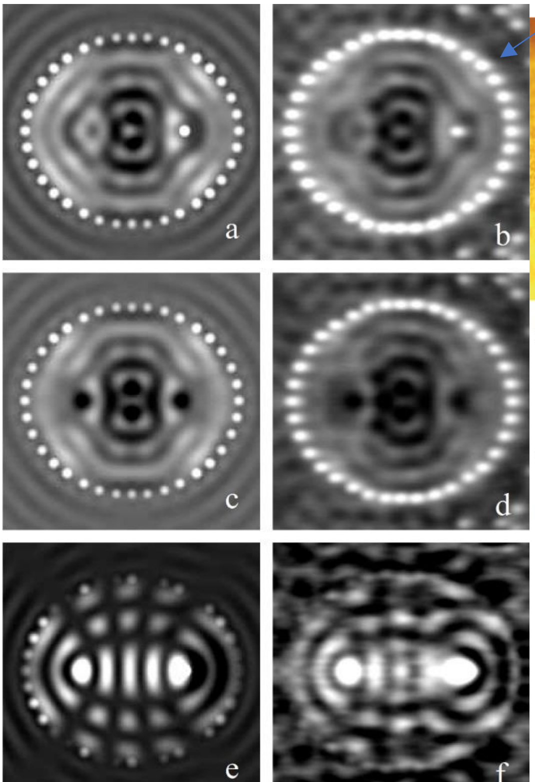
localization (e,f)

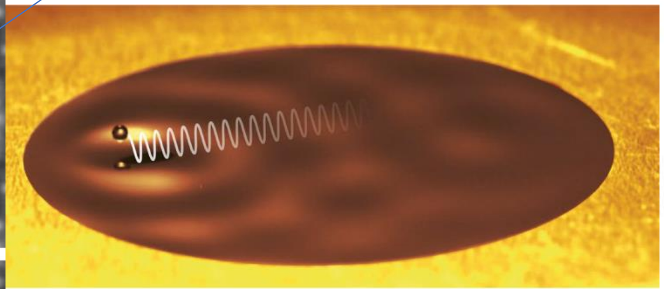
Faraday modes (g,h)

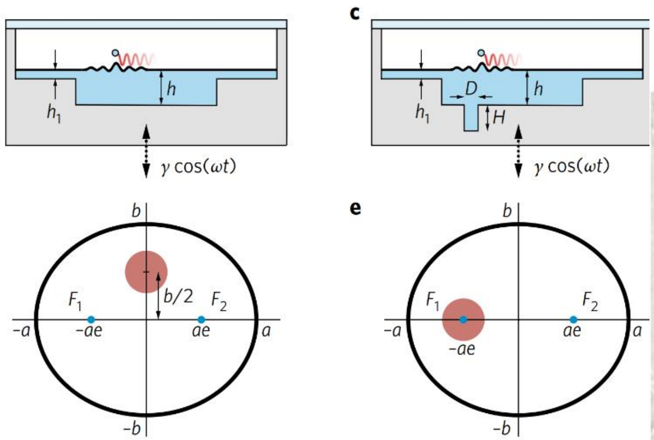


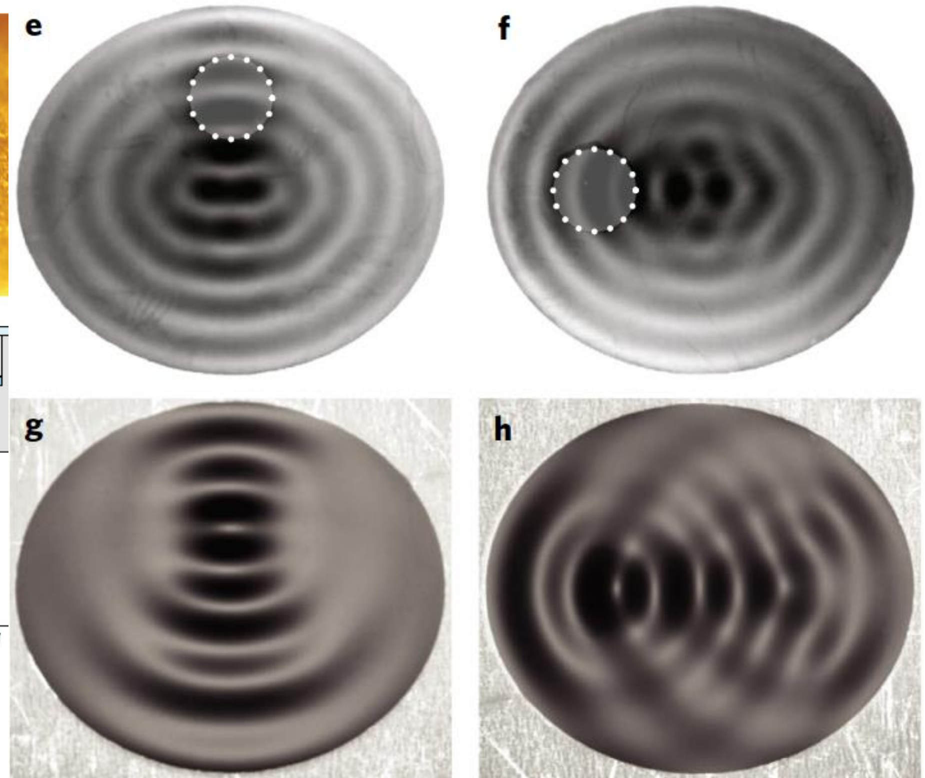
[Quantum corrals](#)

(atoms, scanned with STM)





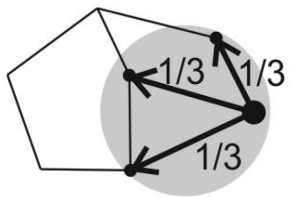




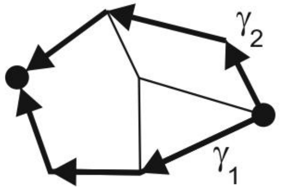
MERW –

time-symmetric uniform/**Boltzmann path ensemble, QM-like diffusion** (Jaynes maximal entropy principle), →QM ground state with Anderson-like localization

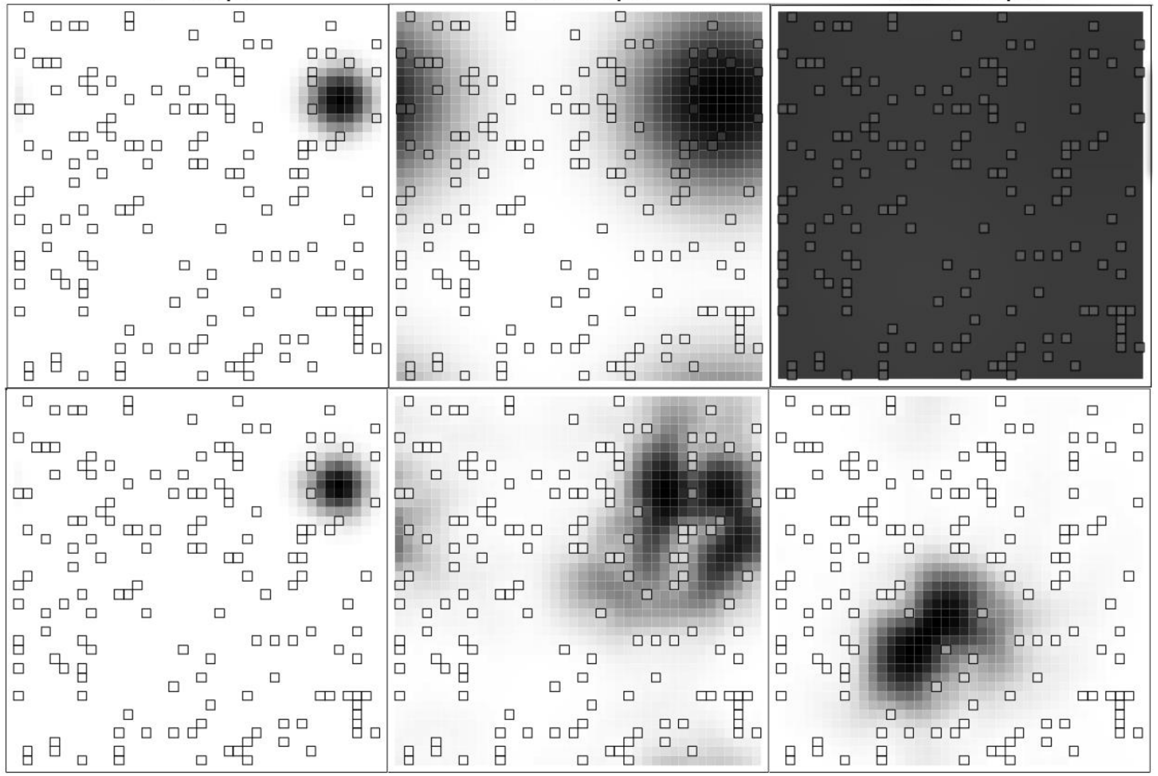
Generic Random Walk (GRW): assume uniform distribution among "the nearest neighbors"



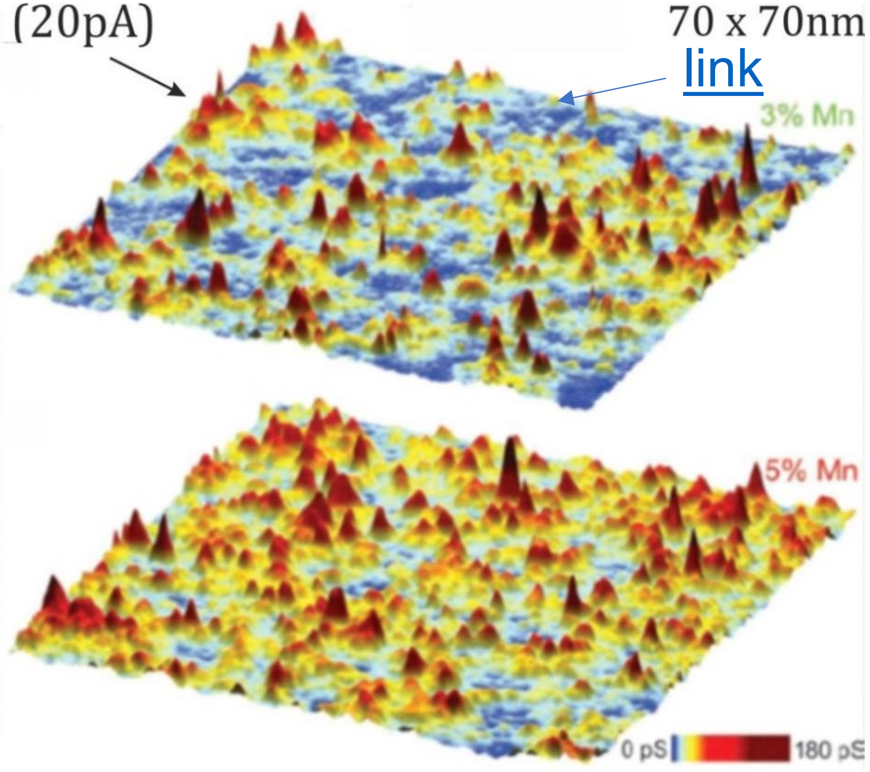
Maximal Entropy Random Walk (MERW): choose that for each two vertices, each path of given length between them is equally probable



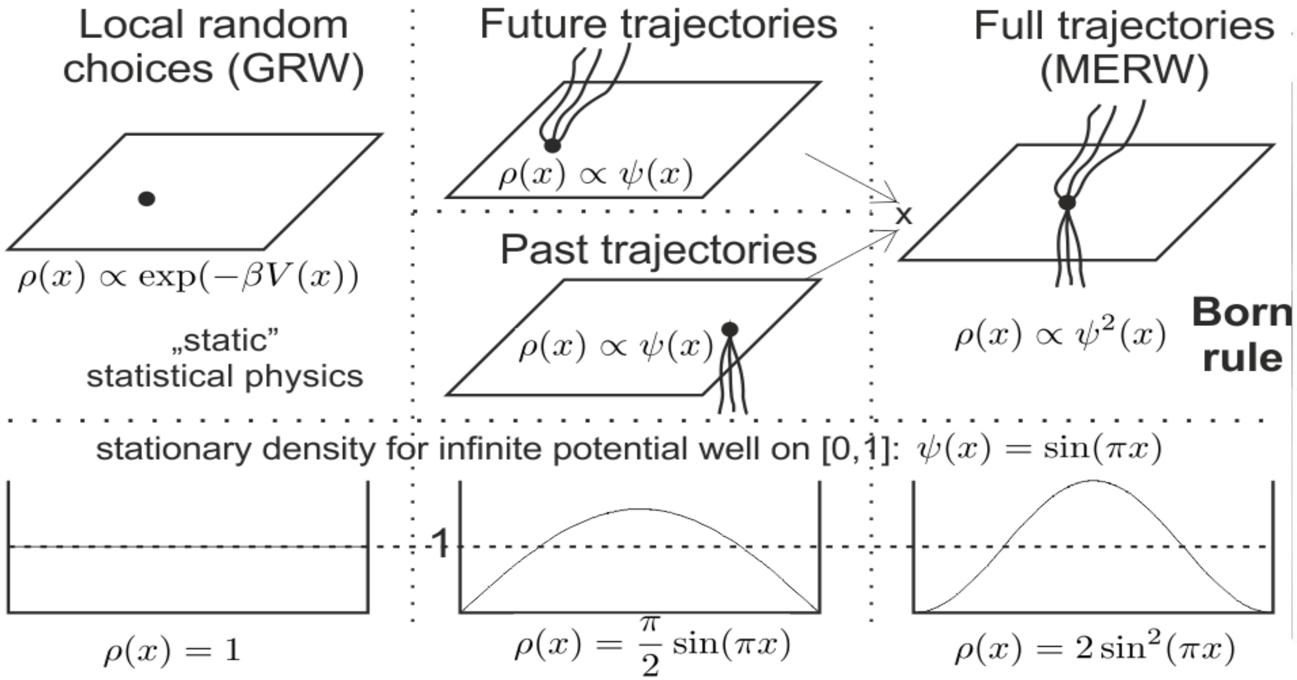
$\Pr(\gamma_1)=\Pr(\gamma_2)$



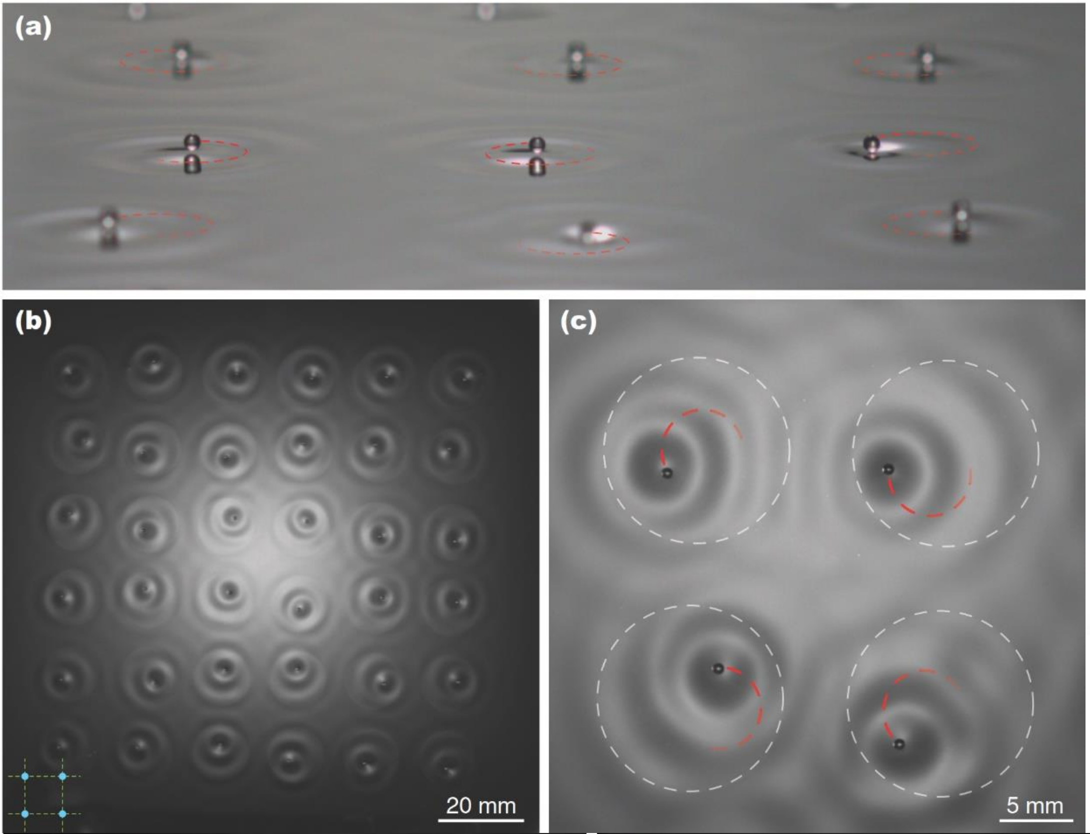
STM measurenments of electron density for $Ga_{1-x}Mn_xAs$ (20pA) 70 x 70nm

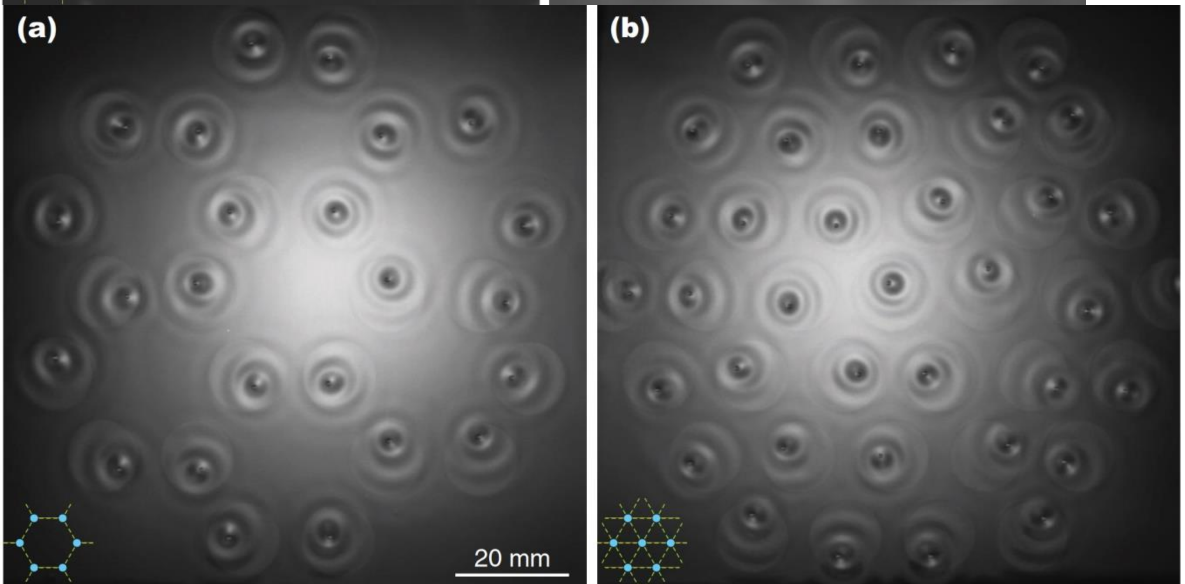


Path ensembles:

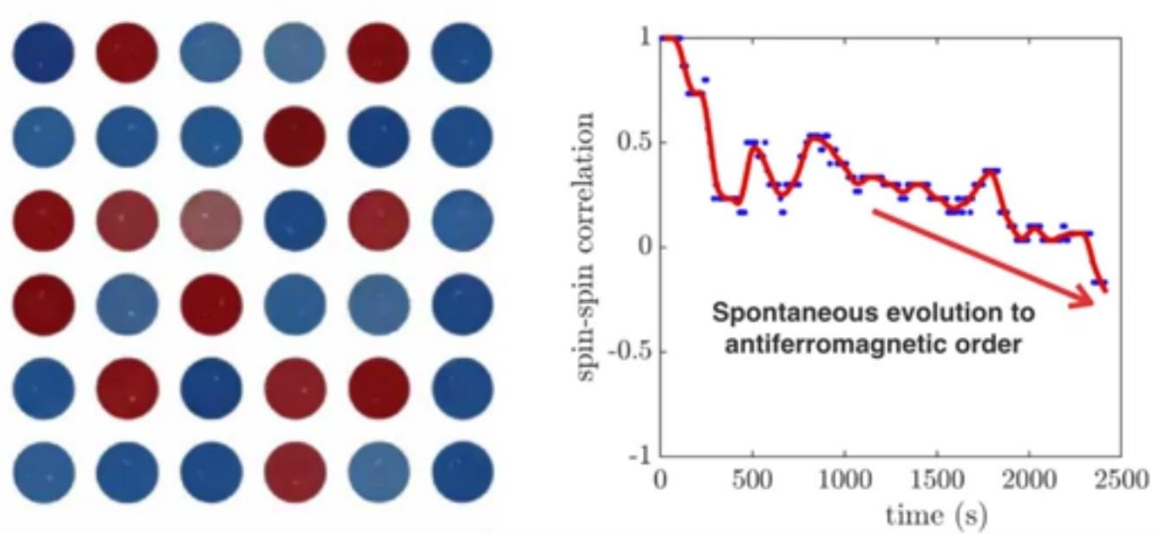


Spin lattices of walking droplets, MIT, PRF, 2018 - video lattice of corrals

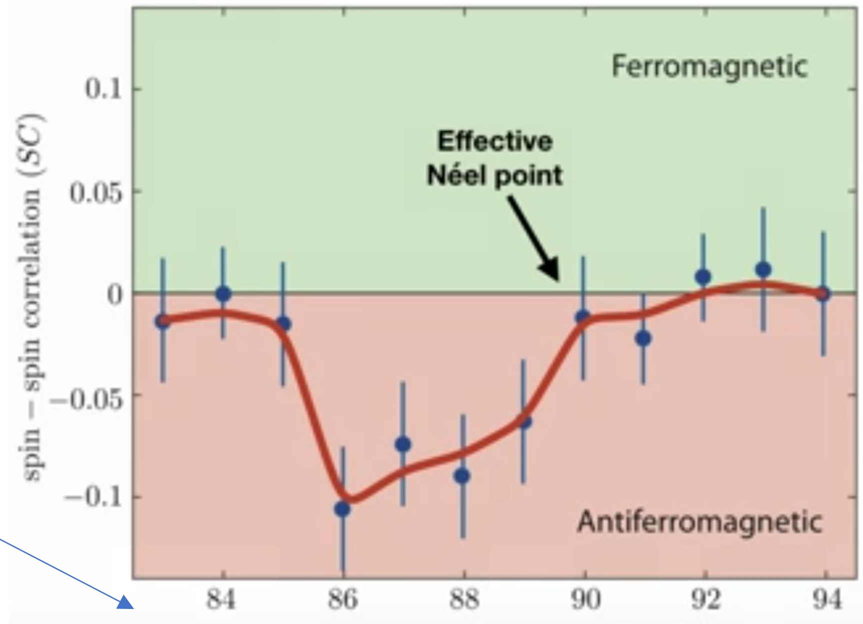




We measure the **time evolution** of the spin-spin correlation



Statistics yield preferred **antiferromagnetic** order



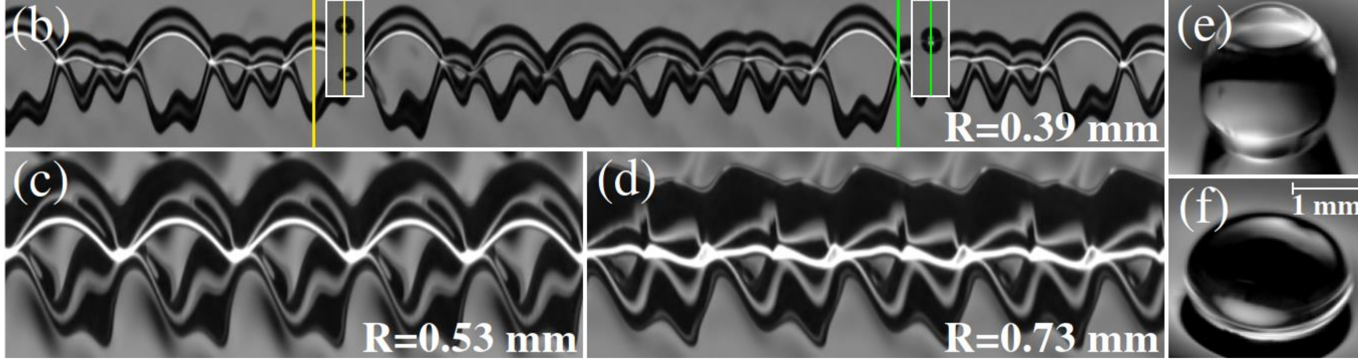
$\frac{\gamma}{\gamma_F}$ %

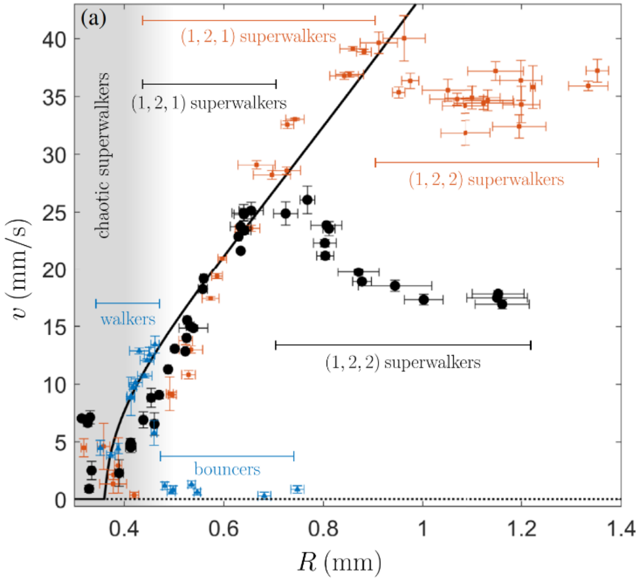
Multiple frequencies ([video]):

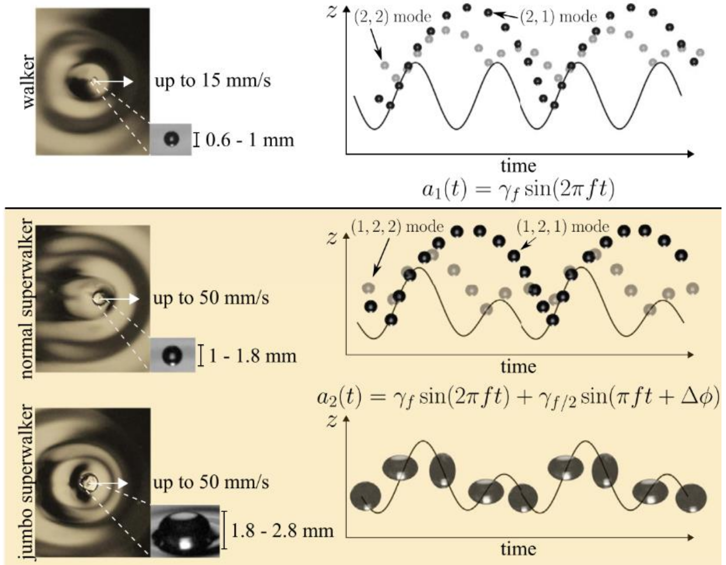
“[Superwalking droplets]”

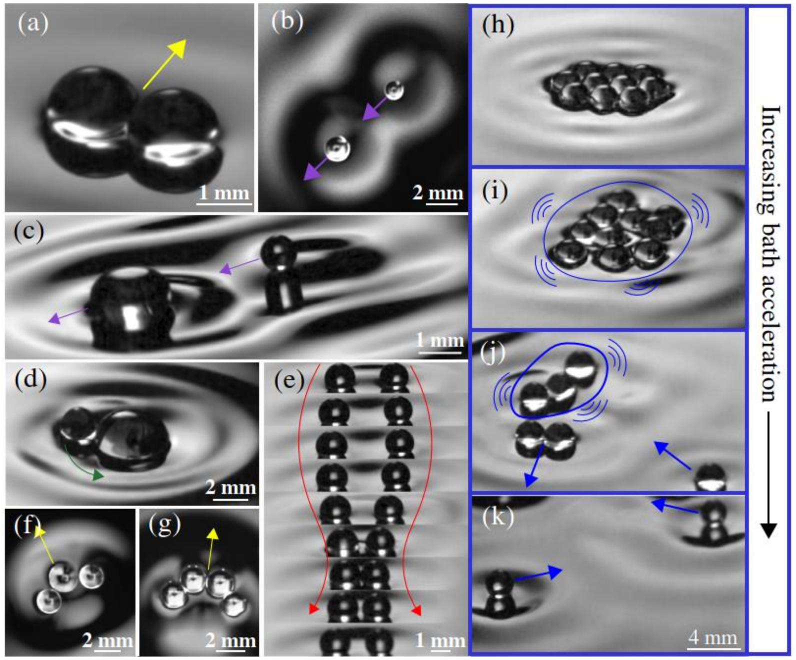
Valani, Slim, Simula, PRL 2019

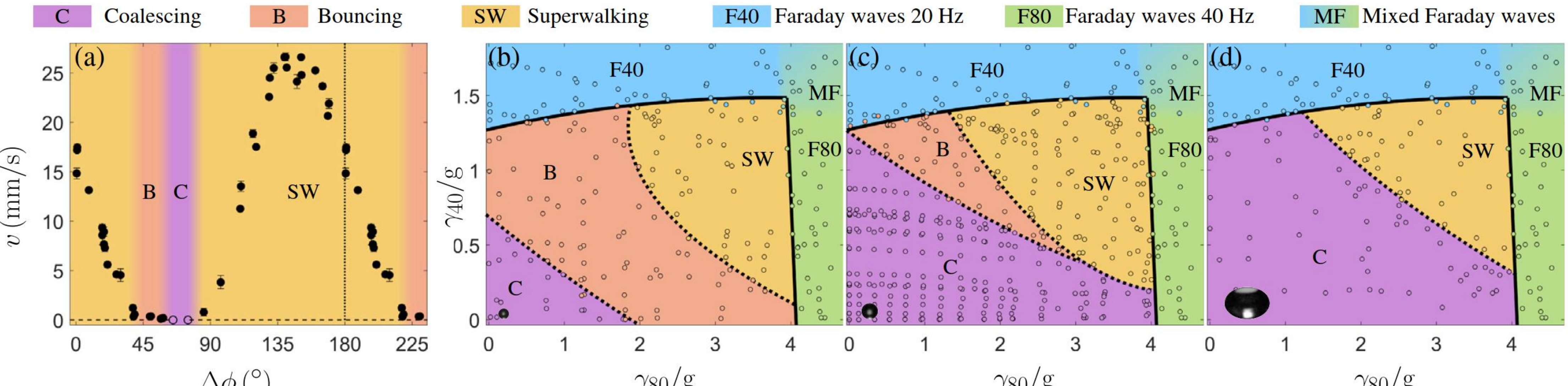
(is $mc^2 = hf = \hbar\omega$ universal?)







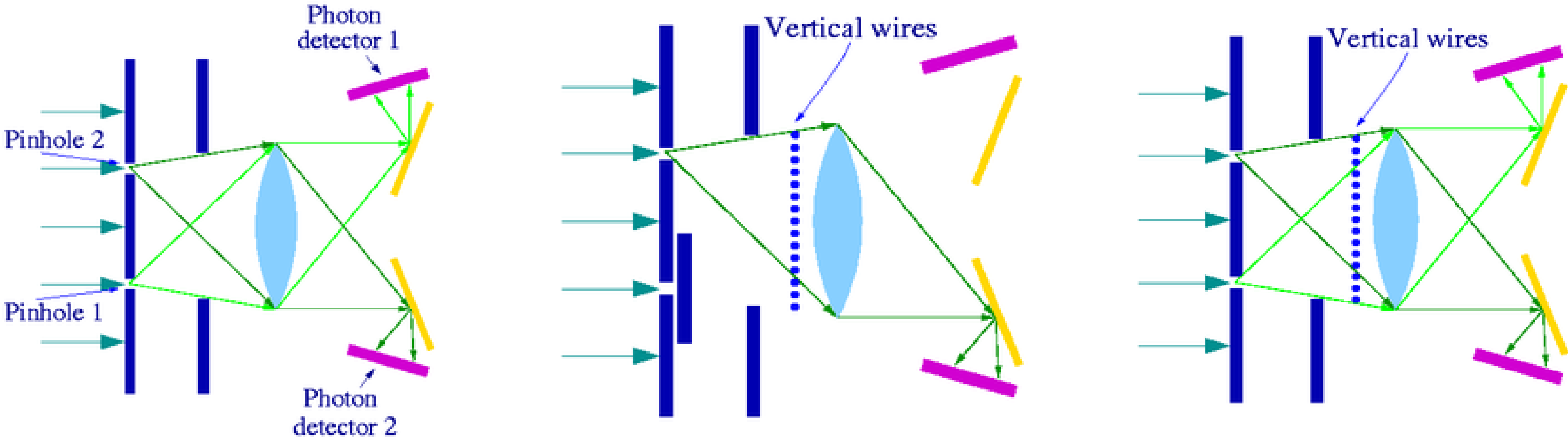




Afshar experiment (Wiki, 2004): **photons simultaneously wave and corpuscle**

How to understand wave-particle duality? Principle of complementarity:

“particle and wave aspects of quantum objects cannot be **observed** at the same time”

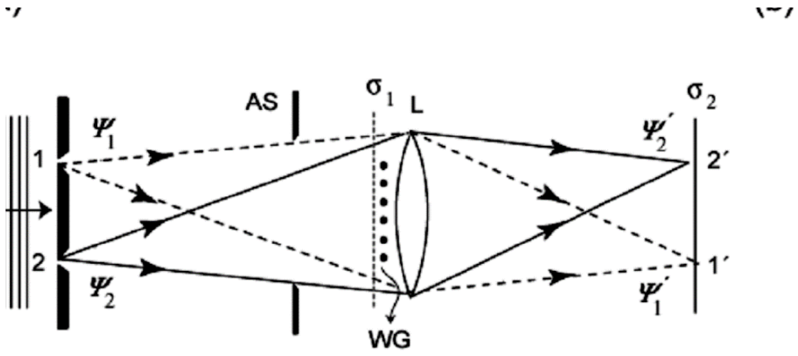


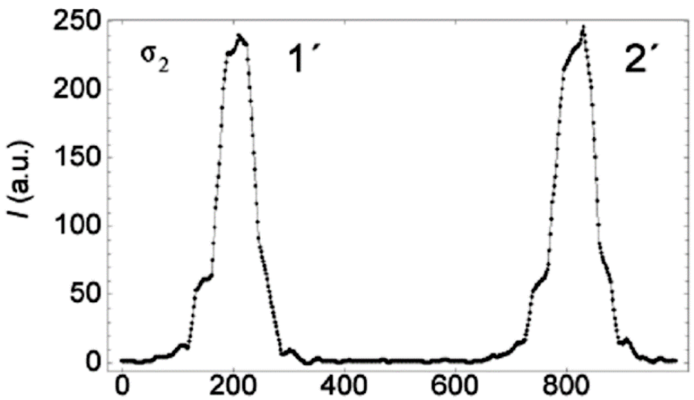
Putting **vertical wires** in **dark fringes** of double-slits **interference**, does not reduce intensity.

Interference – wave nature,

lens – corpuscular nature

contradicts principle of complementarity (?)





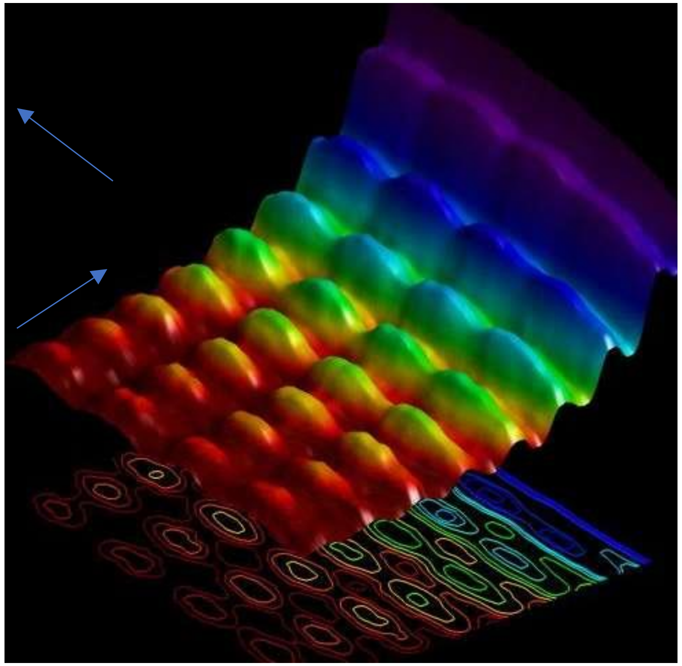
Simultaneous observation of the quantization and the interference pattern of a plasmonic near-field wave (Nature C. 2015) – of surface plasmon polaritons: EM waves in surface of material (news)

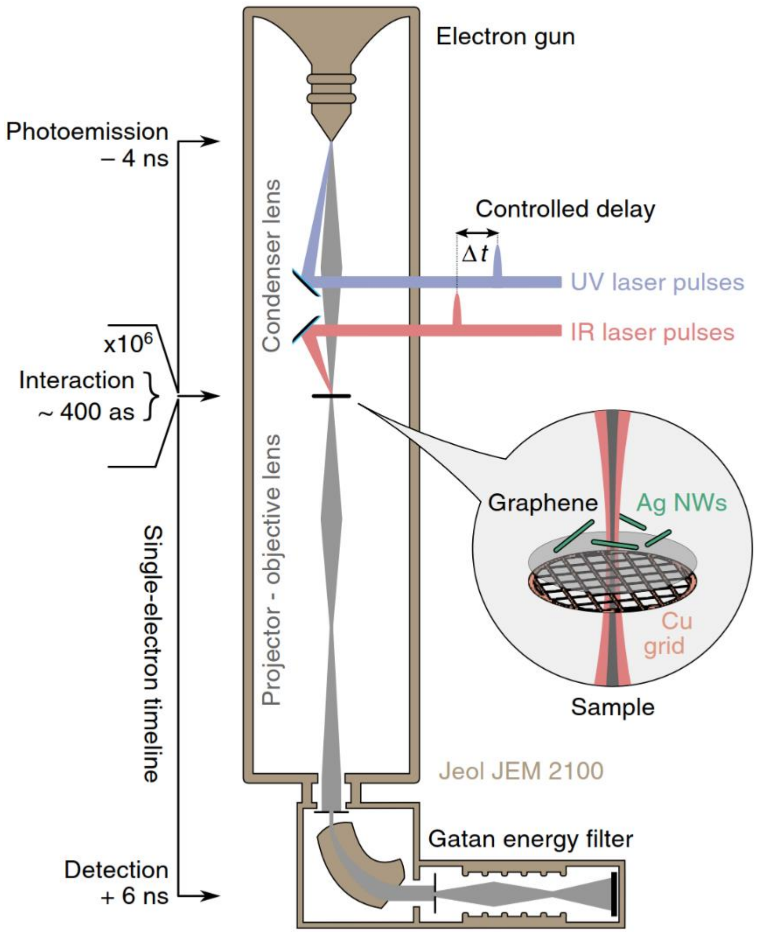
$r \sim 50$nm Ag **nanowires**

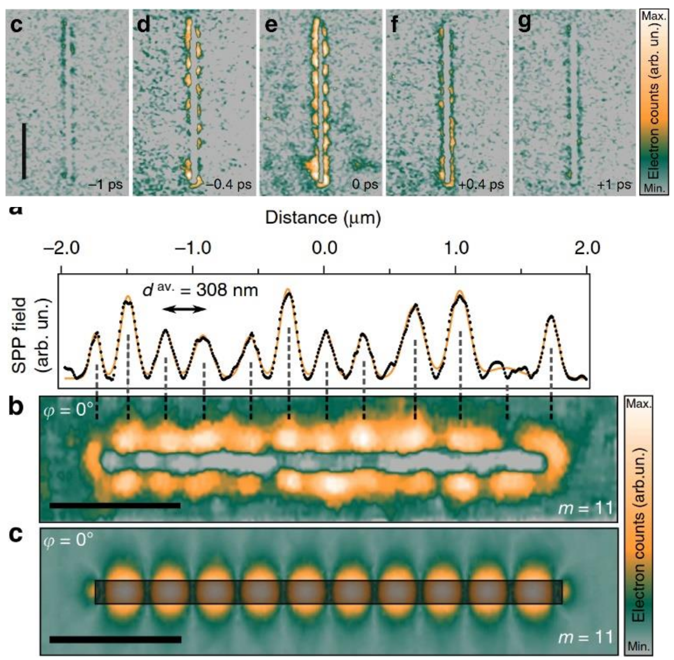
Interference (wave):

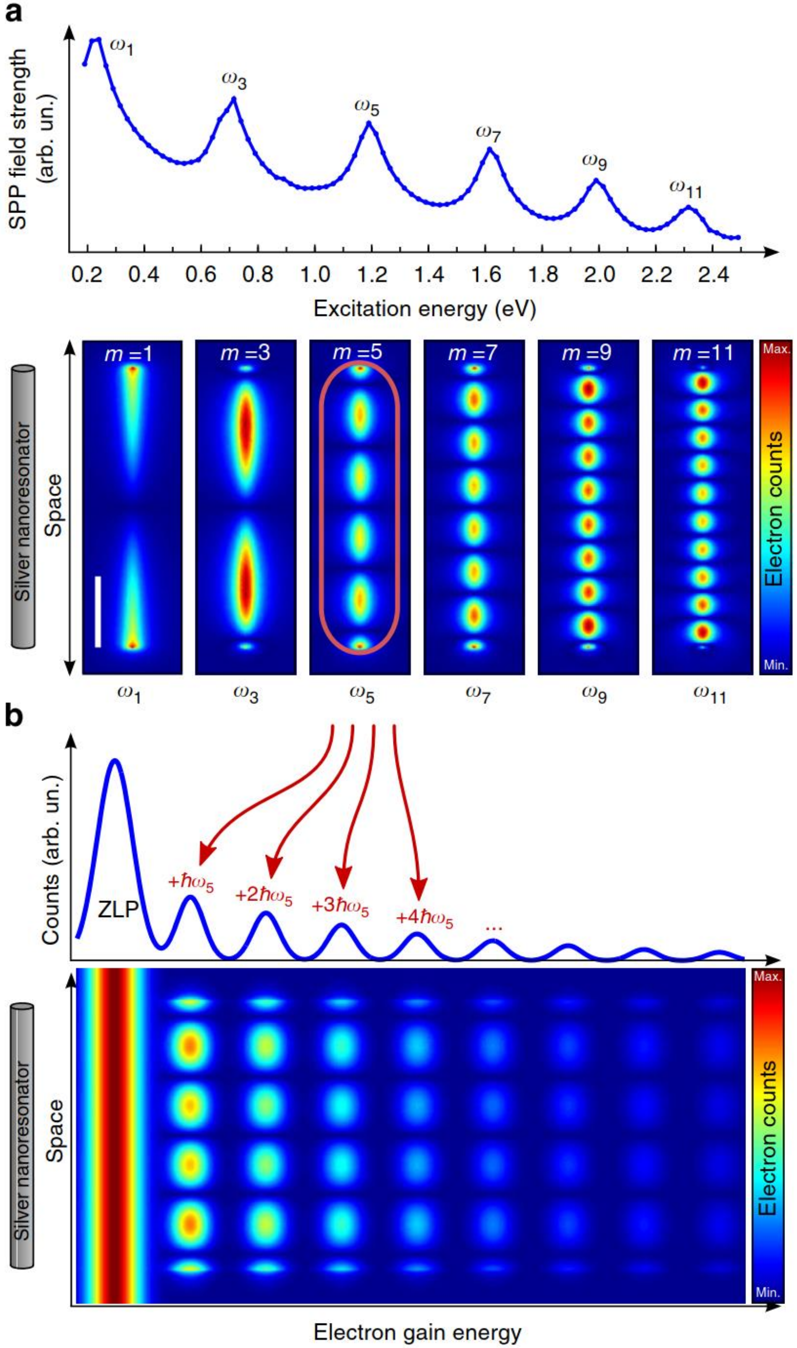
laser creates standing EM waves

quantization (particle): in energy exchange with probing electrons









Observation of **electron's clock** $(\approx 10^{21} Hz)$ – resonating with crystal lattice

De Broglie: $\boldsymbol{E} = mc^2 = hf$ Schrödinger: $\psi = \psi_0 e^{iEt/\hbar} = \psi_0 e^{2\pi i f t}$

Compton wavelength $\lambda_C = \frac{c}{f_0} \approx \frac{r_{Bohr}}{2\pi \cdot 137} \approx 2\text{pm}$ light distance in "one tick"

Time dilation: $f = \frac{f_0}{\gamma}$ for $\gamma = 1/\sqrt{1 - v^2/c^2}$

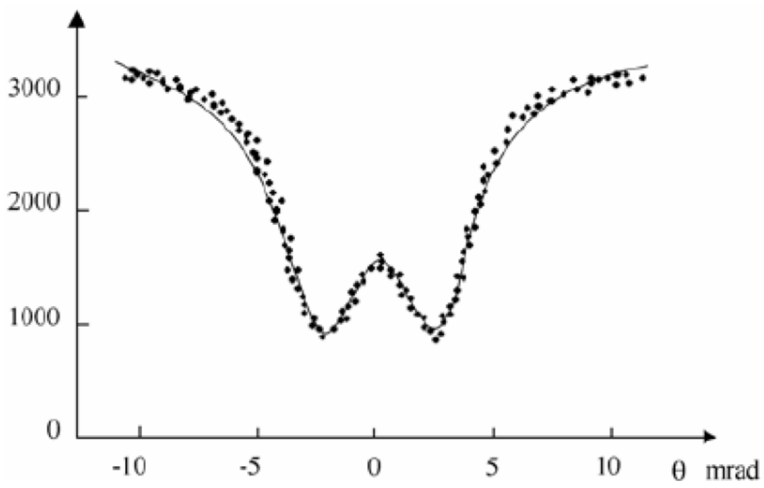
$$L = \frac{v}{f} = \frac{hv\gamma}{m_0 c^2} = \frac{hp}{(m_0 c)^2}$$

<011> axis of silicon crystal has $L = 3.84\ \text{Å}$

$$cp = m_0 \gamma v = \frac{L(m_0 c)^2}{hc} \approx \frac{3.84\text{Å}(0.511\text{MeV})^2}{0.0124\ \text{MeV Å}} \approx 80.87\ \text{MeV}$$

Along <110> axis of silicon crystal atomic spacing correspond to 'internal clock' period for $E \approx 81 MeV$ electrons – observed resonance:

P. Catillon, N. Cue, M.J. Gaillard, R. Genre, M. Gouanère, R.G. Kirsch, J.-C. Poizat, J. Remillieux, L. Roussel, M. Spighel, *A Search for the de Broglie Particle Internal Clock by Means of Electron Channeling*, Found Phys (2008) 38: 659–664



Zitterbewegung ("trembling motion") –

especially for electrons ($\approx 10^{21} Hz$) - from **Dirac equation**:

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi = \left(\alpha_0 mc^2 + \sum_{j=1}^{3} \alpha_j p_j c\right)\psi \qquad \text{where} \quad \alpha_j \in \mathbb{C}^{4\times 4} \qquad \Rightarrow$$

$$x_k(t) = x_k(0) + c^2 p_k H^{-1} t + \frac{1}{2} i\hbar c H^{-1}(\alpha_k - c p_k H^{-1})\left(e^{-\frac{2iHt}{\hbar}} - 1\right)$$

Observation of Zitterbewegung in a spin-orbit-coupled Bose-Einstein condensate, Qu, Hamner, Gong, Zhang, Engels, PRA 2013

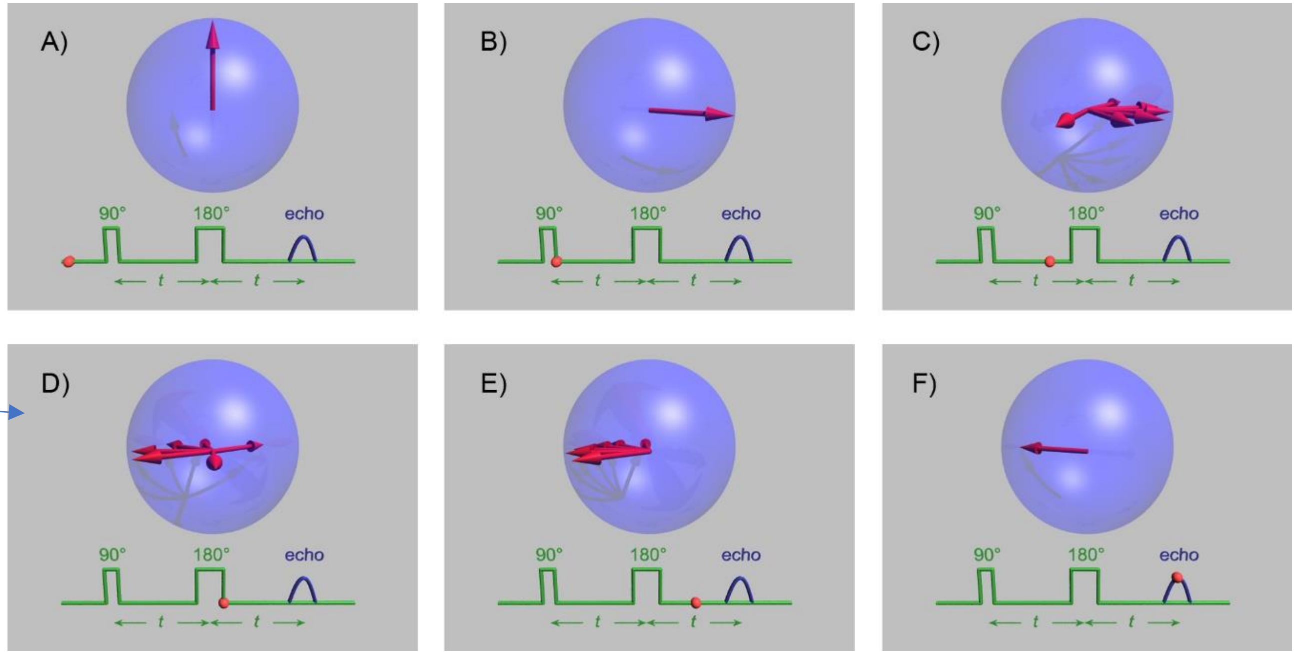
"In the past eight decades, analogs of the ZB oscillation have been predicted to exist in various physical systems [2–8], ranging from solid state (e.g., semiconductor quantum wells) to trapped cold atoms, but experimentally, a ZB analog has only recently been observed using trapped ions as a quantum emulator of the Dirac equation [Quantum simulation of the Dirac equation, Nature, 2010]. A crucial ingredient for the ZB oscillation is the coupling between spin and linear momentum of particles, leading to simultaneous velocity and position oscillations accompanying the spin oscillation, which distinguishes ZB from Rabi oscillations where spin oscillations between two bands do not induce velocity and position oscillations."

Direct observation of zitterbewegung in a Bose–Einstein condensate, LeBlanc, Beeler, Jimenez-Garcia, Perry, Sugawa, Williams, Spielman, New Journal of Physics, 2013

“Trembling motion” as precession of electron’s spin?

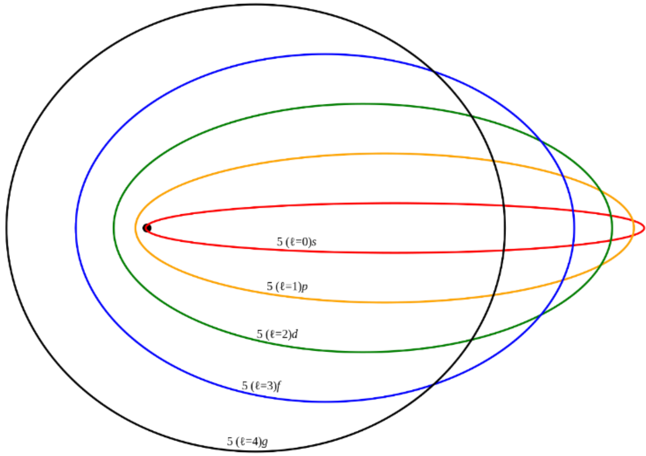
Larmor frequency: $\omega = \frac{eg}{2m} B$,

e.g. in EPR (Electron Paramagnetic Resonance) … spin echo



Spin-dynamical theory of the wave-corpuscular duality, IJoTP, 1987

Postulates **precession of spin $\hat{s}$ for traveling electron**:



$$\frac{d\hat{s}}{dt} = \left(\frac{mv}{\hbar}\right) \hat{s} \times v \;\Rightarrow\; \frac{d\psi_s}{dt} = \frac{mv^2}{\hbar} \qquad \psi_s - \text{phase of } \hat{s}$$

$$\frac{d\psi_s}{dl} = \frac{mv}{\hbar} \qquad \oint \frac{mv}{\hbar} dl = 2\pi n$$

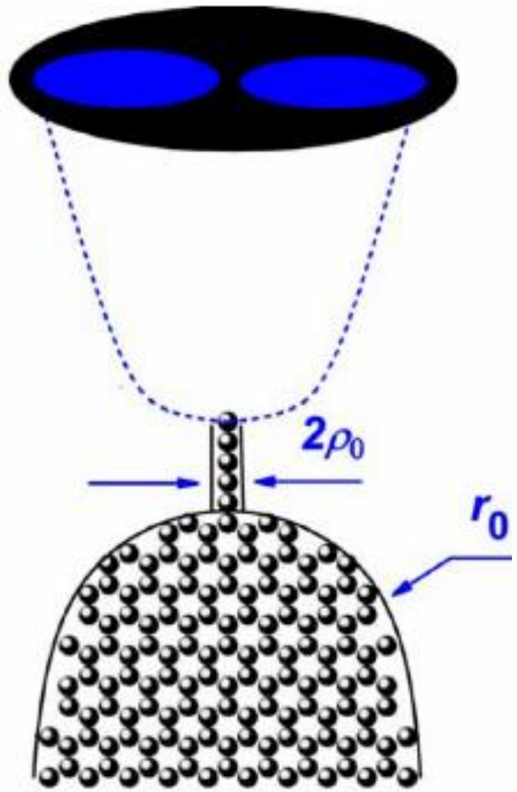
$$\oint mv^2 dt = \oint p dl = h$$ Bohr-Sommerfeld quantization

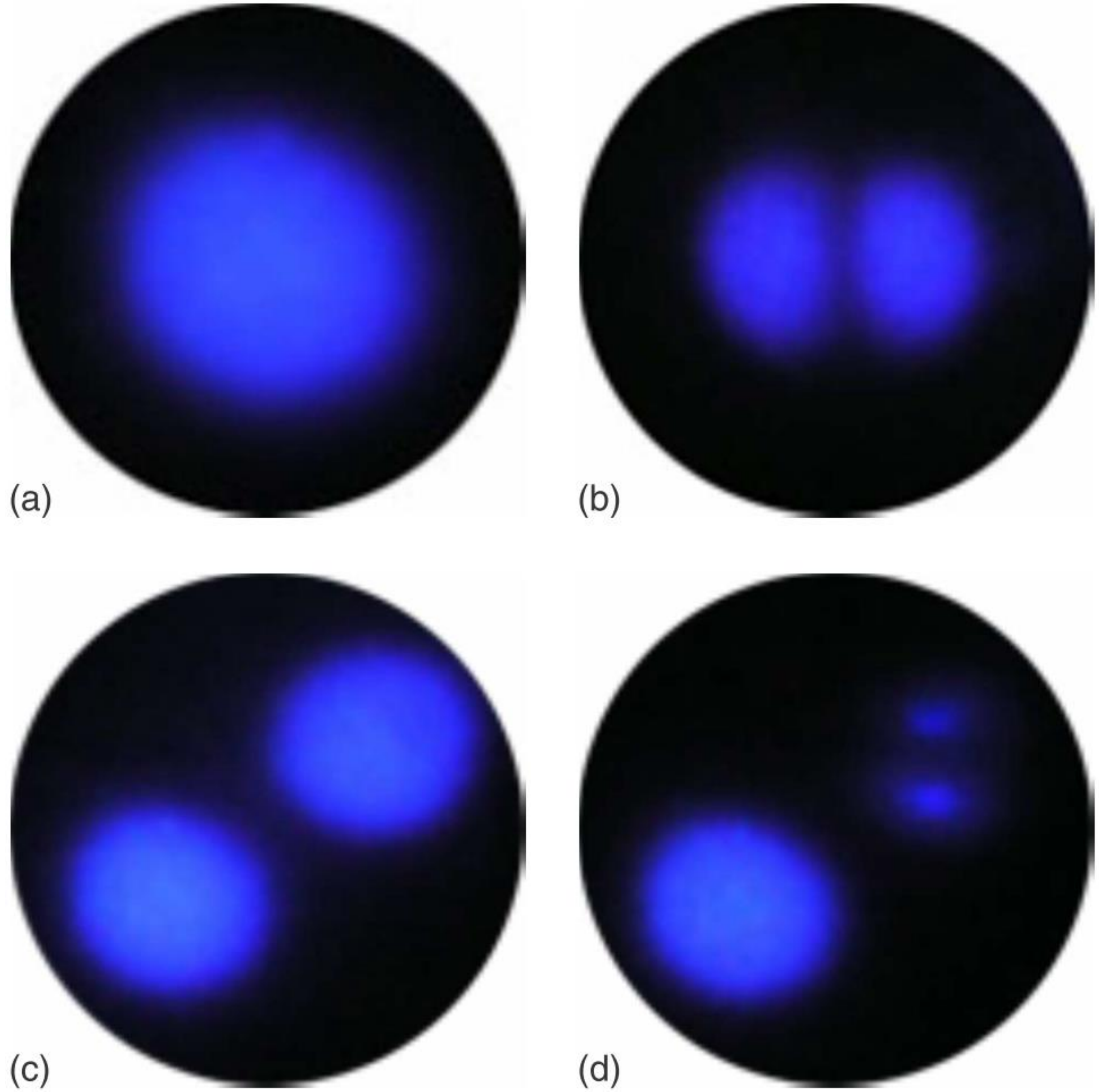
$$E = -\frac{13.6eV}{(k+l)^2} \qquad lh = \oint L d\theta$$

SR corrections?: $fh = m_0 c^2 \sqrt{1 - v^2/c^2} \approx \left(m_0 c^2 - \frac{1}{2} m_0 v^2\right)$

Imaging the atomic orbitals *of carbon atomic chains with* ***field-emission electron microscopy***,

I. M. Mikhailovskij, E. V. Sadanov, T. I. Mazilova, V. A. Ksenofontov, O. A. Velicodnaja (Phys. Rev. B, 2009)





"**Photos of orbitals**" obtained by **averaging** over **positions of single electrons,** pulled from the last atom due to electric potential.

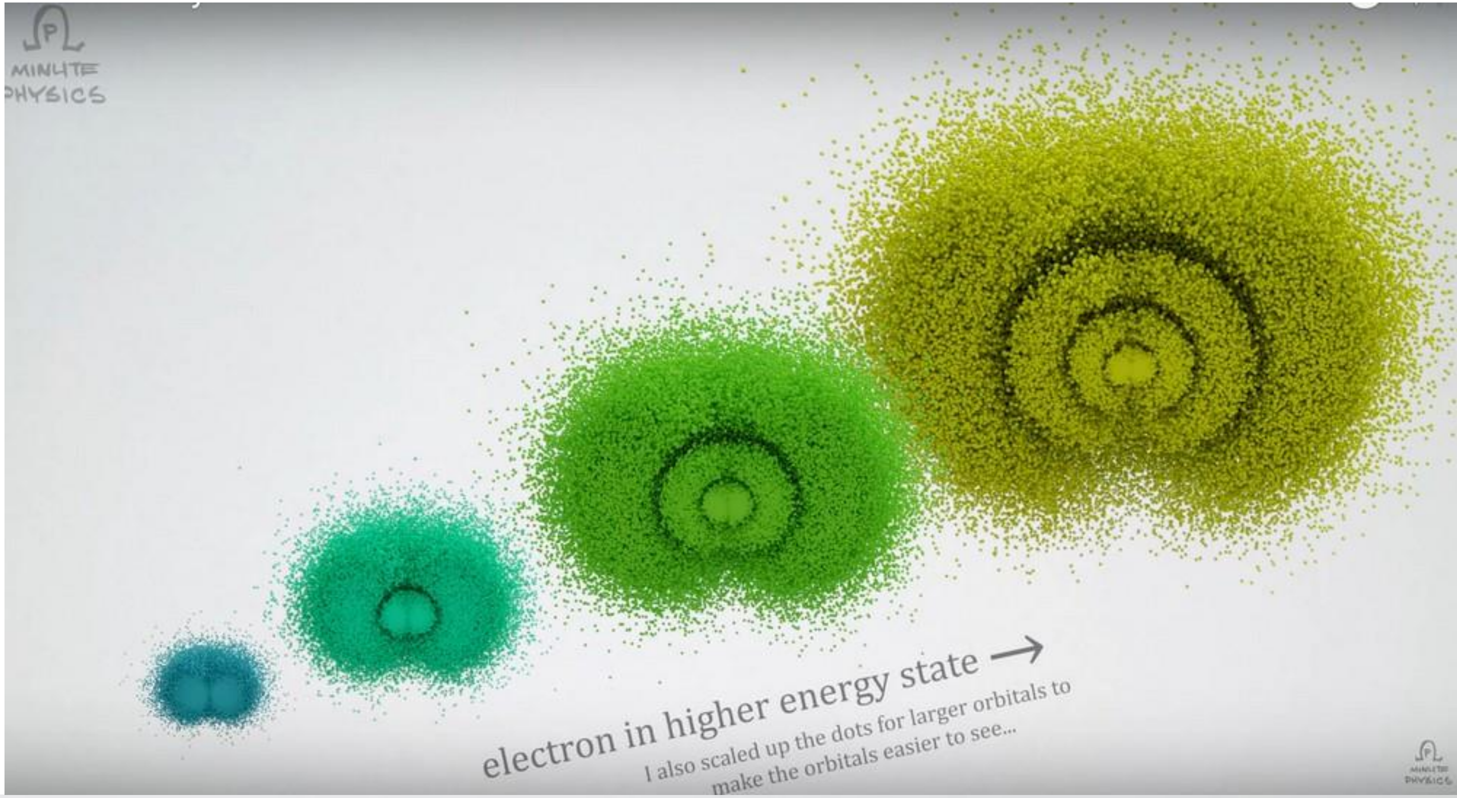
Bohmian electron trajectories

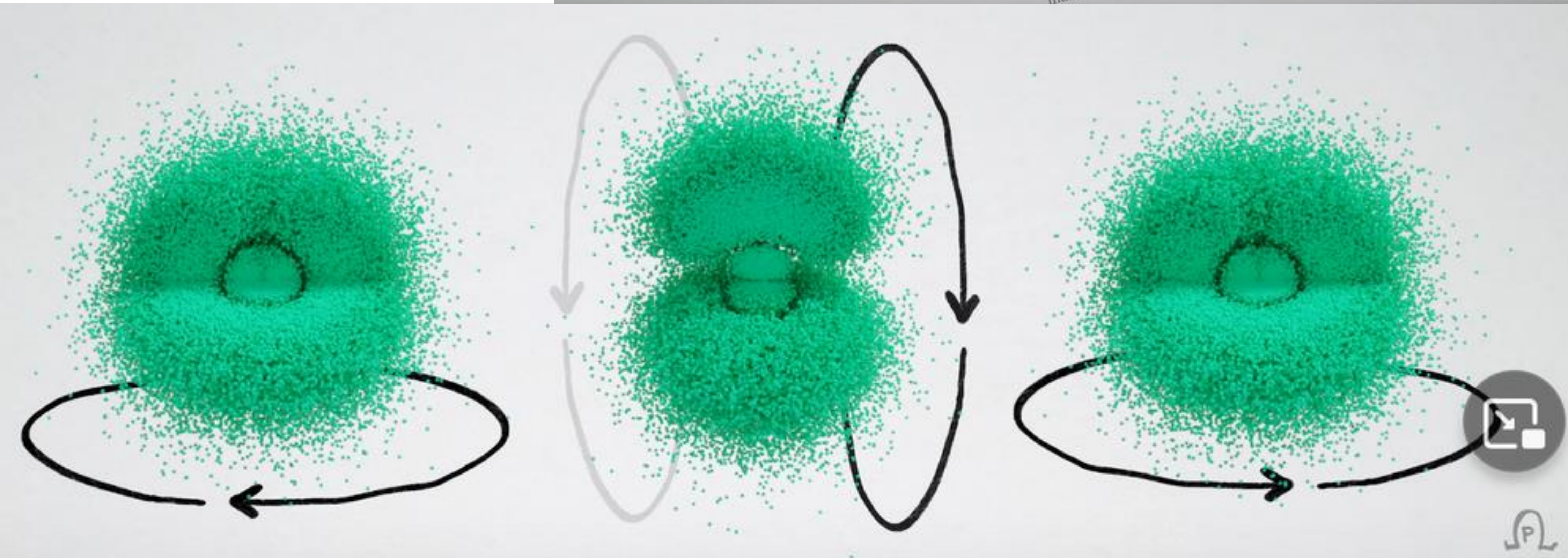
in atomic orbitals –

evolution due to momentum:

“A Better Way To Picture Atoms”

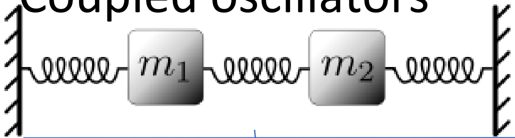
https://youtu.be/W2Xb2GFK2yc





Is there CM-QM boundary? "Classical" ~~vs~~ perspective "Quantum"

Coupled oscillators



$m\ddot{u}_1 = -Cu_1 + C(u_2 - u_1)$

$m\ddot{u}_2 = -Cu_2 + C(u_1 - u_2)$

"unitary evolution" of normal modes

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \mathrm{Re}\left(c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_1 t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_2 t}\right)$$

Crystal

1D lattice of oscillators $m\ddot{u}_x = C(u_{x+1} - 2u_x + u_{x+1})$

Phonons – quasiparticles in QFT

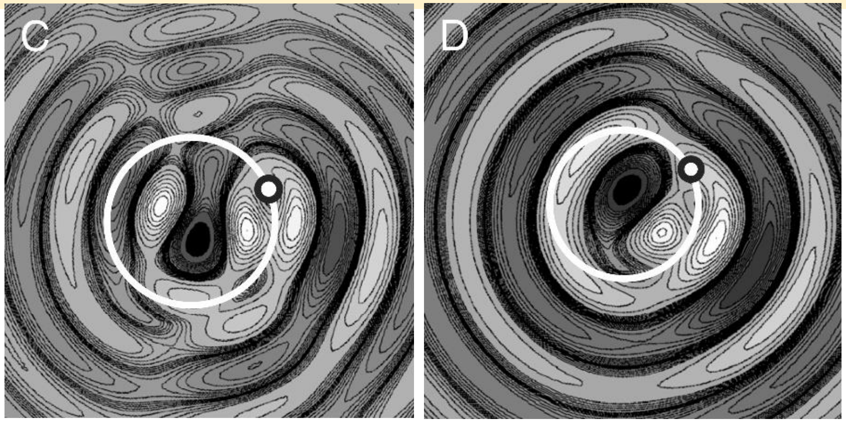
$u_x = \mathrm{Re}\left(\sum_k c_k\, e^{i(kx - \omega_k t)}\right)$

Infinitesimal lattice limit → field, like in **hydrodynamics**

e.g Couder's

Quantized

orbits



Schrödinger equation:

1) coupled **standing wave** $\psi_0 e^{iEt/\hbar}$, resonant with electron's clock,

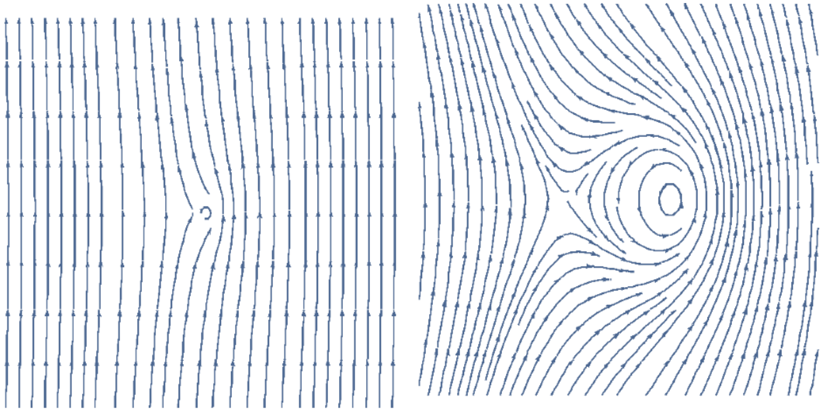
2) $\rho = |\psi|^2$ **statistics**, e.g. MERW?, $\arg\ \psi$: expected $\Delta\psi$?

Stable field configurations: **solitons** (topological?)

Varying

number of

particles

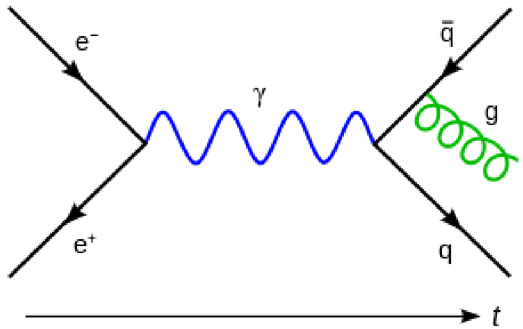


perturbative QFT "algebra for particles"

ensemble of scenarios: Feynman diagrams

$\xrightarrow{description?}$

$\xleftarrow[fields?]{}$



Experimental boundaries for size of electron?

Dehmelt 1988, Penning trap (Nobel in 1989):

$R < 10^{-22} m$ **extrapolating from g-factor**

"(...) electron as formed by very tightly binding together **three smaller and much heavier new fermions** [Brodsky, Drell, 1980] (...)"

RMS: root mean square radius

Neutron (udd): $g \approx -3.8 \qquad < r_\mathrm{n}^2 > \approx -0.1\ \mathrm{fm}^2$

Classically: $g = \frac{2m}{q}\frac{\mu}{L} = \frac{2m}{q}\frac{\int A dI}{\omega I} = \frac{m}{q}\frac{\int \rho_q(r)\, r^2 dr}{\int \rho_m(r)\, r^2\, dr}$

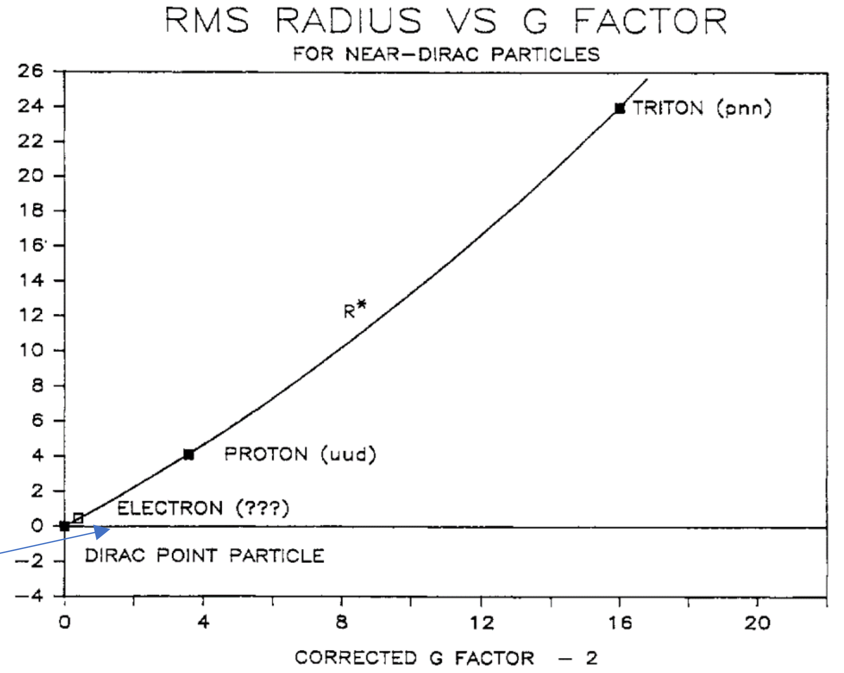


Fig. 8. Normalized RMS radius $R^* = R/\lambda\!\!\!^{-}_C$ vs. corrected g-factor minus 2 for near-Dirac particles [3]. A parabola has been fitted to the data points. Recent theories conjecture that the electron, similar to proton and triton, is composed of three smaller fermions. The data point at the origin represents a Dirac point particle of finite arbitrary charge and mass.

Electron-positron cross-section:

Which energy should we use???

Lorentz contraction, $\sigma \propto \gamma^{-2}$

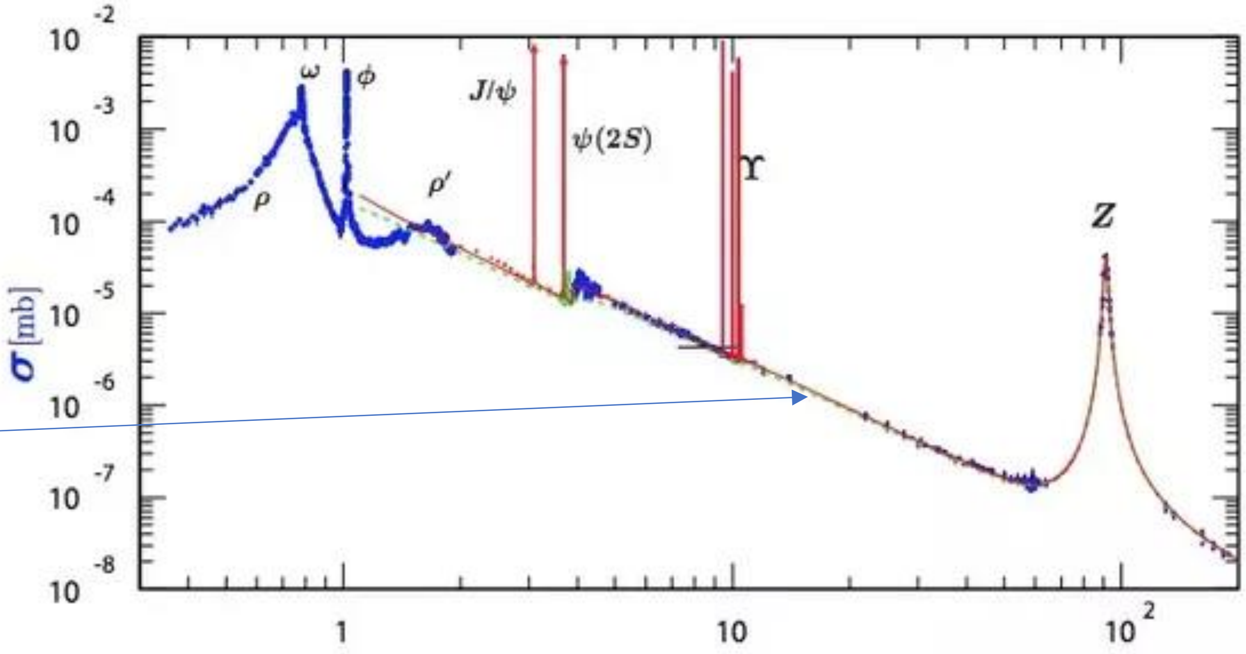
line in log-log, plus resonances

$\approx$ 1nb for 10GeV $\qquad \gamma \approx 10000$

$\approx$ 100nb for 1GeV $\qquad \gamma \approx 1000$

Extrapolating to resting: $\gamma = 1$

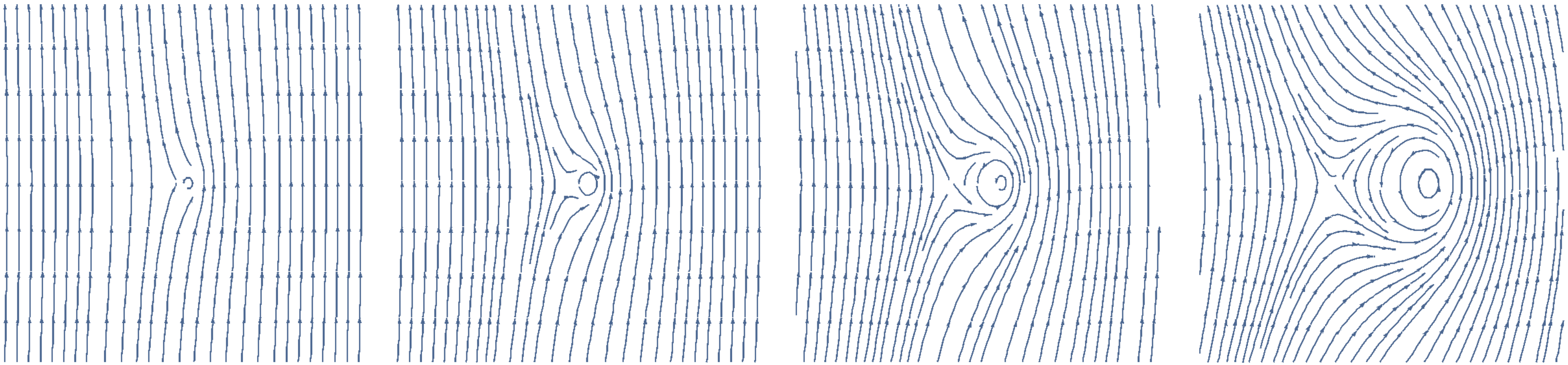
we get 100 mb: $r \approx 2$ fm circle?

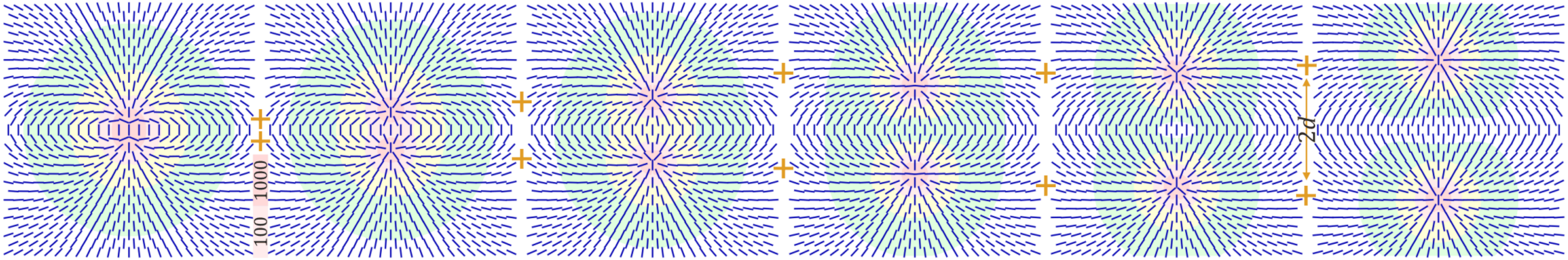


electromagnetism	**topological charge/index/degree** (slides)
charge conservation	the number of **covering** $\boldsymbol{S^{d-1}} \to \boldsymbol{S^{d-1}}$ **Hopf thm**: f, g are homotopic $\Leftrightarrow$ $deg\ f = deg\ g$ **Poincare-Hopf thm**.: $\sum_i deg_{x_i} = \chi(S)$ **Riemann-Hurwitz thm**.: $\chi(k \text{ covering of } S) = k\,\chi(S)$
Gauss law: **charge** inside $S = \oint_S E \cdot dS$ charge is real number	**argument principle in 2D:** $\oint_C (\ln f)' dz = 2\pi i(N-P)$ general - quantized **Gauss-Bonnet theorem**: $\oint_S K dS = 2\pi\,\chi(S)$ (K – curvature)
Coulomb: same charges repel opposite attract $\propto 1/r^2$	to **reduce stress of the field** entire field causes attraction/repulsion

$\vec{\Gamma}_i = (\partial_i \vec{u}) \times \vec{u}$ $\quad \vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu$ $\quad \mathcal{L}_{EM} = -\frac{\alpha\hbar c}{16\pi} \vec{R}_{\mu\nu}\vec{R}^{\mu\nu}$ Faber's quantized EM

Local rotation axis, curvature, EM Lagrangian with $F_{\mu\nu} \sim R^*_{\mu\nu}$ $(E \leftrightarrow B)$

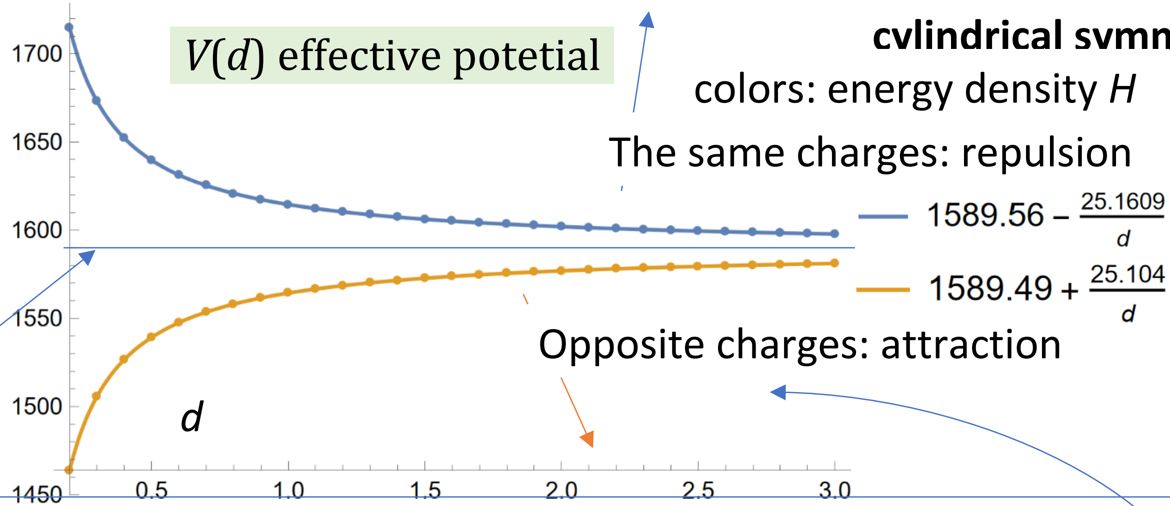




Field energy: ~Coulomb potent.

$$E \approx \frac{m_0}{\sqrt{1-v_1^2}} + \frac{m_0}{\sqrt{1-v_2^2}} + V(d)$$

cutoff ϵ around singularities: be **regularized** to rest masses invariant: SRT scaling



cylindrical symmetry

colors: energy density H

The same charges: repulsion

Opposite charges: attraction

```
cos = 1 + (z - d) / Sqrt[(z - d)^2 + r^2] - (z + d) / Sqrt[(z + d)^2 + r^2]; (*Manfried Faber dipole ansatz*)
n = {Sqrt[1 - cos^2] x / r, Sqrt[1 - cos^2] y / r, cos} /. r → Sqrt[x^2 + y^2]; (* cylindrical symmetry *)
M = KroneckerProduct[n, n]; dM = {D[M, x], D[M, y], D[M, z]}; (*vector n → matrix M field*)
H = Simplify[Sum[Total[(dM[[i]].dM[[j]] - dM[[j]].dM[[i]])^2, 2], {i, 2}, {j, i + 1, 3}]]; (*Hamilitonian*)
Es = Table[{d, NIntegrate[4 Pi * x (H /. {y → 0}) * Boole[x^2 + (z - d)^2 > 0.001], (* 0.001 cutoff *)
    {x, 0, Infinity}, {z, 0, Infinity}]}, {d, 0.1, 3, 0.1}]; (*integrate H: total field energy*)
ft = Fit[Es, {1, 1 / d}, d]; Show[Plot[ft, {d, 0.2, 3}], ListPlot[Es]] (*fit Coulomb potentential*)
```